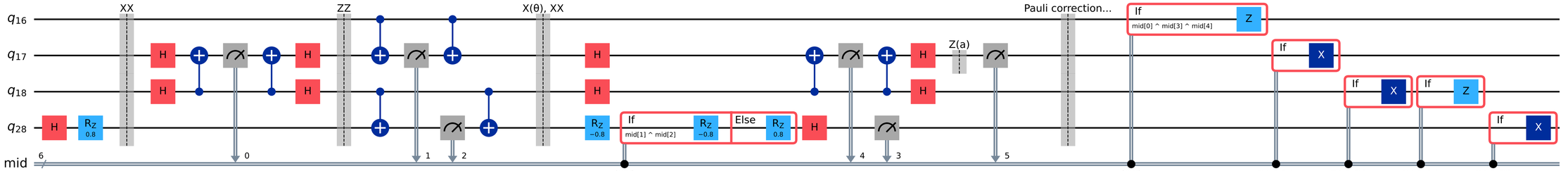
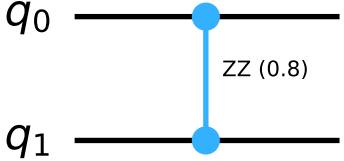
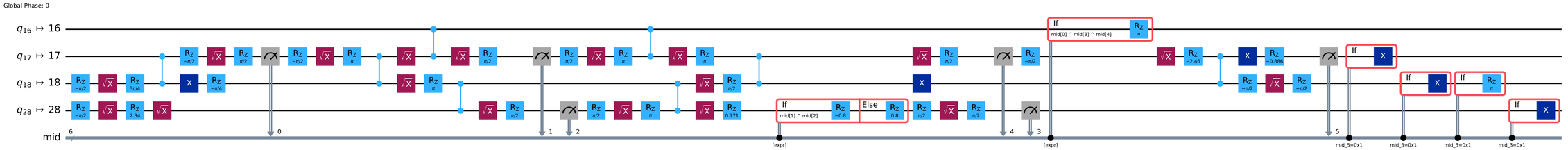
[Z2] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 6, 6 mid-circuit measurements, logical→physical [16, 18]



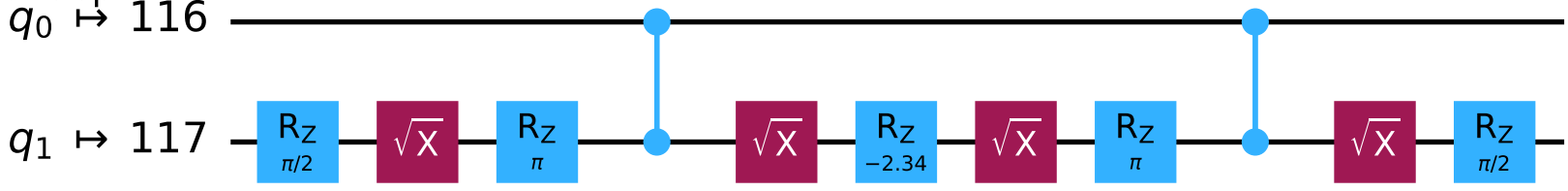
[Z2] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 1



[Z2] DYNAMIC circuit, fault-tolerant (|0> ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:27 sx:18 cz:8 measure:6 if_else:6 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1 (2 logical qubits)
2q depth 6, total depth 39

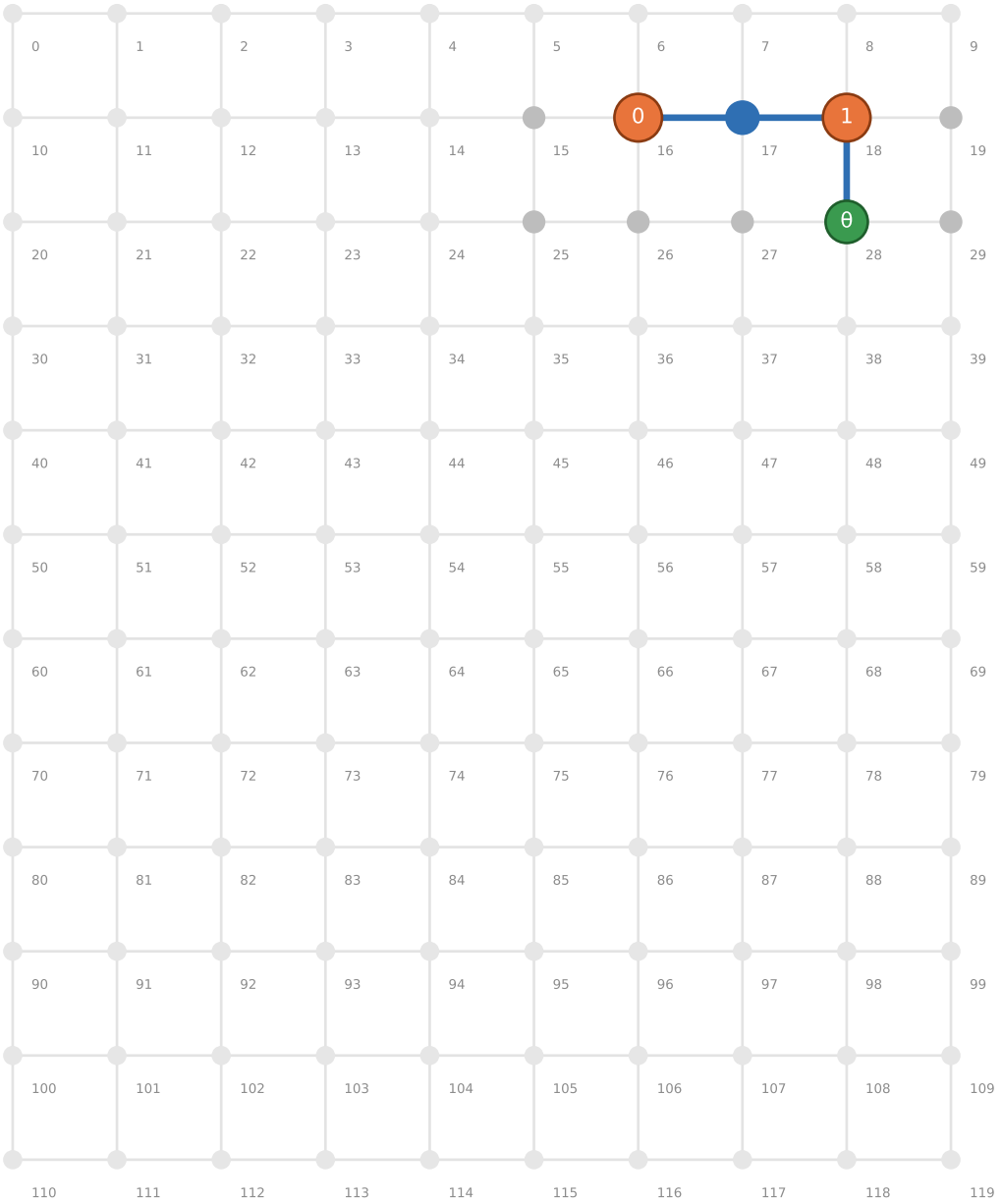


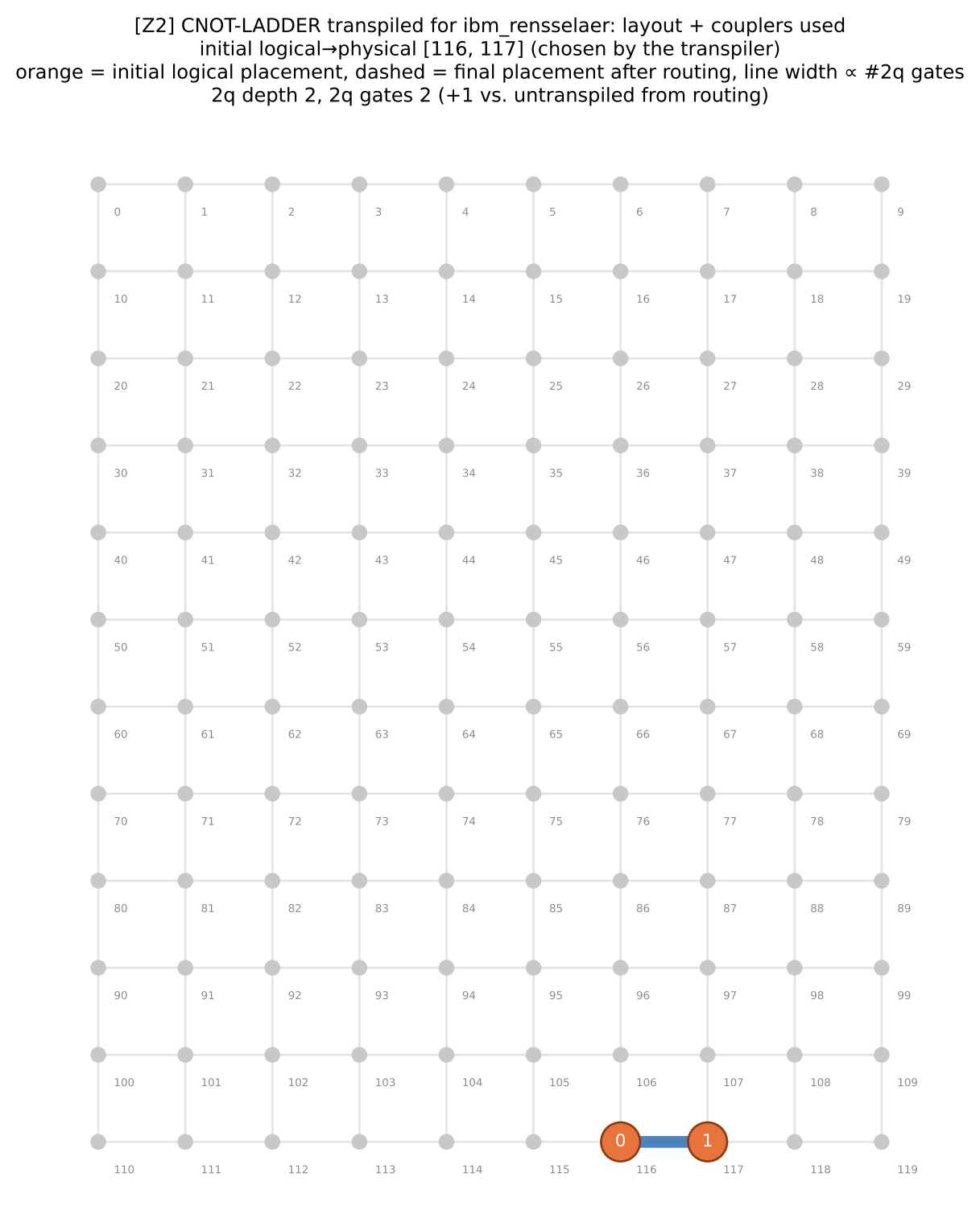
[Z2] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:5 sx:4 cz:2)
Global Phase: 0
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 2, total depth 11



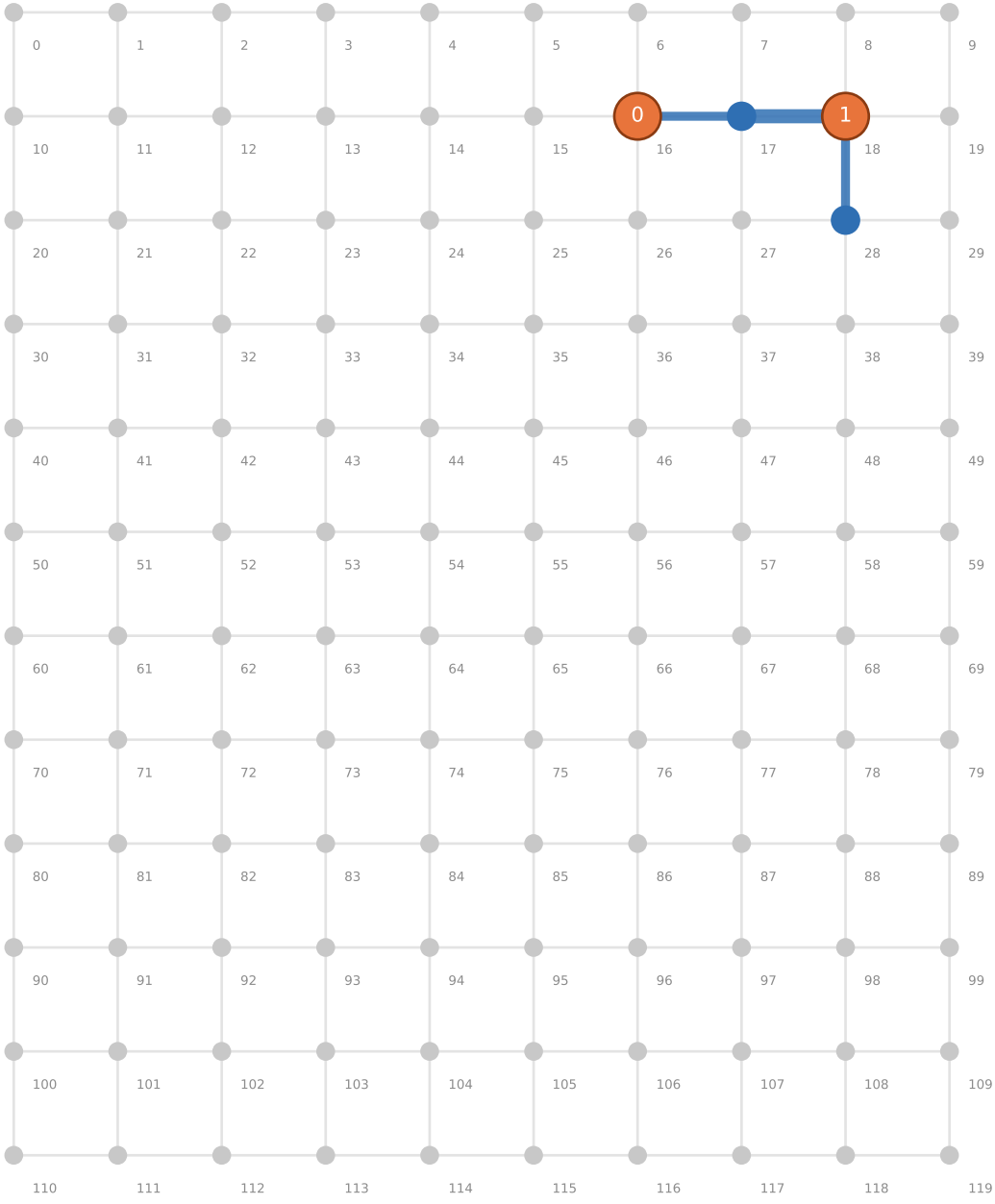
[Z2] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [16, 18]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1
chain length 3, 1 ancillas, 0 |0> fillers

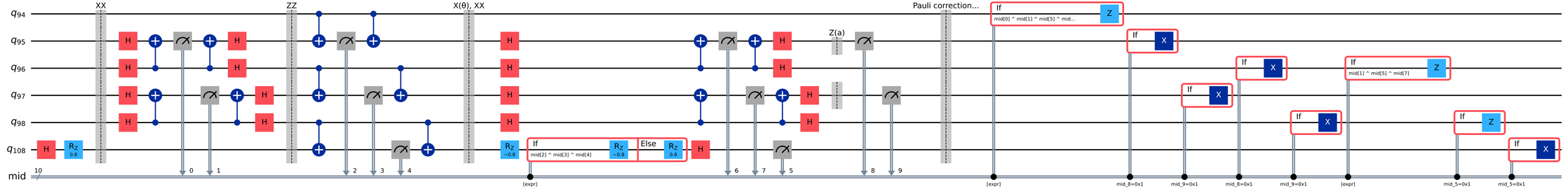



[Z2] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 2, 2q gates 2 (+1 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1

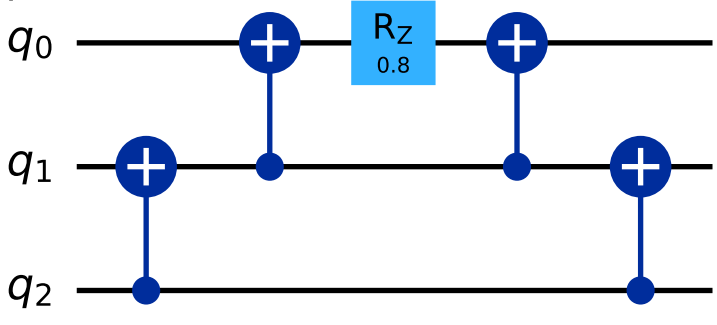
[Z2] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [16, 18]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 8



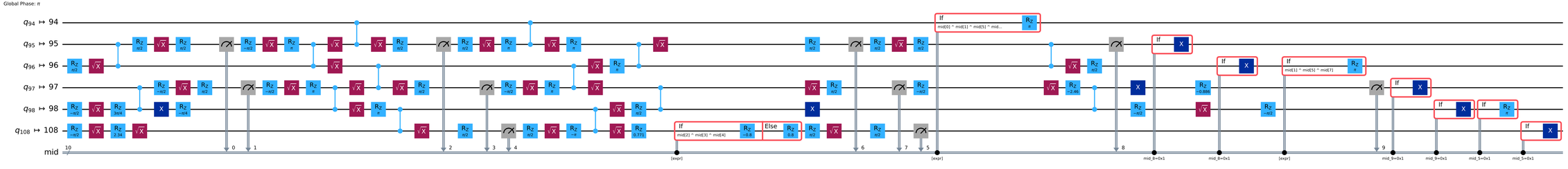
[Z3] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 6, 10 mid-circuit measurements, logical→physical [94, 96, 98]



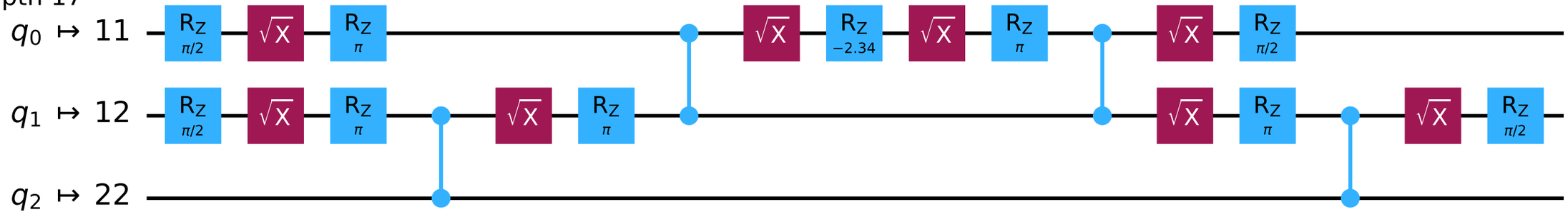
[Z3] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4



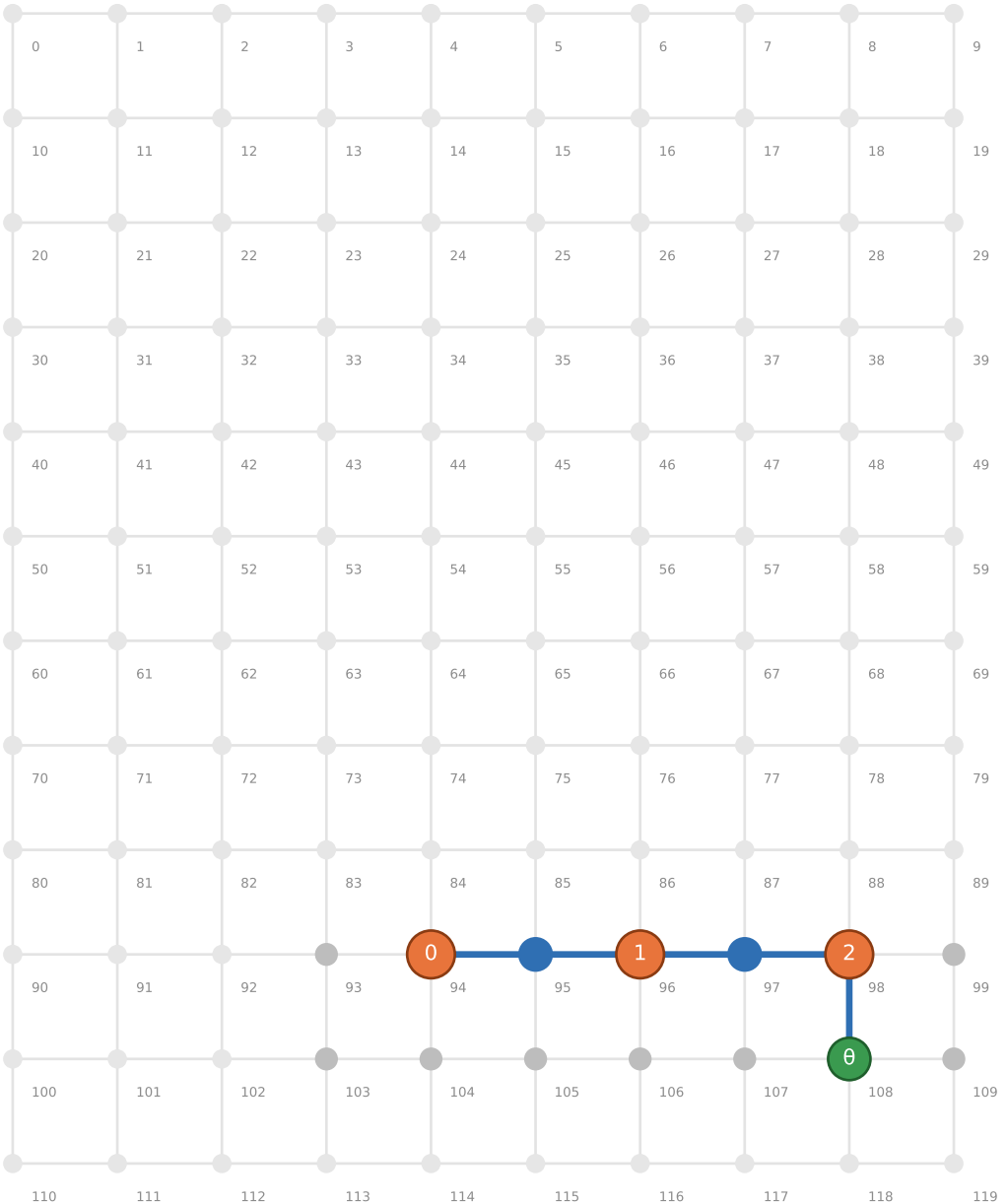
[Z3] DYNAMIC circuit, fault-tolerant (|0⟩ ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:40 sx:30 cz:14 measure:10 if_else:9 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 6, total depth 39



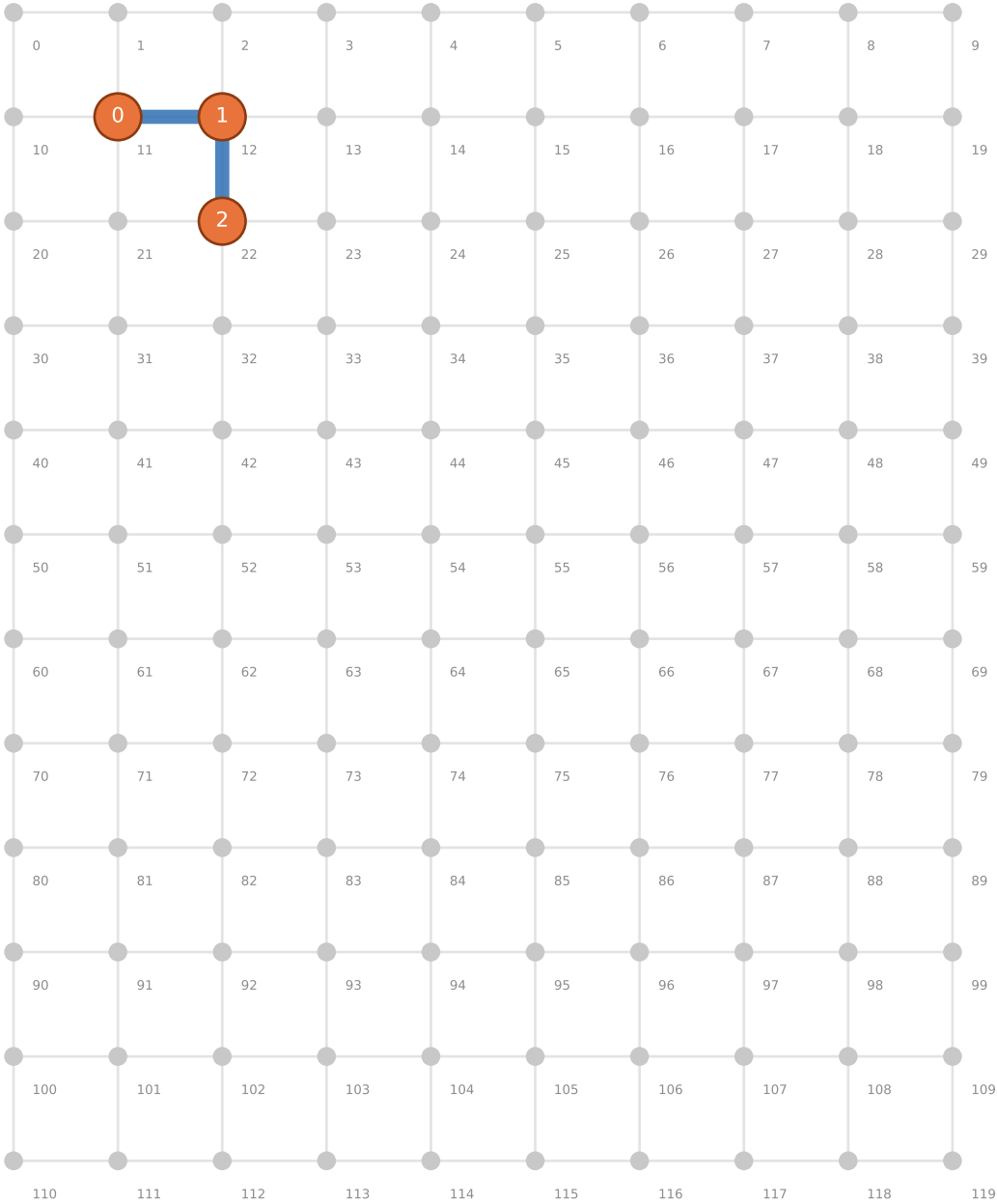
[Z3] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:10 sx:8 cz:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 (3 logical qubits)
2q depth 4, total depth 17



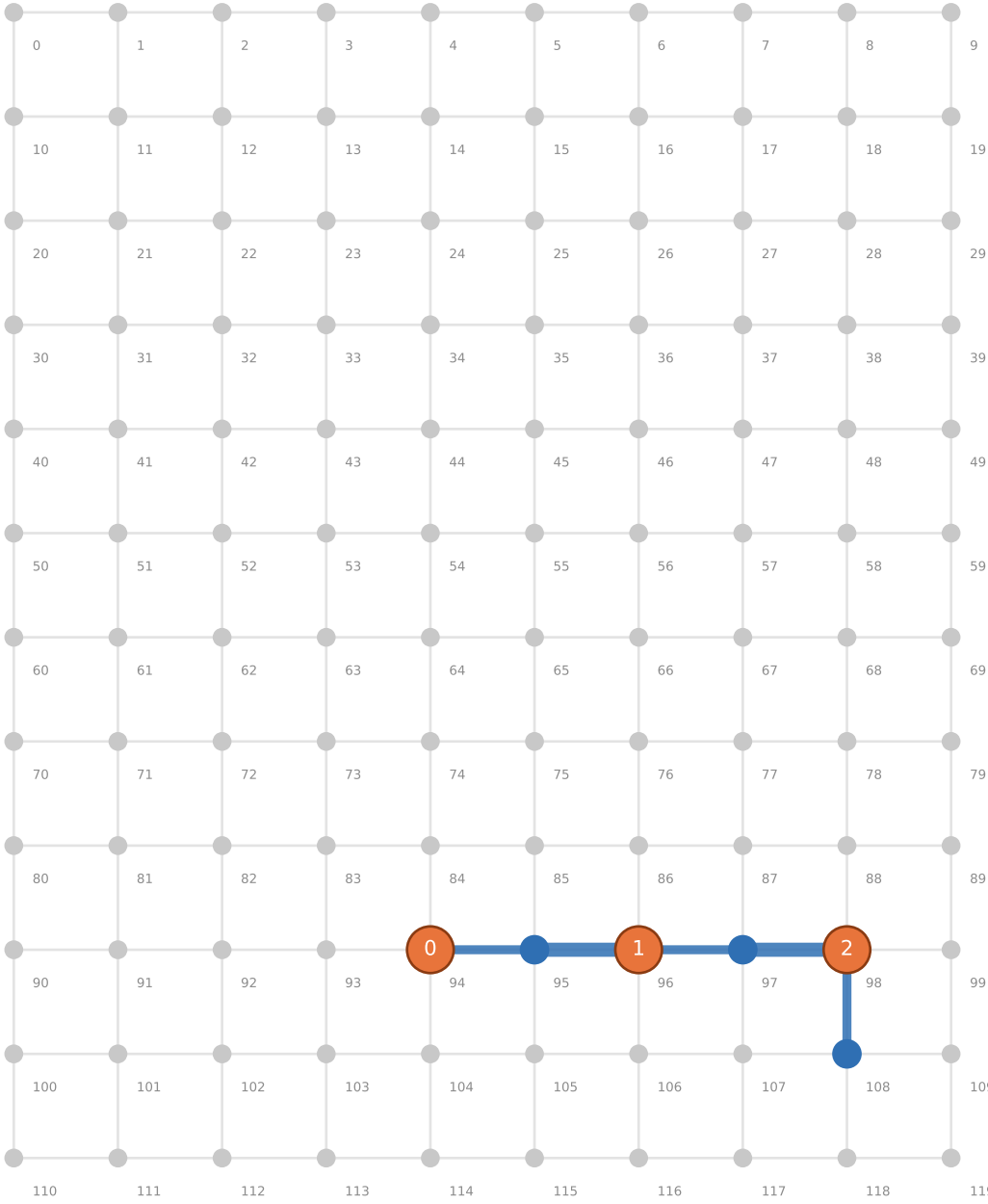
[Z3] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [94, 96, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain
term 0: Z0 Z1 Z2
chain length 5, 2 ancillas, 0 |0> fillers



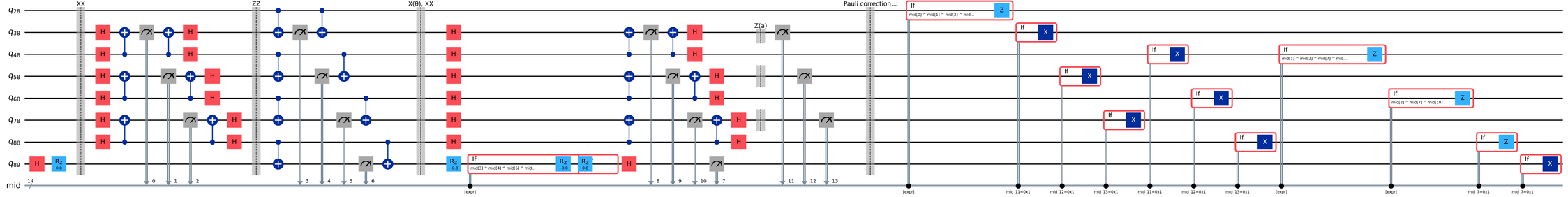
[Z3] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [11, 12, 22] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 4, 2q gates 4 (+0 vs. untranspiled from routing)



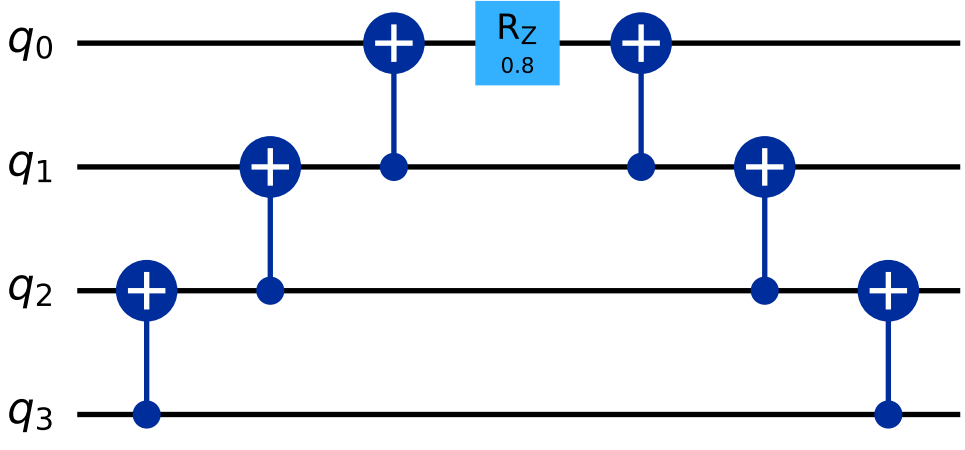
[Z3] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [94, 96, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 14



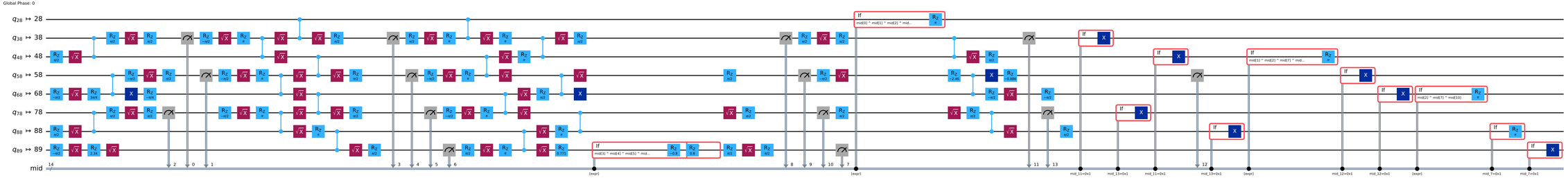
[Z4] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, 14 mid-circuit measurements, logical→physical [28, 48, 68, 88]



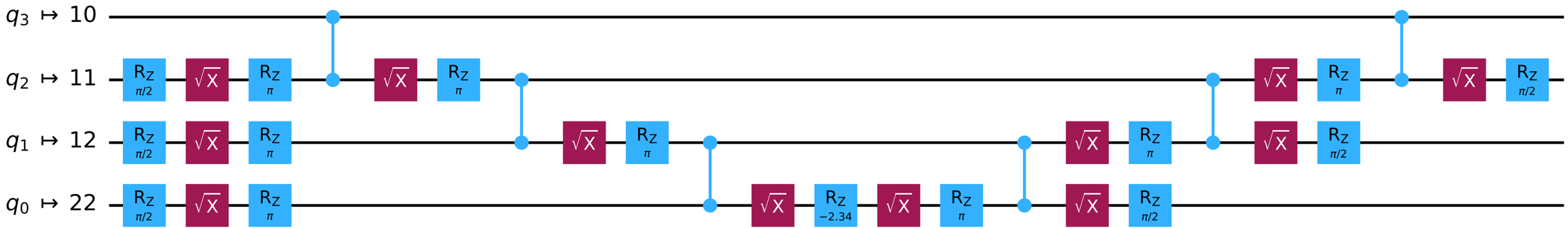
[Z4] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6



[Z4] DYNAMIC circuit, fault-tolerant (|0> ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:53 sx:42 cz:20 measure:14 if_else:12 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, total depth 39

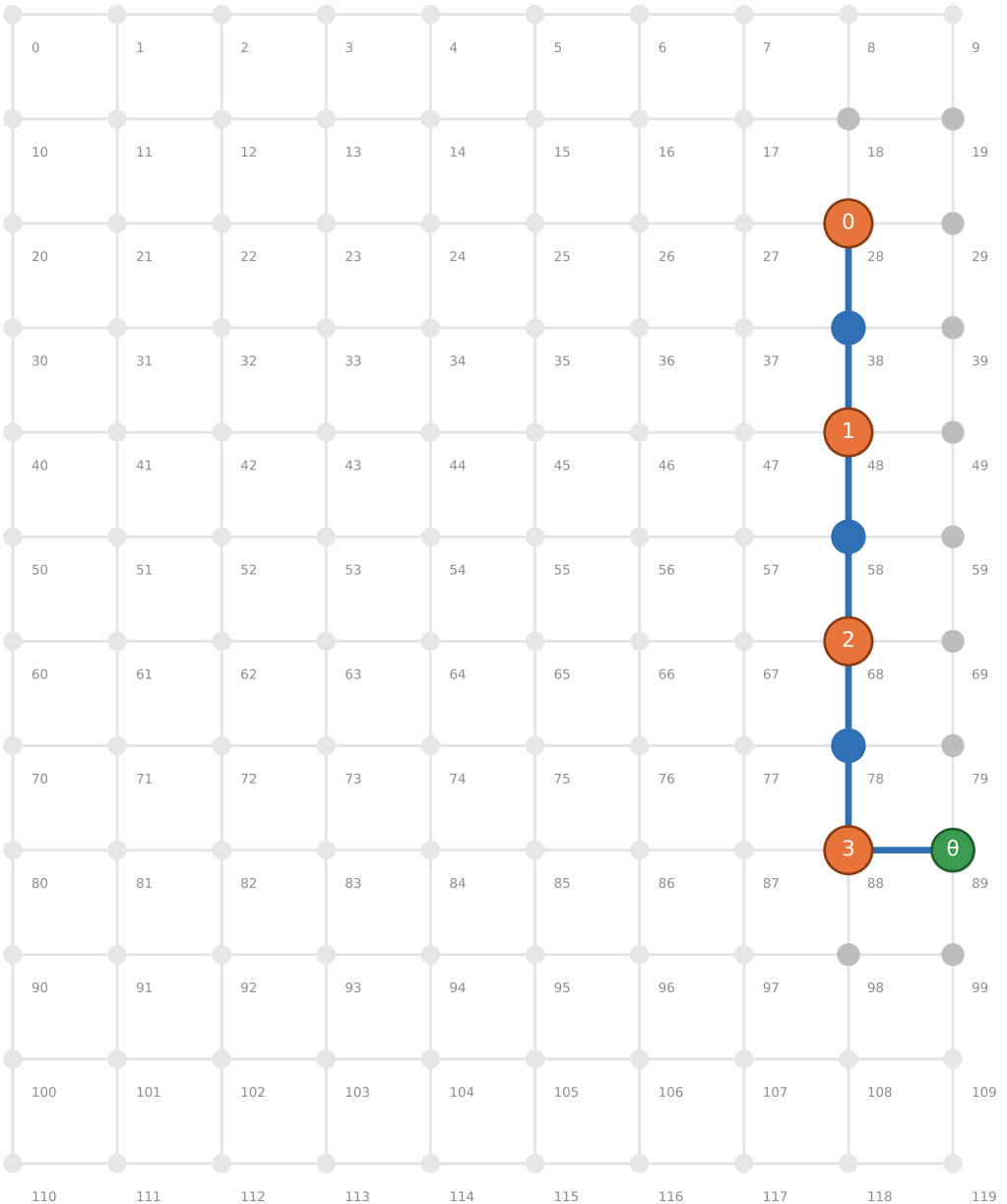


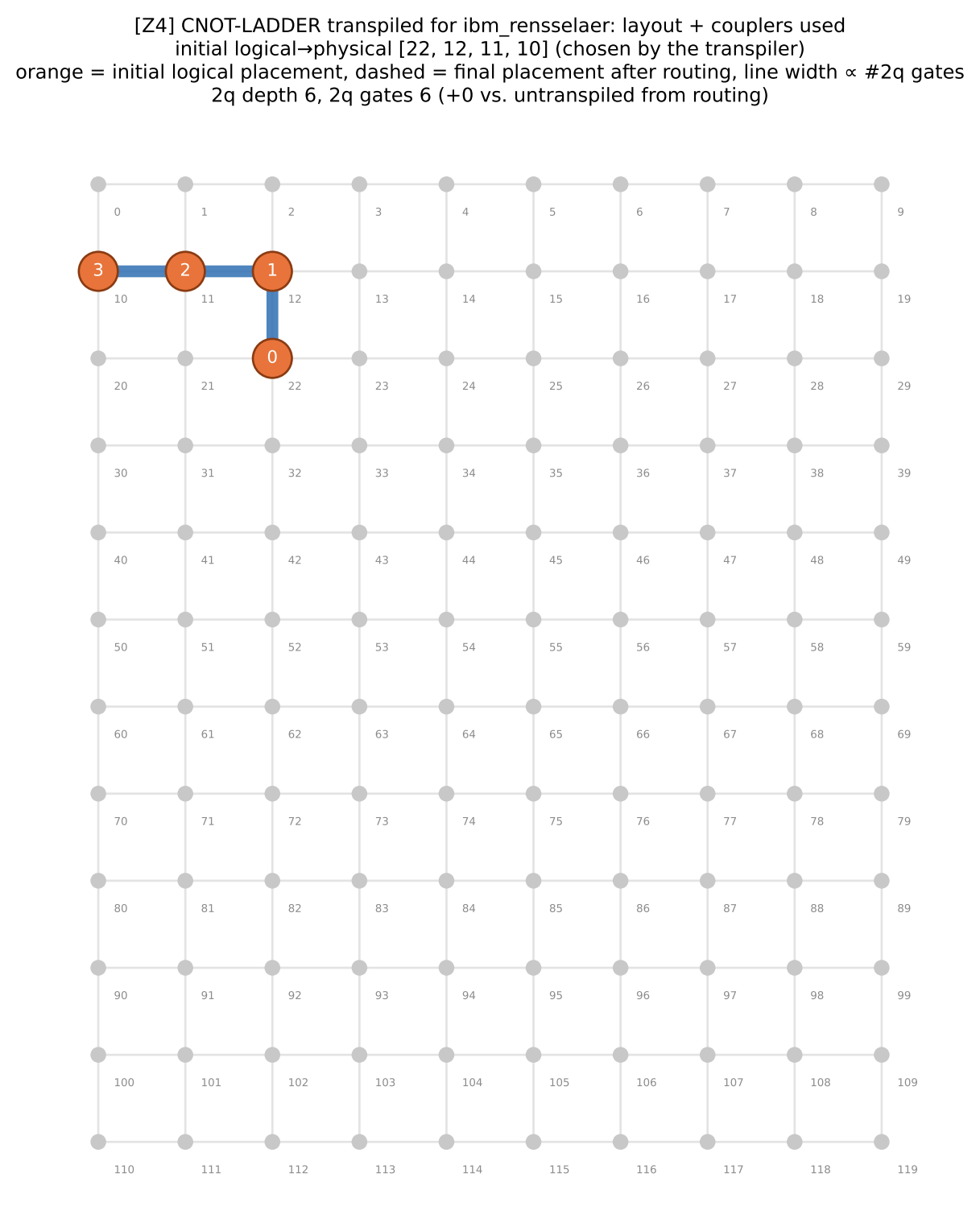
[Z4] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:15 sx:12 cz:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
Global Phase: 0
2q depth 6, total depth 23



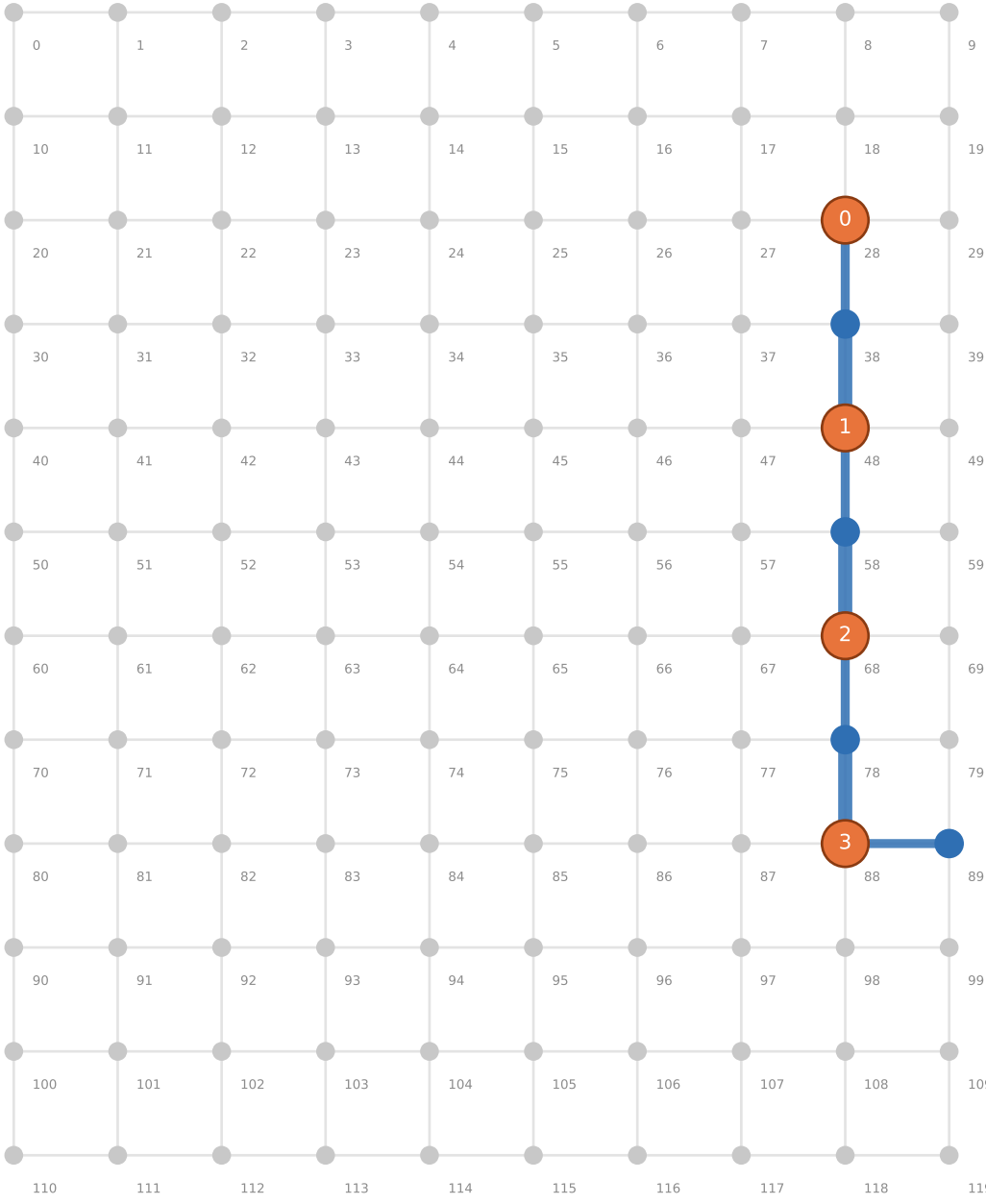
[Z4] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [28, 48, 68, 88]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3
chain length 7, 3 ancillas, 0 |0> fillers

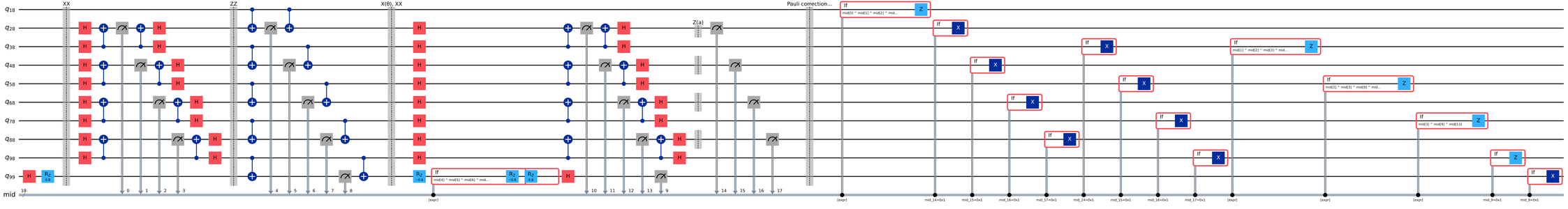



[Z4] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 12, 11, 10] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
3
2
1
0
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

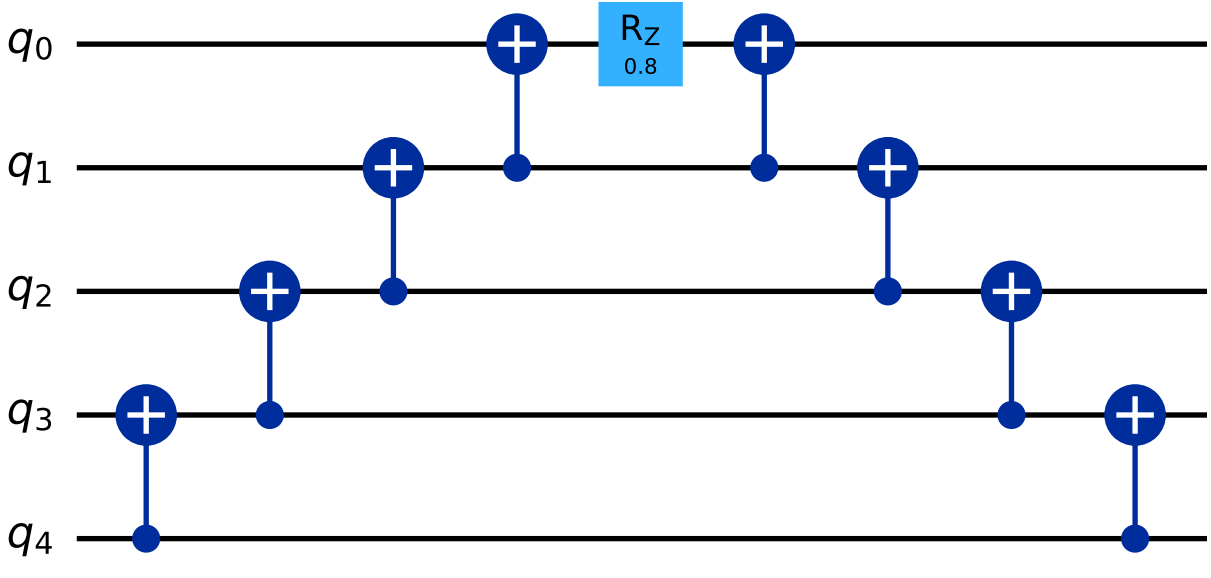
[Z4] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [28, 48, 68, 88]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 20



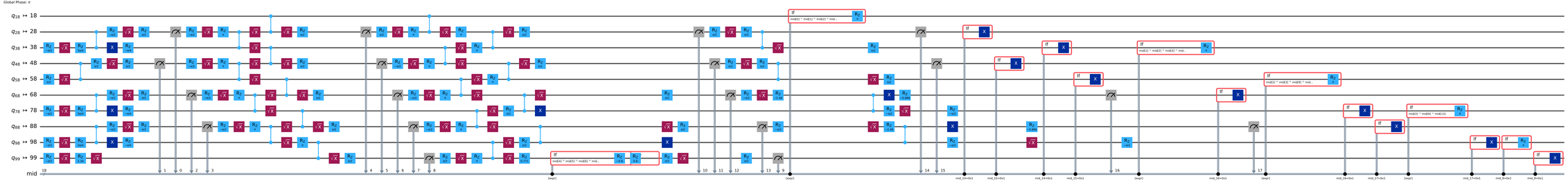
[Z5] DYNAMIC constant-depth circuit, fault-tolerant (|0> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 6, 18 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



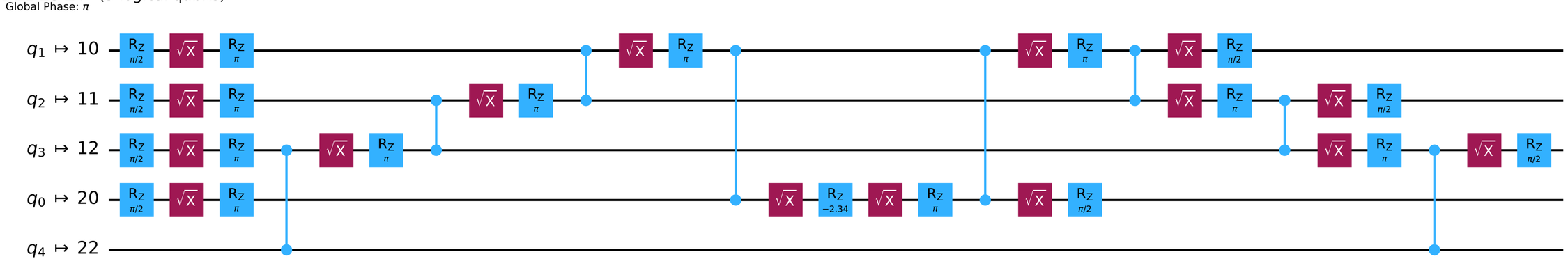
[Z5] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8



[Z5] DYNAMIC circuit, fault-tolerant (|0> ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:72 sx:54 cz:26 measure:18 if_else:15 x:7)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 6, total depth 39

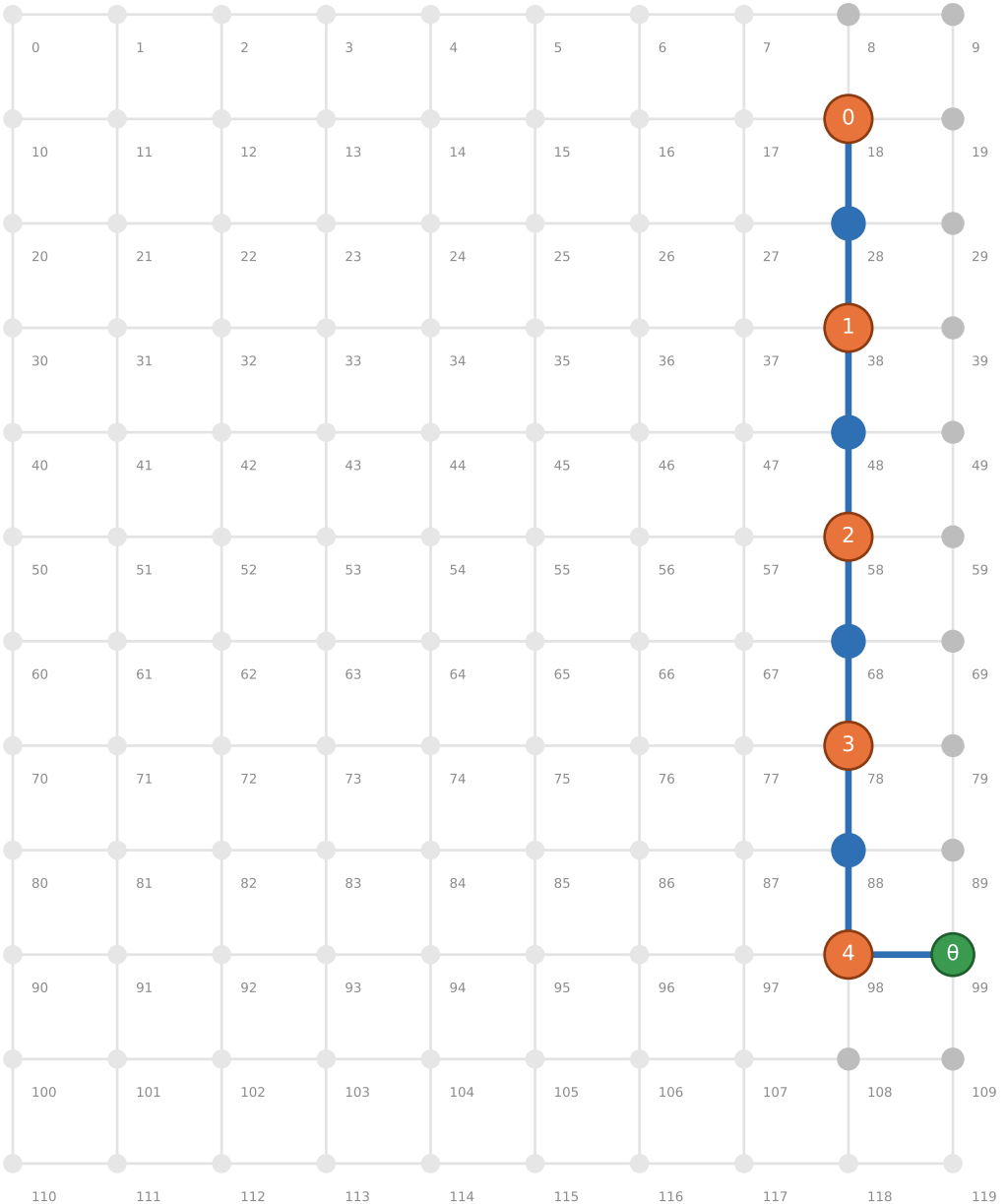


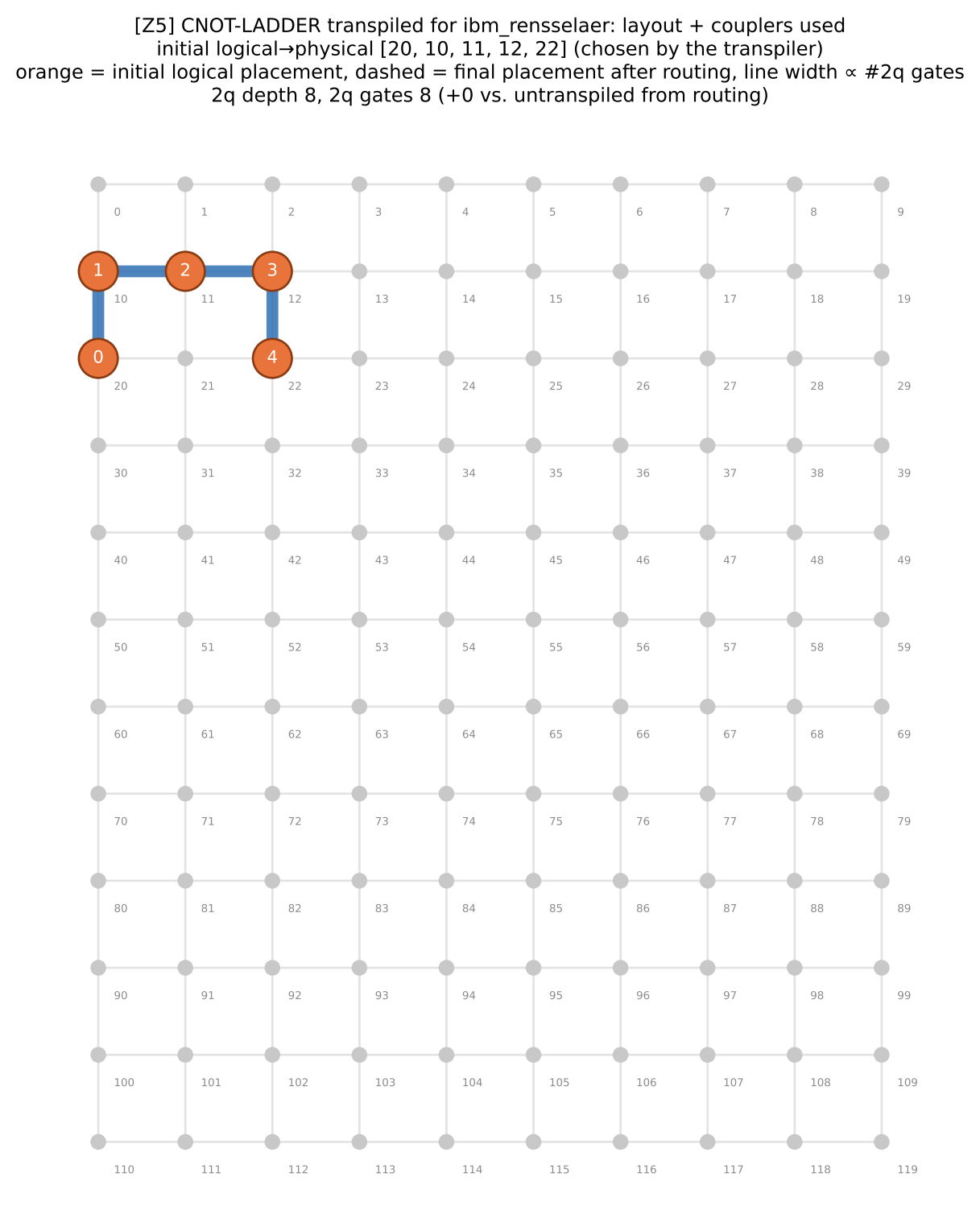
[Z5] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:20 sx:16 cz:8)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8, total depth 29



[Z5] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4
chain length 9, 4 ancillas, 0 |0> fillers




[Z5] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [20, 10, 11, 12, 22] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 8, 2q gates 8 (+0 vs. untranspiled from routing)
0
1
2
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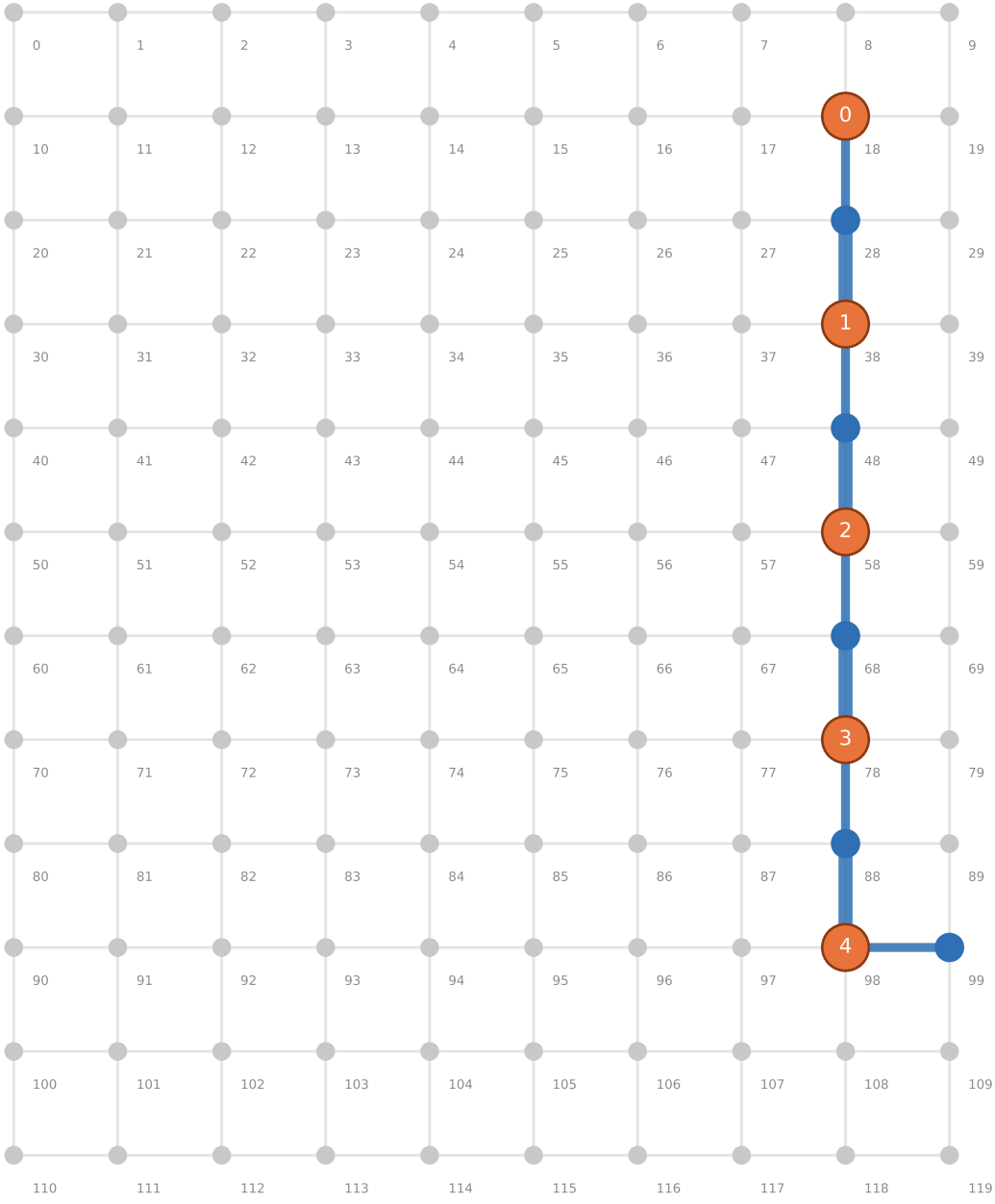
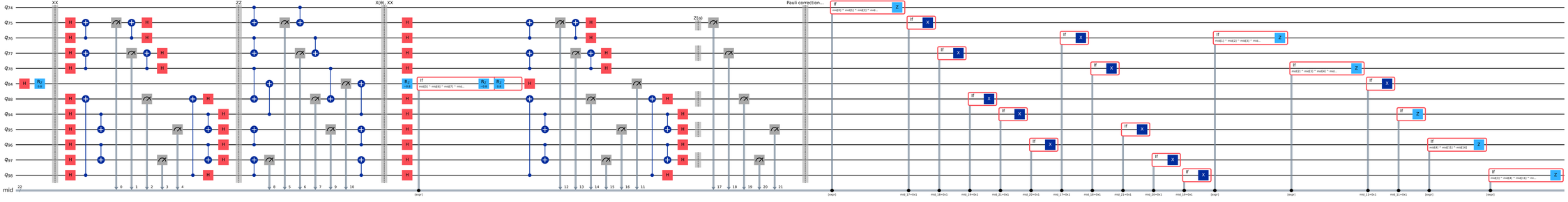
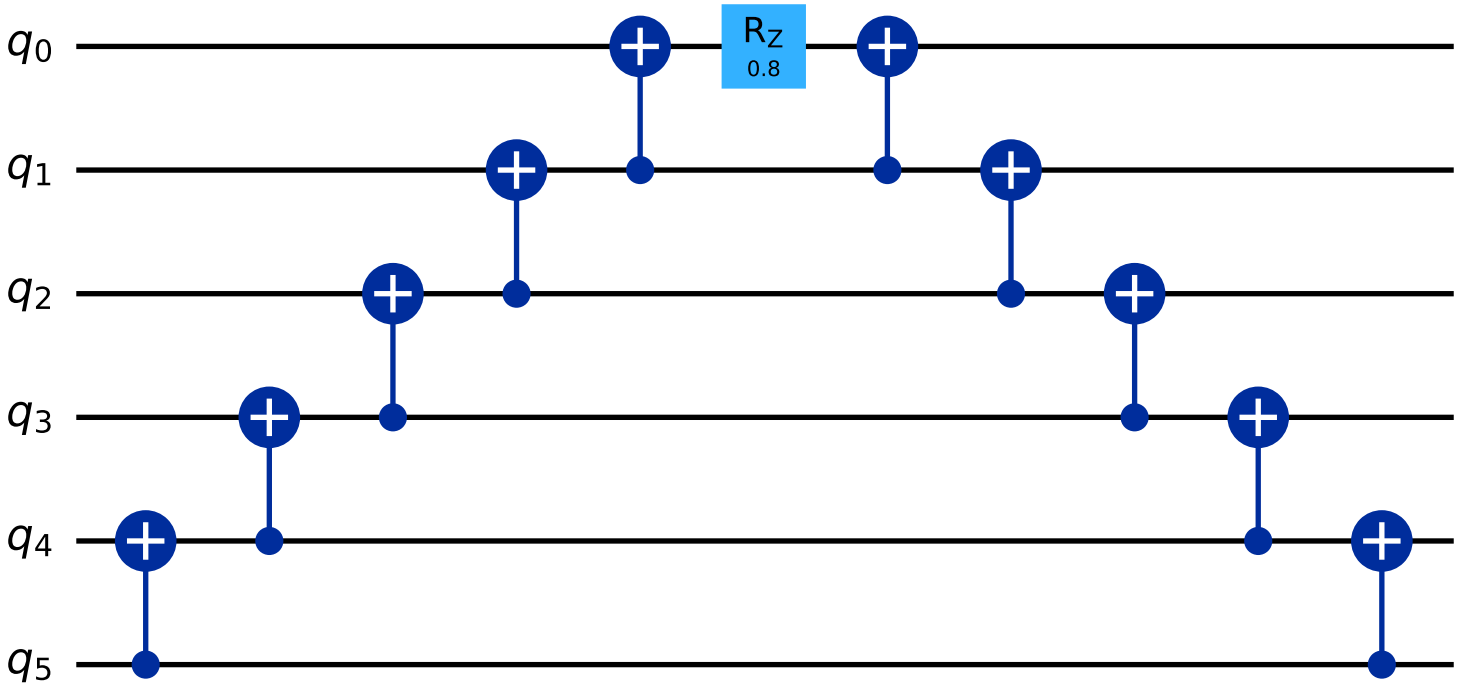
[Z5] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 26



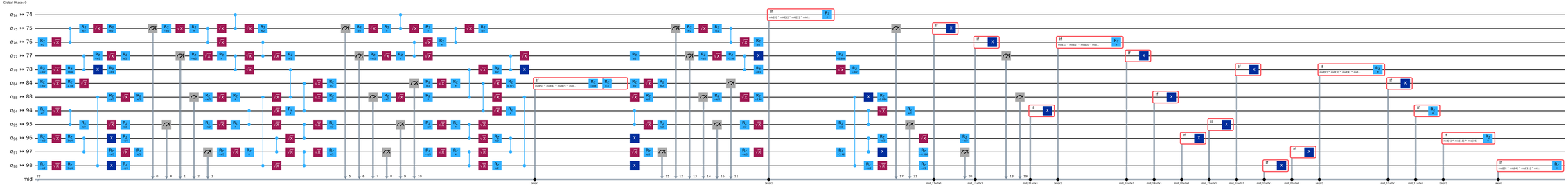
[Z6] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 6, 22 mid-circuit measurements, logical→physical [74, 76, 78, 98, 96, 94]



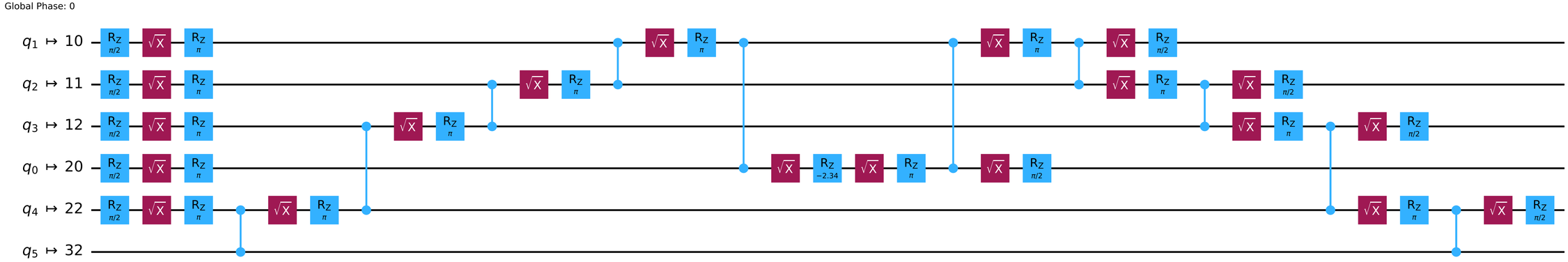
[Z6] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10



[Z6] DYNAMIC circuit, fault-tolerant (|0> ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:87 sx:66 cz:32 measure:22 if_else:18 x:9)
exp(-i θ P), θ=0.4: P = ZQ Z1 Z2 Z3 Z4 Z5 (6 logical qubits)
2q depth 6, total depth 41

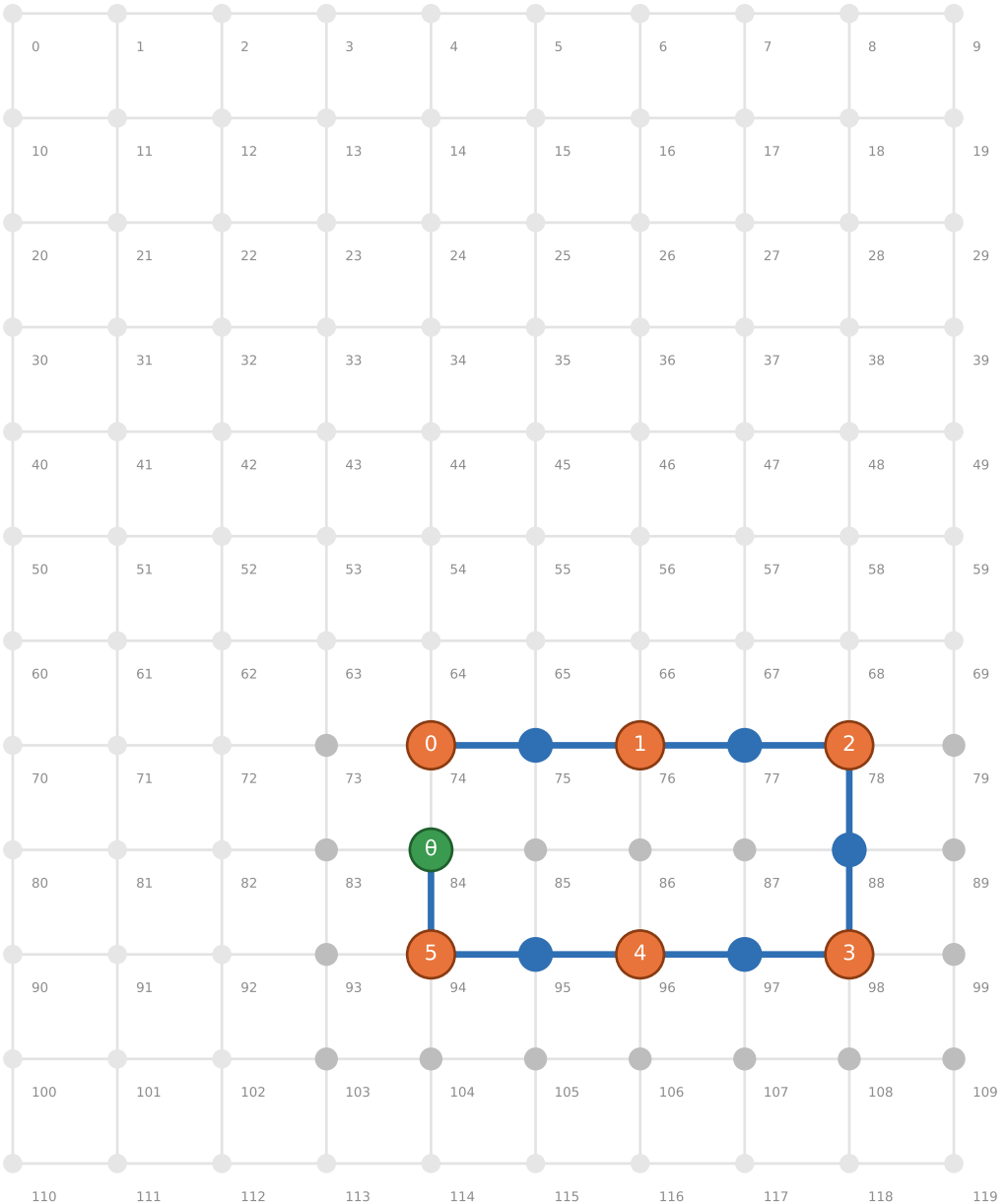


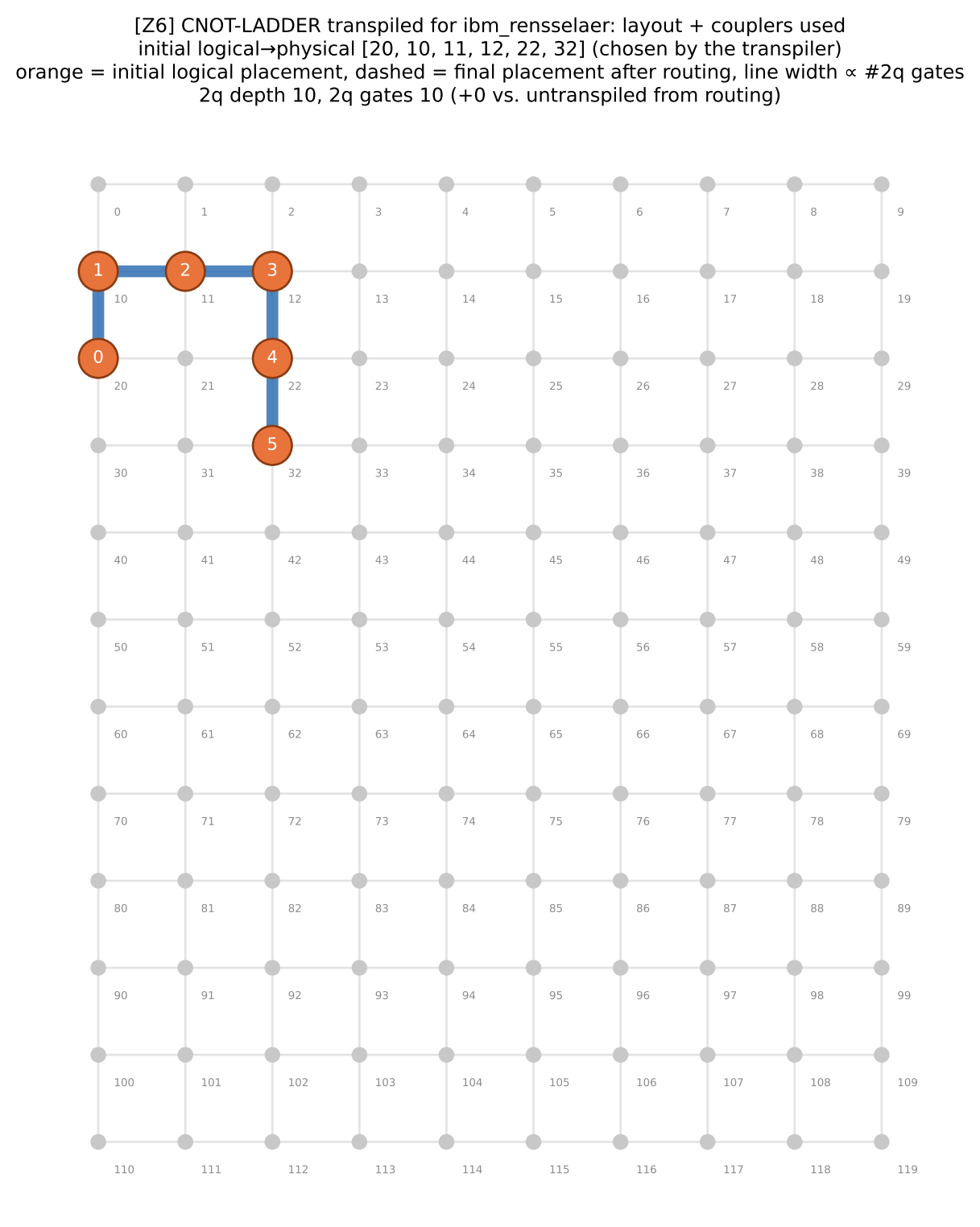
[Z6] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:25 sx:20 cz:10)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 (6 logical qubits)
2q depth 10, total depth 35



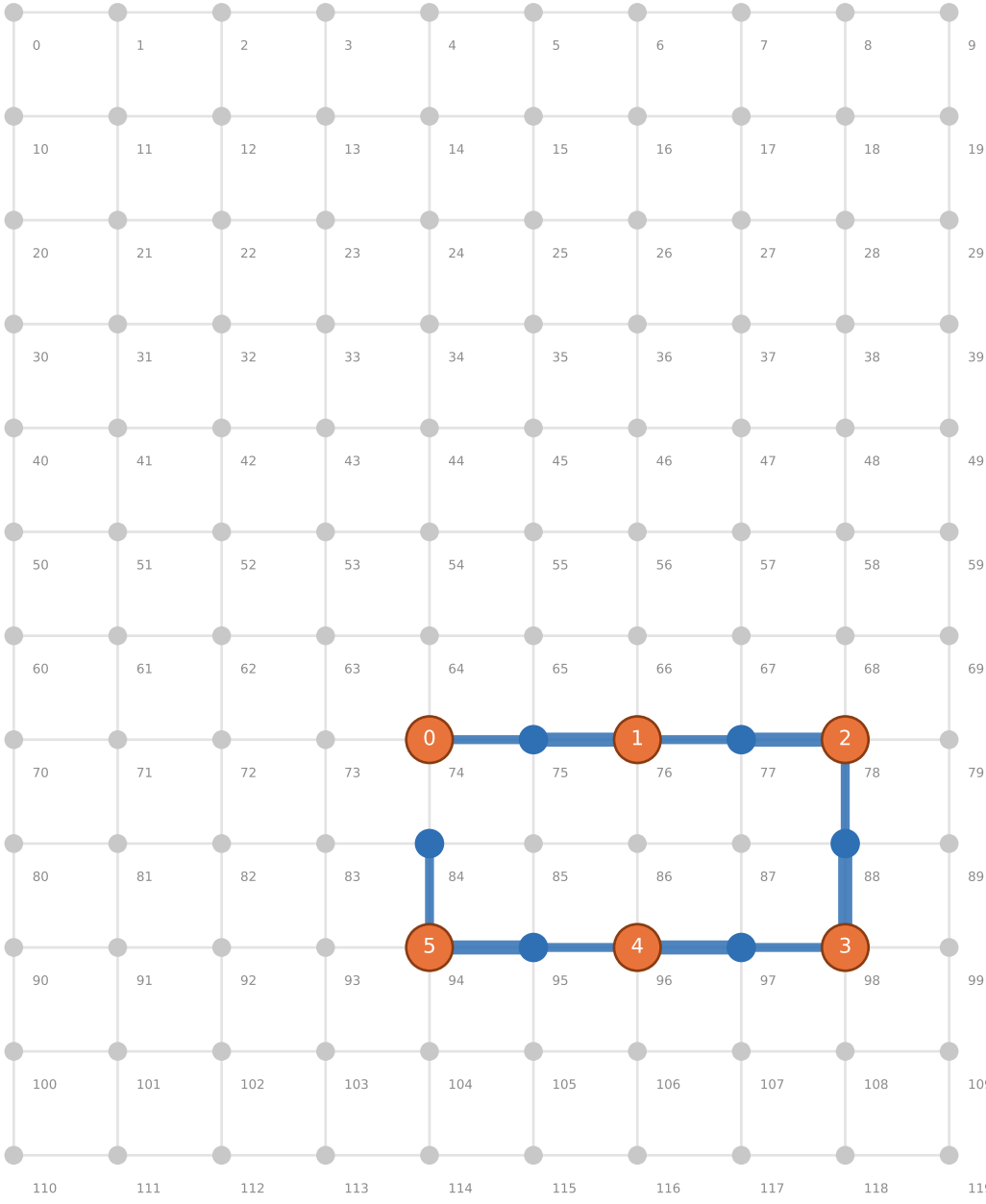
[Z6] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [74, 76, 78, 98, 96, 94]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5
chain length 11, 5 ancillas, 0 |0> fillers

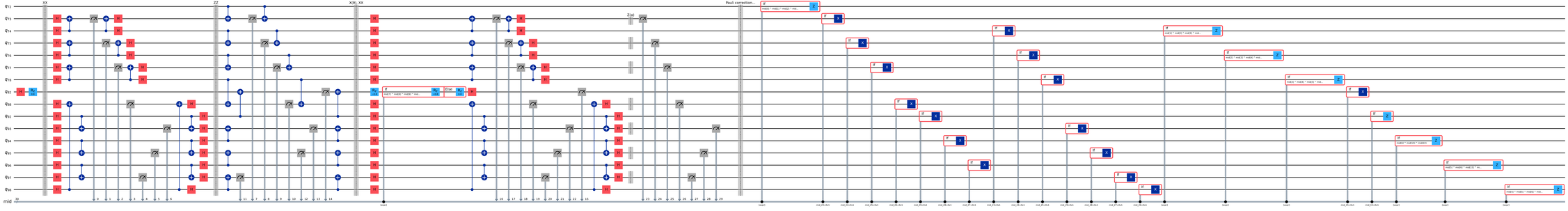



[Z6] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [20, 10, 11, 12, 22, 32] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 10, 2q gates 10 (+0 vs. untranspiled from routing)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
105
106
107
108
109
110
111
112
113
114
115
116
117
118
119

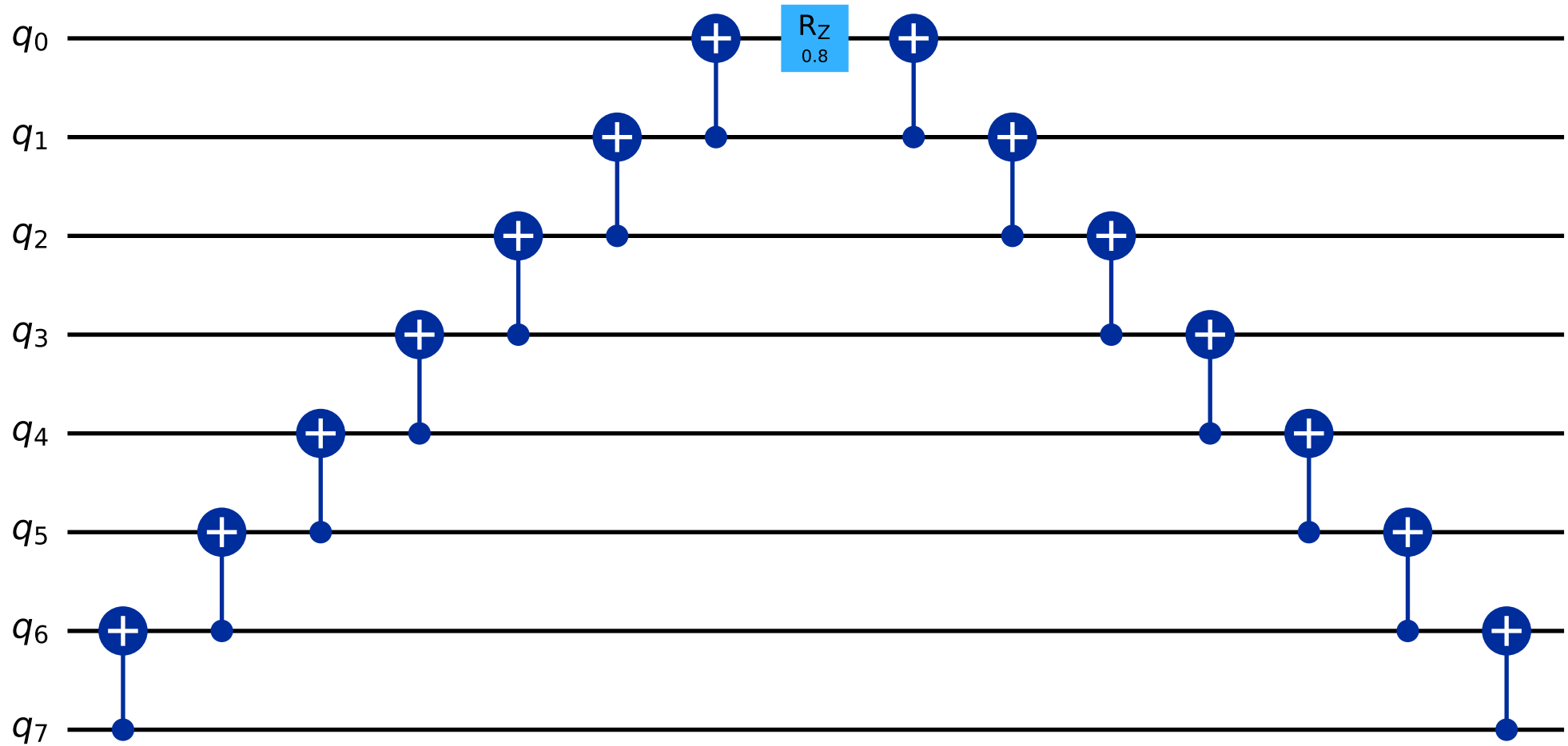
[Z6] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [74, 76, 78, 98, 96, 94]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 32



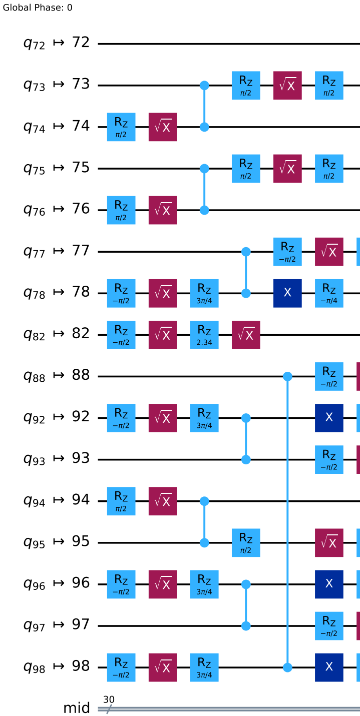
[ZB] DYNAMIC constant-depth circuit, fault-tolerant |0⟩+ ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 : (8 logical qubits)
2q depth 6, 30 mid-circuit measurements, logical–physical [72, 74, 76, 78, 98, 96, 94, 92]

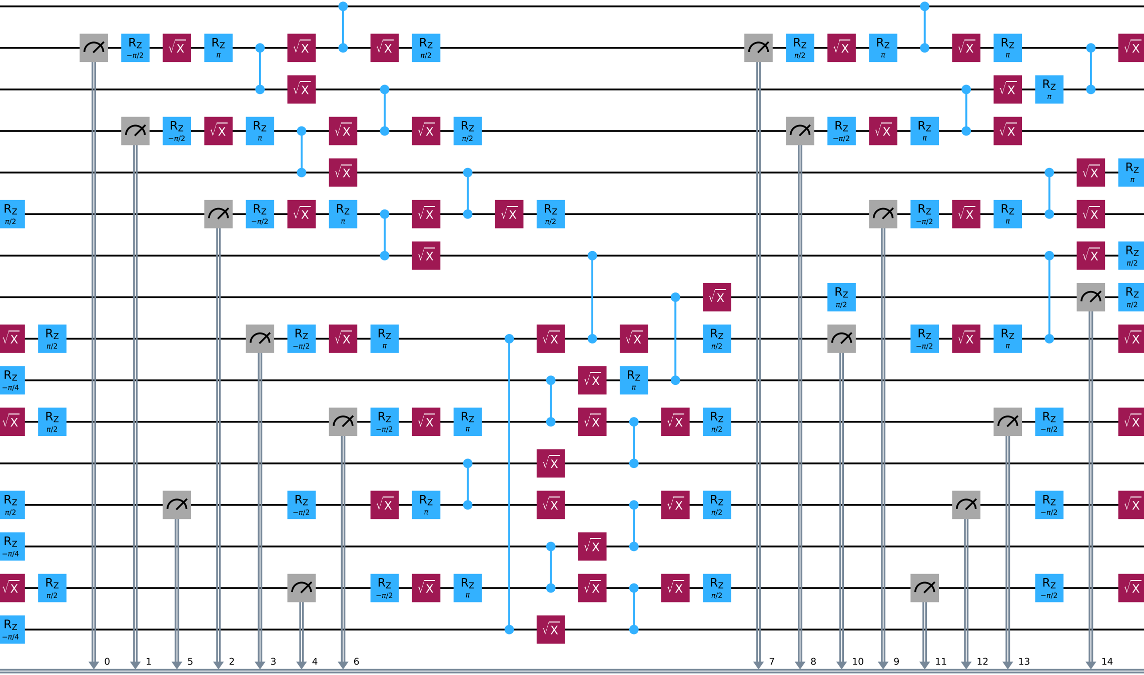


[Z8] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14

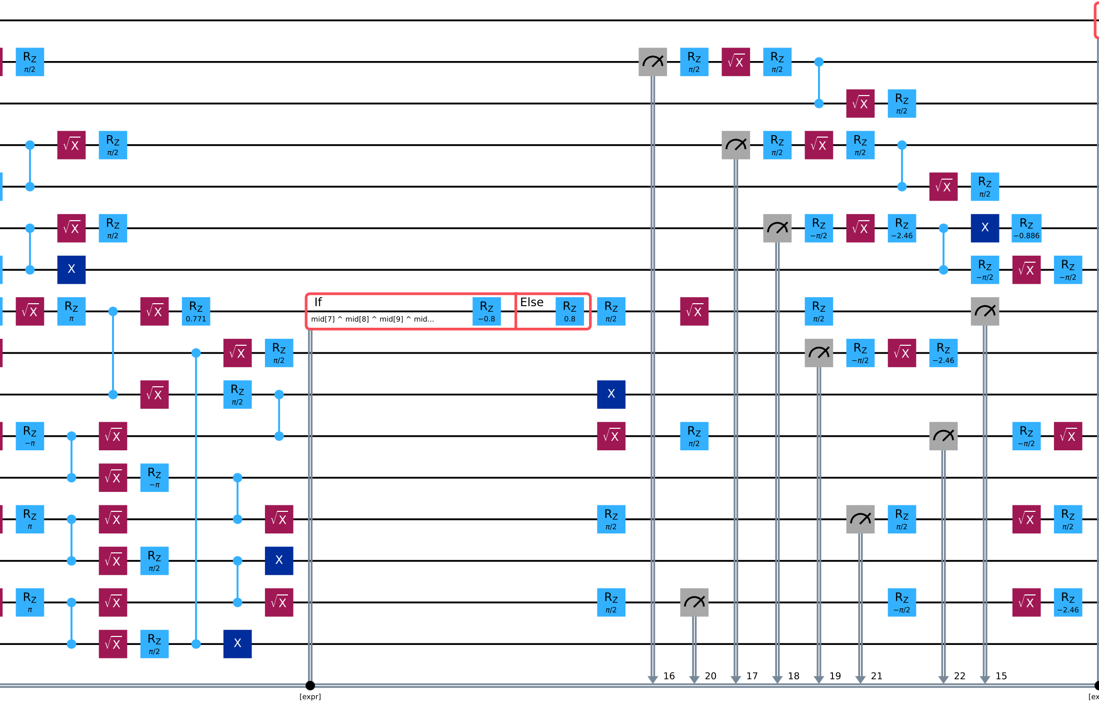


[Z8] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:117 sx:90 cz:44 measure:30 if_else:24 x:12)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 6, total depth 42

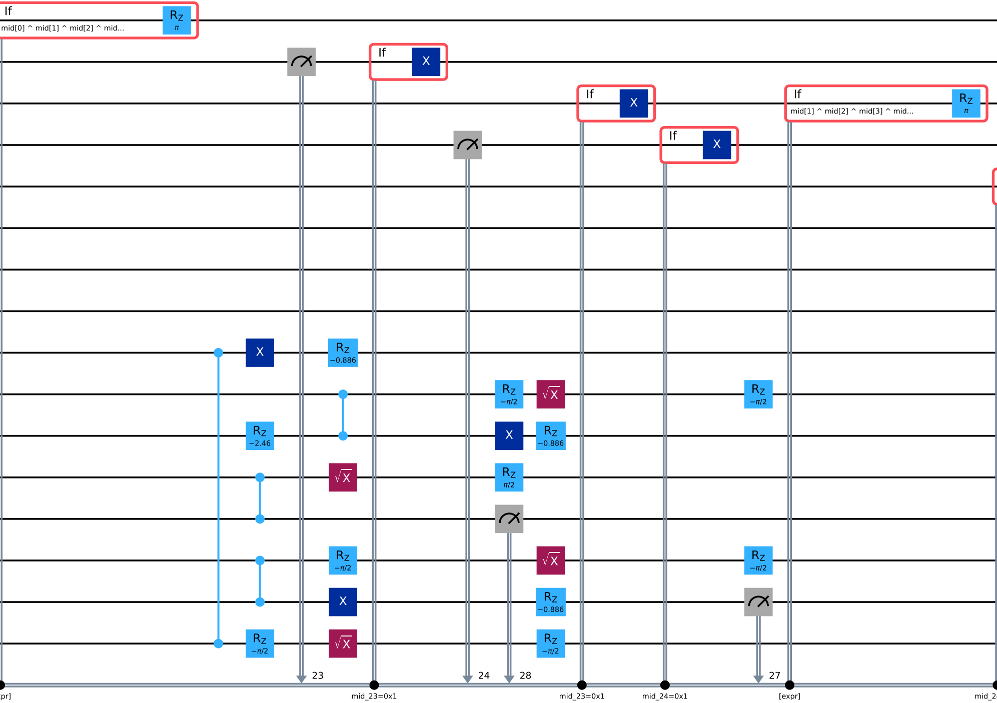

Global Phase: 0
q72 ↦ 72
q73 ↦ 73
q74 ↦ 74
q75 ↦ 75
q76 ↦ 76
q77 ↦ 77
q78 ↦ 78
q82 ↦ 82
q88 ↦ 88
q92 ↦ 92
q93 ↦ 93
q94 ↦ 94
q95 ↦ 95
q96 ↦ 96
q97 ↦ 97
q98 ↦ 98
mid
30
RZ π/2
√X
RZ −π/2
RZ 3π/4
RZ 2.34
RZ −π/4
X

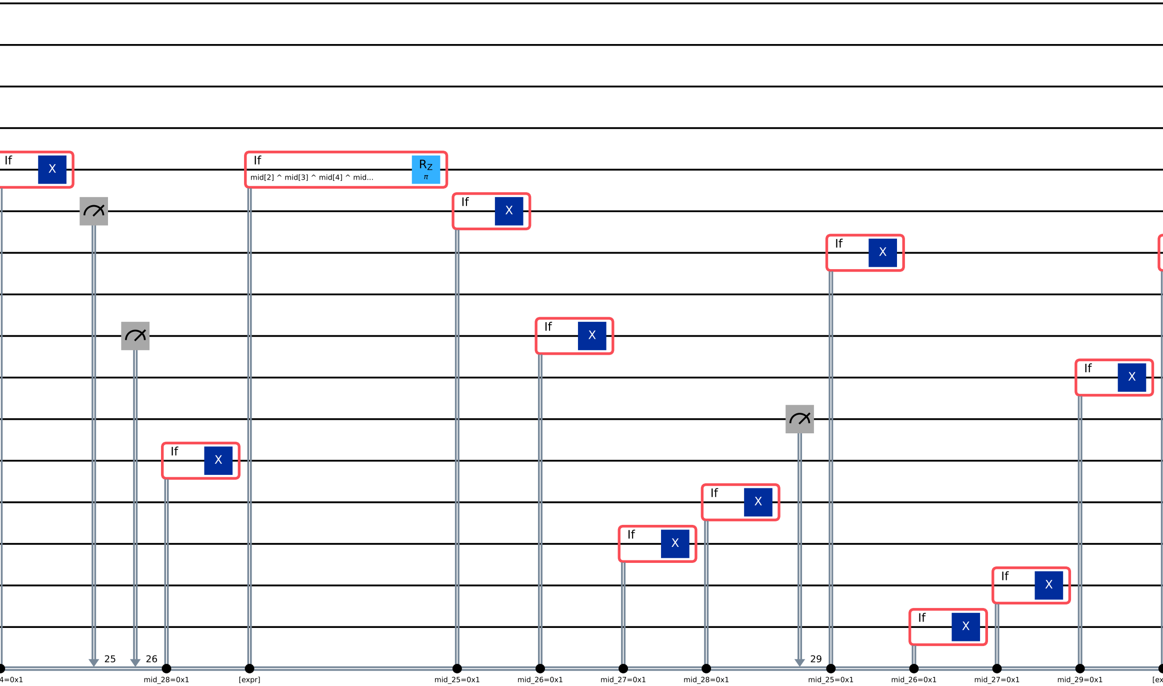


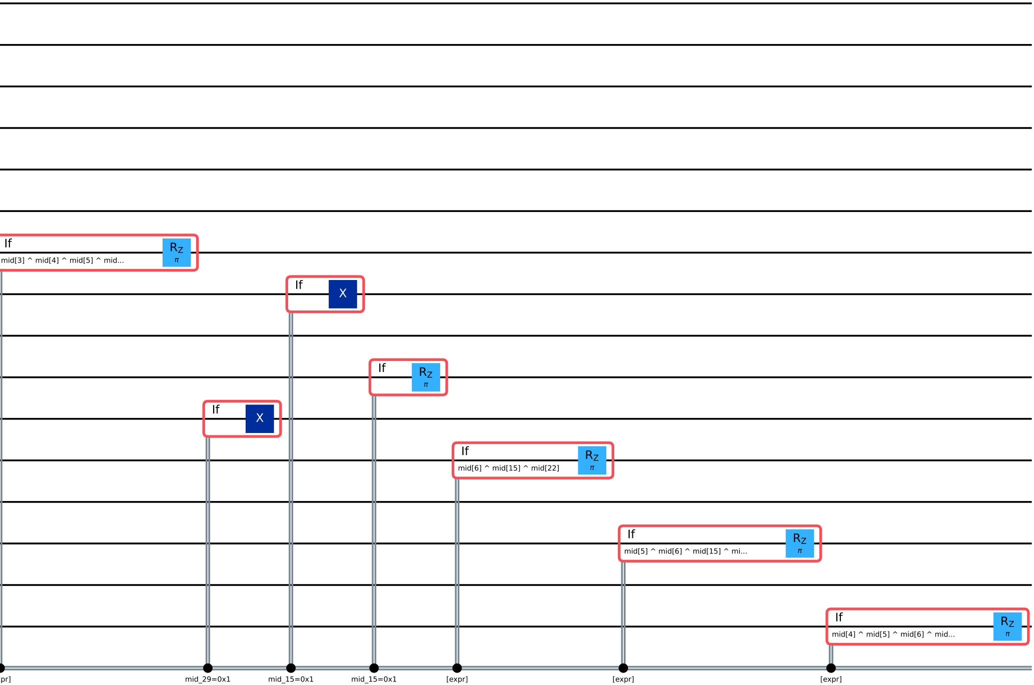
RZ
π/2
−π/2
π
−π/4
√X
0
1
5
2
3
4
6
7
8
10
9
11
12
13
14



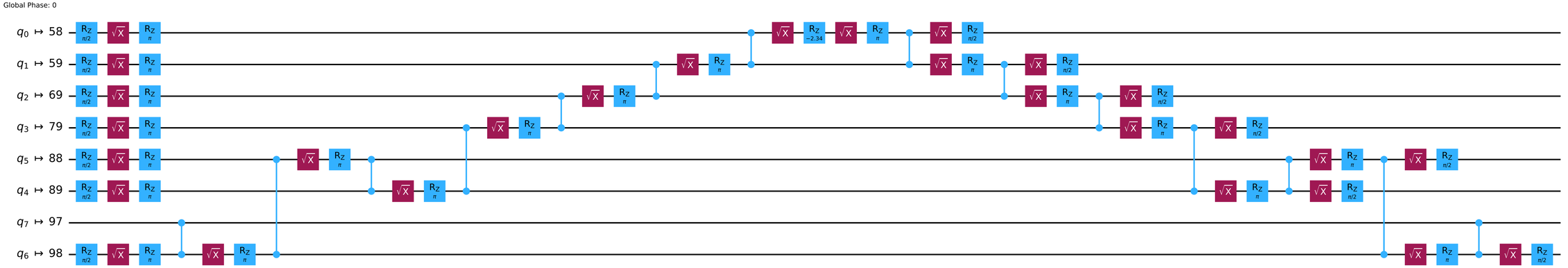
Rz
π/2
√X
X
Rz
π
Rz
−π
Rz
0.771
If
mid[7] ^ mid[8] ^ mid[9] ^ mid...
Rz
−0.8
Else
Rz
0.8
Rz
−π/2
Rz
−2.46
Rz
−0.886
16
20
17
18
19
21
22
15
[expr]
[ex


If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
Rz
π
If
X
If
X
If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
Rz
π
If
X
X
Rz
−0.886
Rz
−2.46
√X
Rz
−π/2
X
√X
Rz
−π/2
Rz
−π/2
√X
X
Rz
−0.886
Rz
π/2
√X
Rz
−0.886
Rz
−π/2
Rz
−π/2
Rz
−π/2
23
24
28
27
pr]
mid_23=0x1
mid_23=0x1
mid_24=0x1
[expr]
mid_2


If
X
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
R_Z
π
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
4=0x1
25
26
mid_28=0x1
[expr]
mid_25=0x1
mid_26=0x1
mid_27=0x1
mid_28=0x1
29
mid_25=0x1
mid_26=0x1
mid_27=0x1
mid_29=0x1
[ex

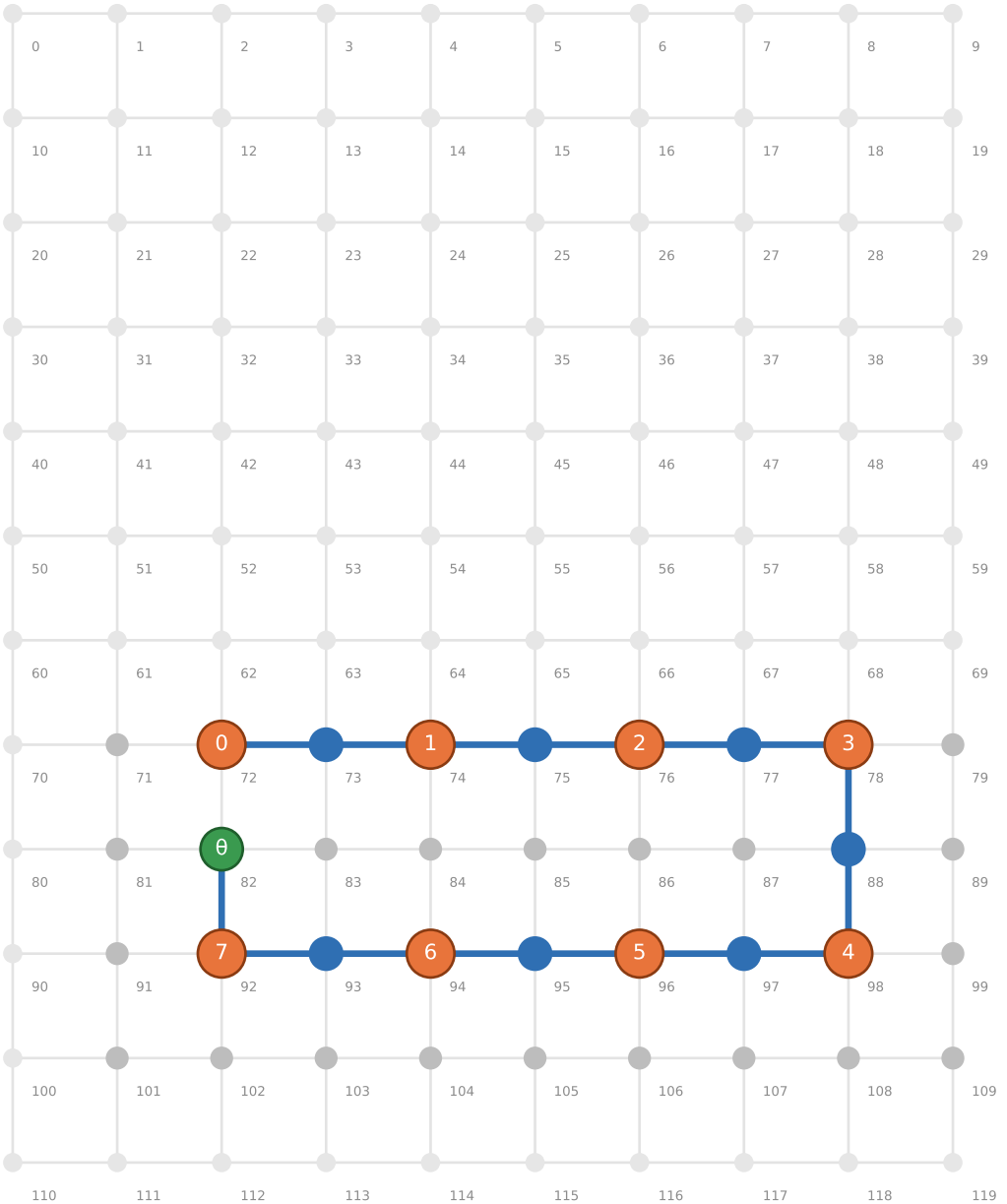

If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
RZ
π
If
X
If
X
If
RZ
π
If
mid[6] ^ mid[15] ^ mid[22]
RZ
π
If
mid[5] ^ mid[6] ^ mid[15] ^ mi...
RZ
π
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
RZ
π
pr]
mid_29=0x1
mid_15=0x1
mid_15=0x1
[expr]
[expr]
[expr]

[Z8] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:35 sx:28 cz:14)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14, total depth 47

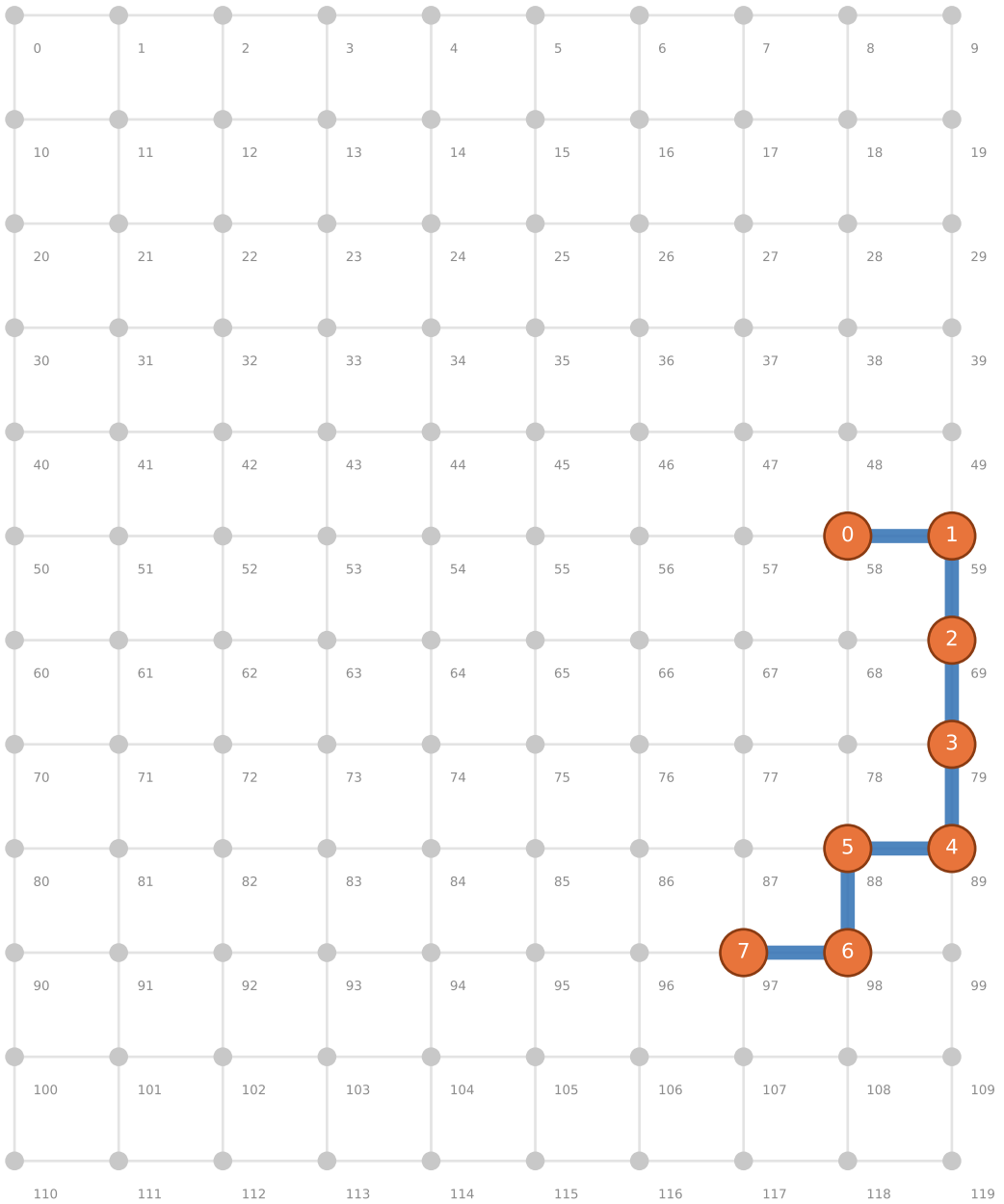


[Z8] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

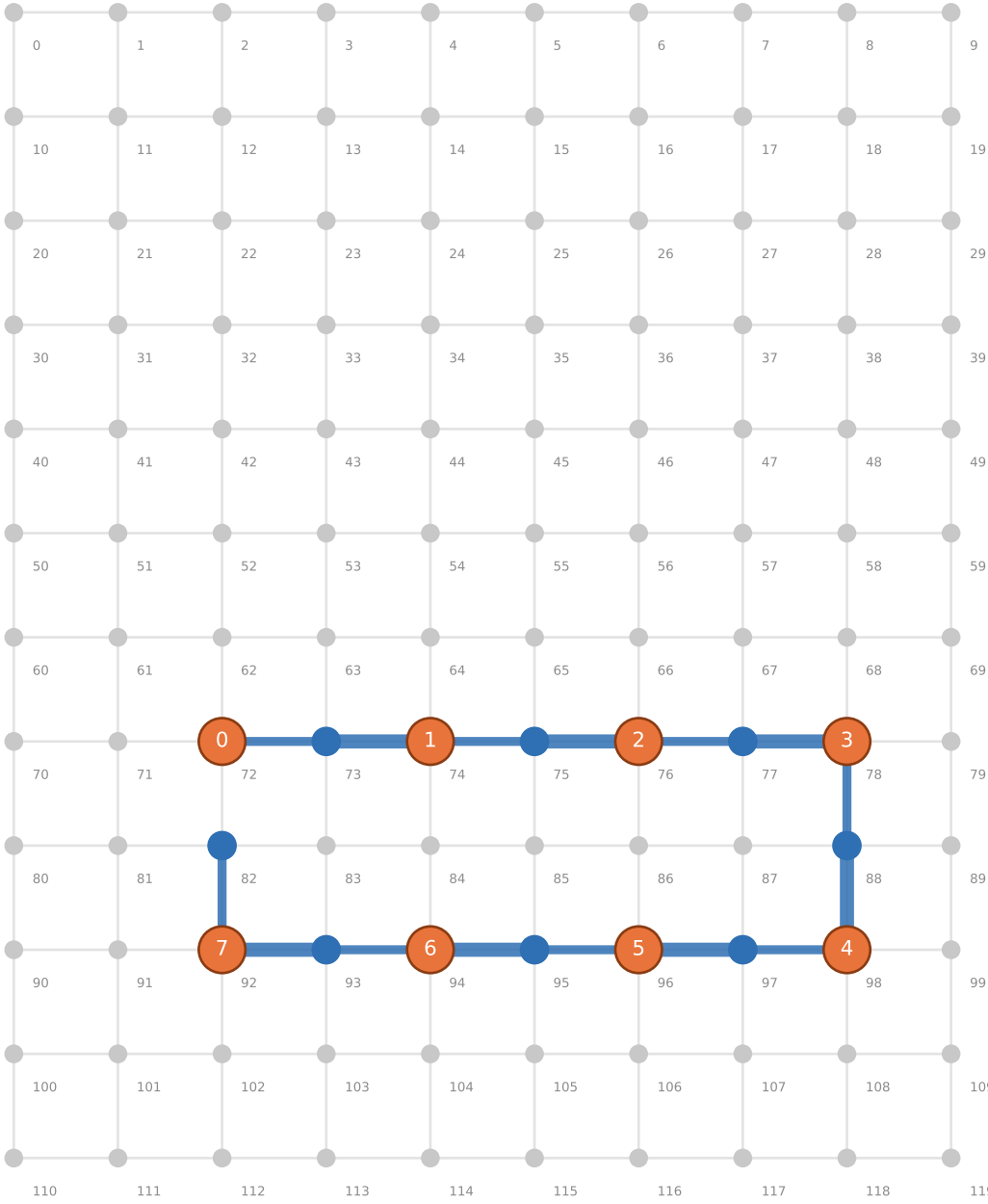
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7
chain length 15, 7 ancillas, 0 |0> fillers



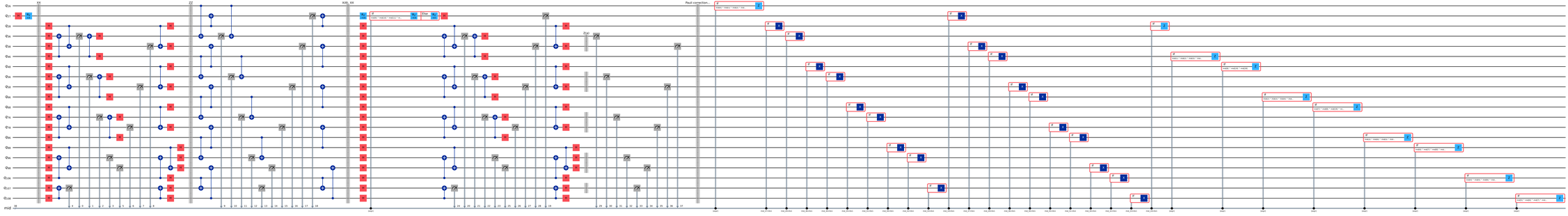
[Z8] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [58, 59, 69, 79, 89, 88, 98, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



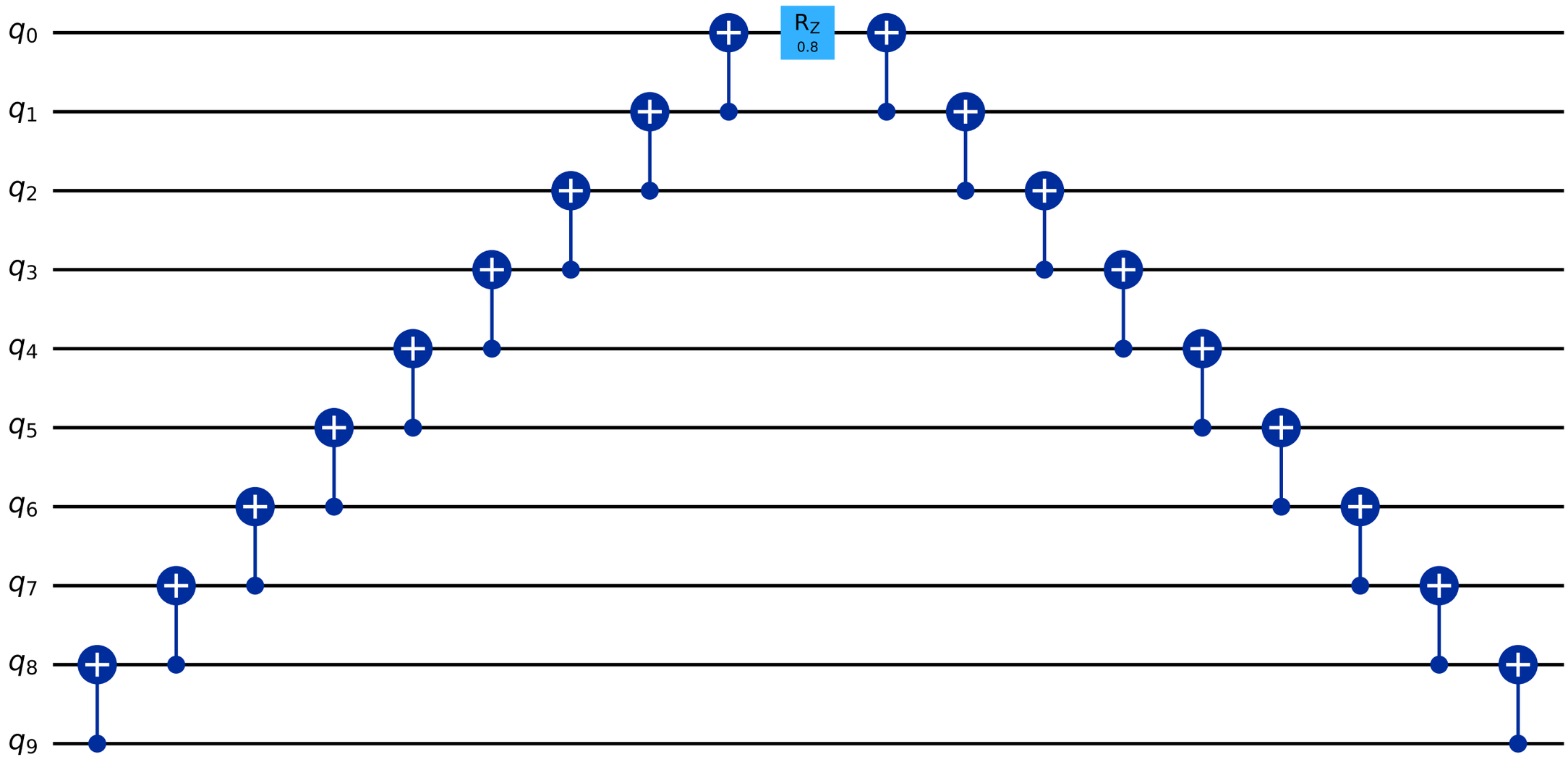
[Z8] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 44



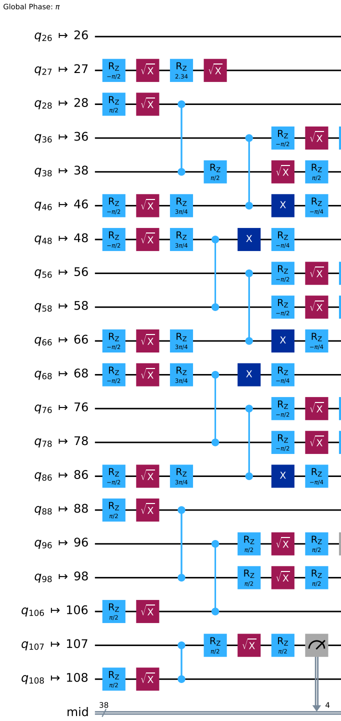
[Z16] DYNAMIC constant-depth circuit, fault-tolerant (8n+ ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = ZI Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 (10 logical qubits)
2q depth 6, 38 mid-circuit measurements, logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]

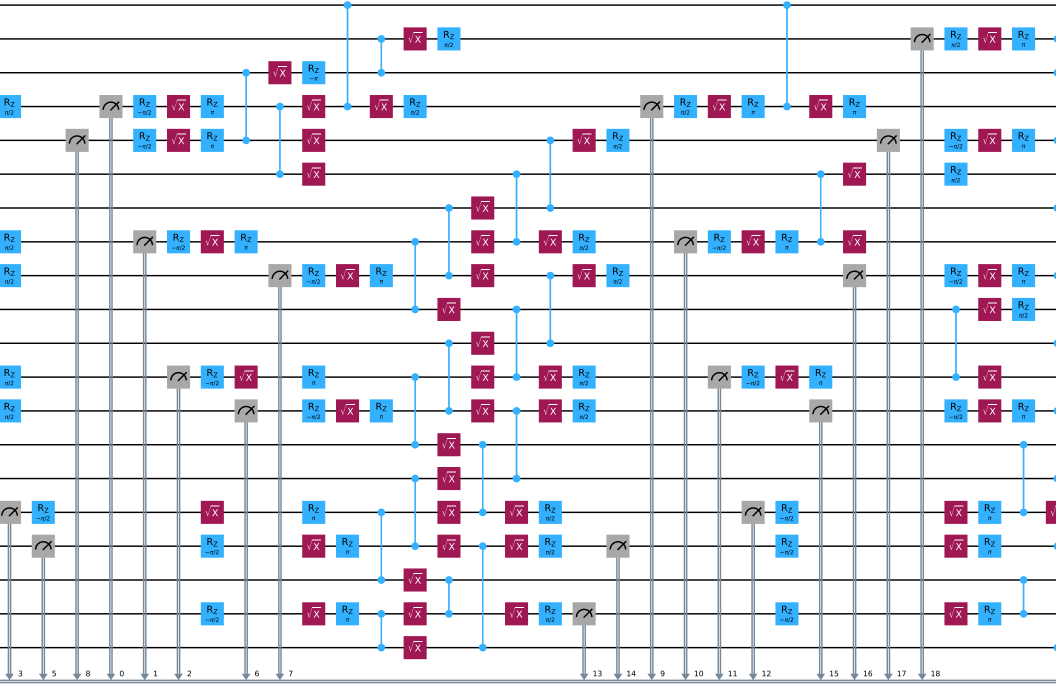


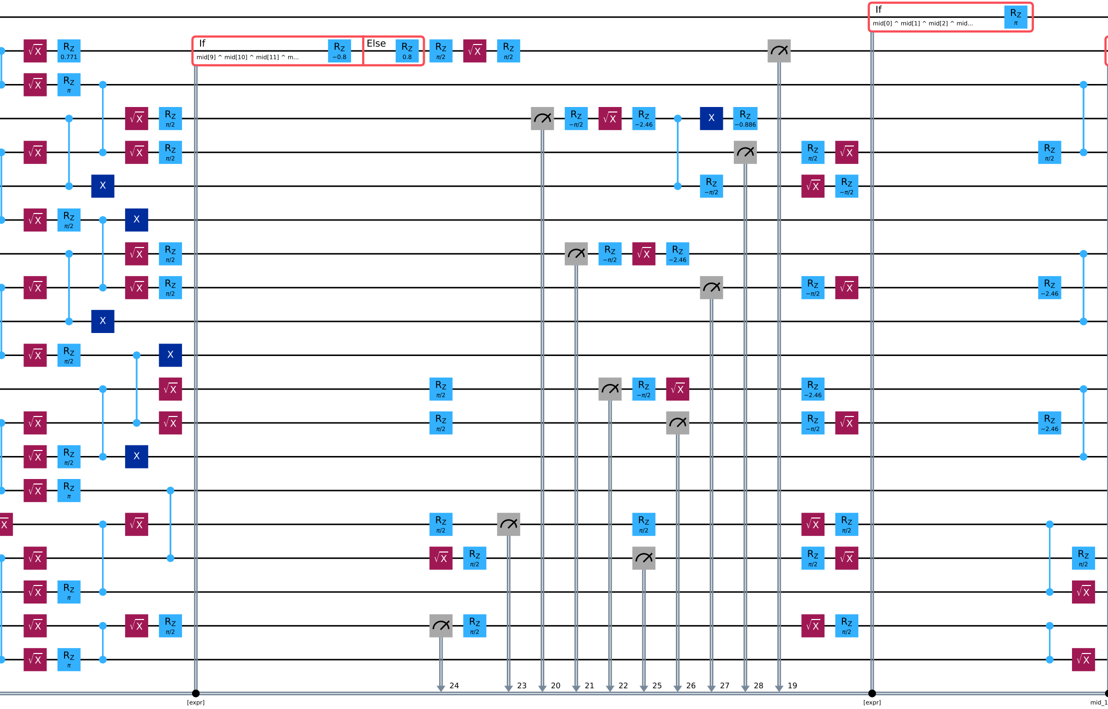
[Z10] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18

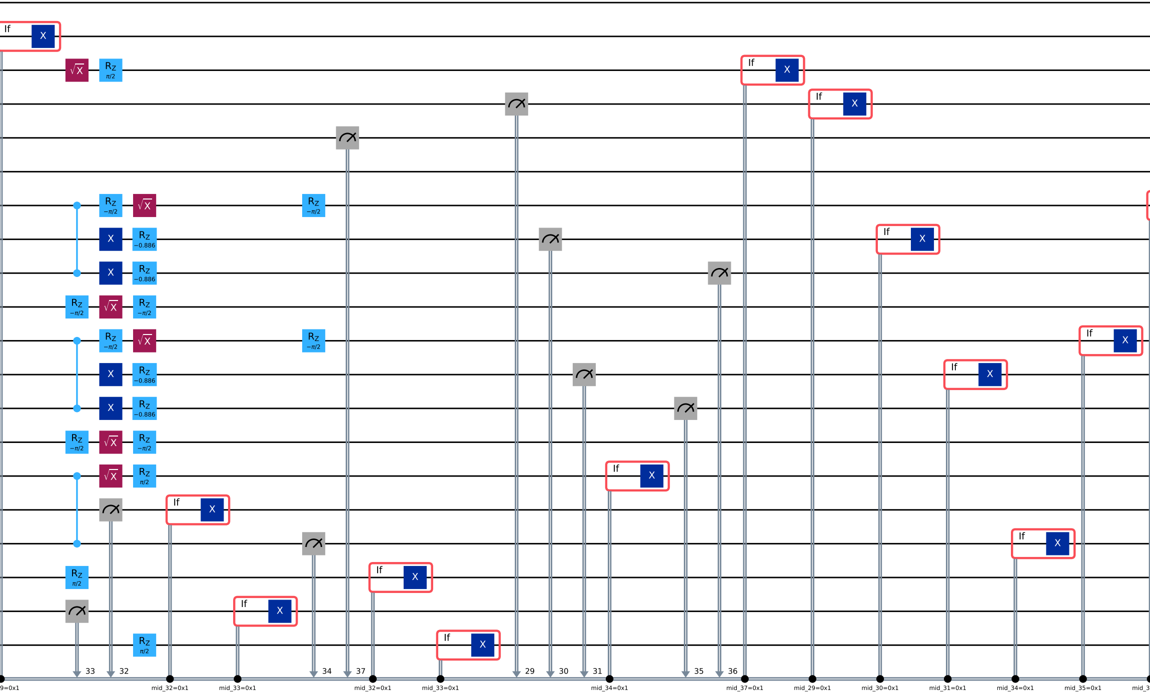


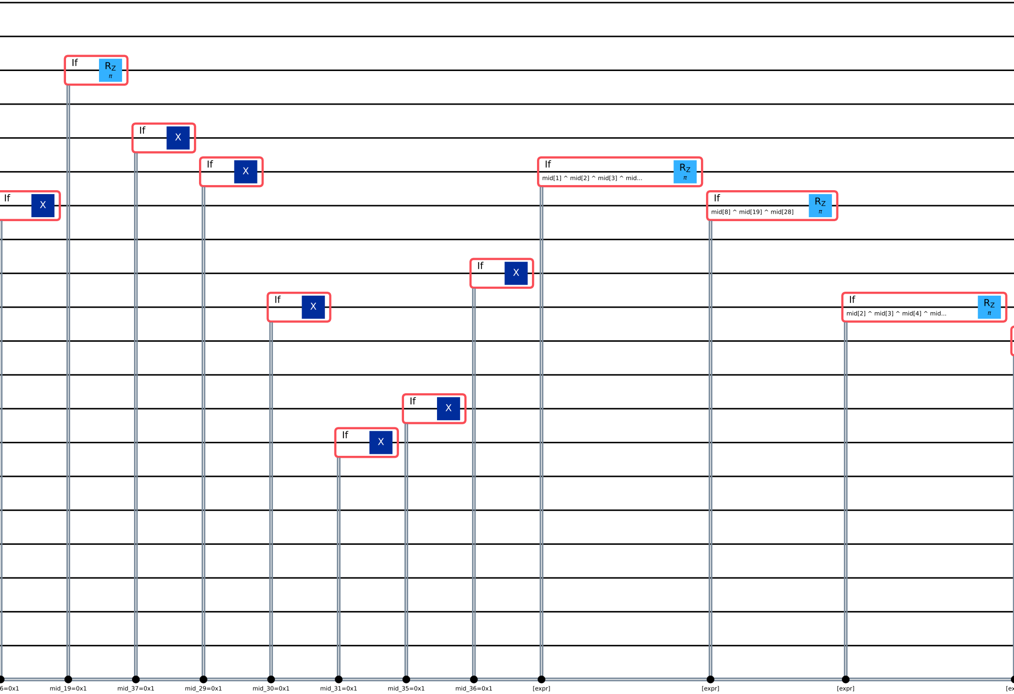
[Z10] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:147 sx:114 cz:56 measure:38 if_else:30 x:15)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 6, total depth 44

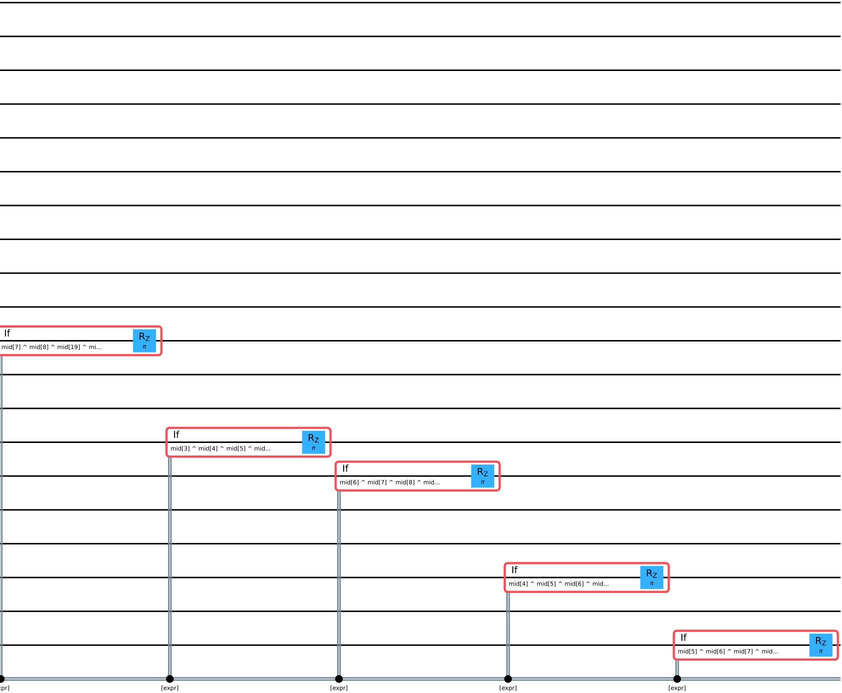

Global Phase: π
q₂₆ ↦ 26
q₂₇ ↦ 27
q₂₈ ↦ 28
q₃₆ ↦ 36
q₃₈ ↦ 38
q₄₆ ↦ 46
q₄₈ ↦ 48
q₅₆ ↦ 56
q₅₈ ↦ 58
q₆₆ ↦ 66
q₆₈ ↦ 68
q₇₆ ↦ 76
q₇₈ ↦ 78
q₈₆ ↦ 86
q₈₈ ↦ 88
q₉₆ ↦ 96
q₉₈ ↦ 98
q₁₀₆ ↦ 106
q₁₀₇ ↦ 107
q₁₀₈ ↦ 108
mid
38
4
Rz −π/2
Rz π/2
Rz 2.34
Rz 3π/4
Rz −π/4
√X
X


Rz
π/2
−π/2
π
−π
√X
3
5
8
0
1
2
6
7
13
14
9
10
11
12
15
16
17
18

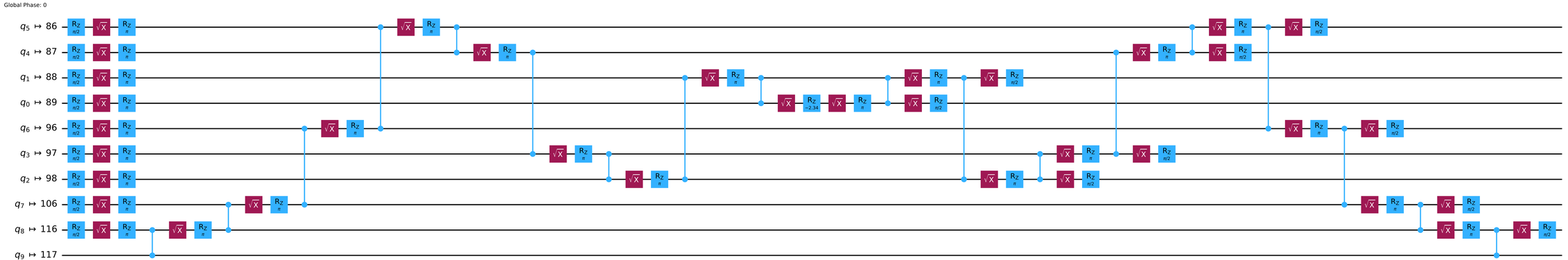

If
mid[9] ^ mid[10] ^ mid[11] ^ m...
Else
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
√X
X
RZ
0.771
π
π/2
−π/2
−0.8
0.8
−2.46
−0.886
24
23
20
21
22
25
26
27
28
19
[expr]
[expr]
mid_1


If
X
√X
R_Z
π/2
−π/2
−0.886
33
32
34
37
29
30
31
35
36
9=0x1
mid_32=0x1
mid_33=0x1
mid_32=0x1
mid_33=0x1
mid_34=0x1
mid_37=0x1
mid_29=0x1
mid_30=0x1
mid_31=0x1
mid_34=0x1
mid_35=0x1
mid_3


If
X
If
R_Z
π
If
X
If
X
If
X
If
X
If
X
If
X
If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
R_Z
π
If
mid[8] ^ mid[19] ^ mid[28]
R_Z
π
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
R_Z
π
6=0x1
mid_19=0x1
mid_37=0x1
mid_29=0x1
mid_30=0x1
mid_31=0x1
mid_35=0x1
mid_36=0x1
[expr]
[expr]
[expr]
[ex

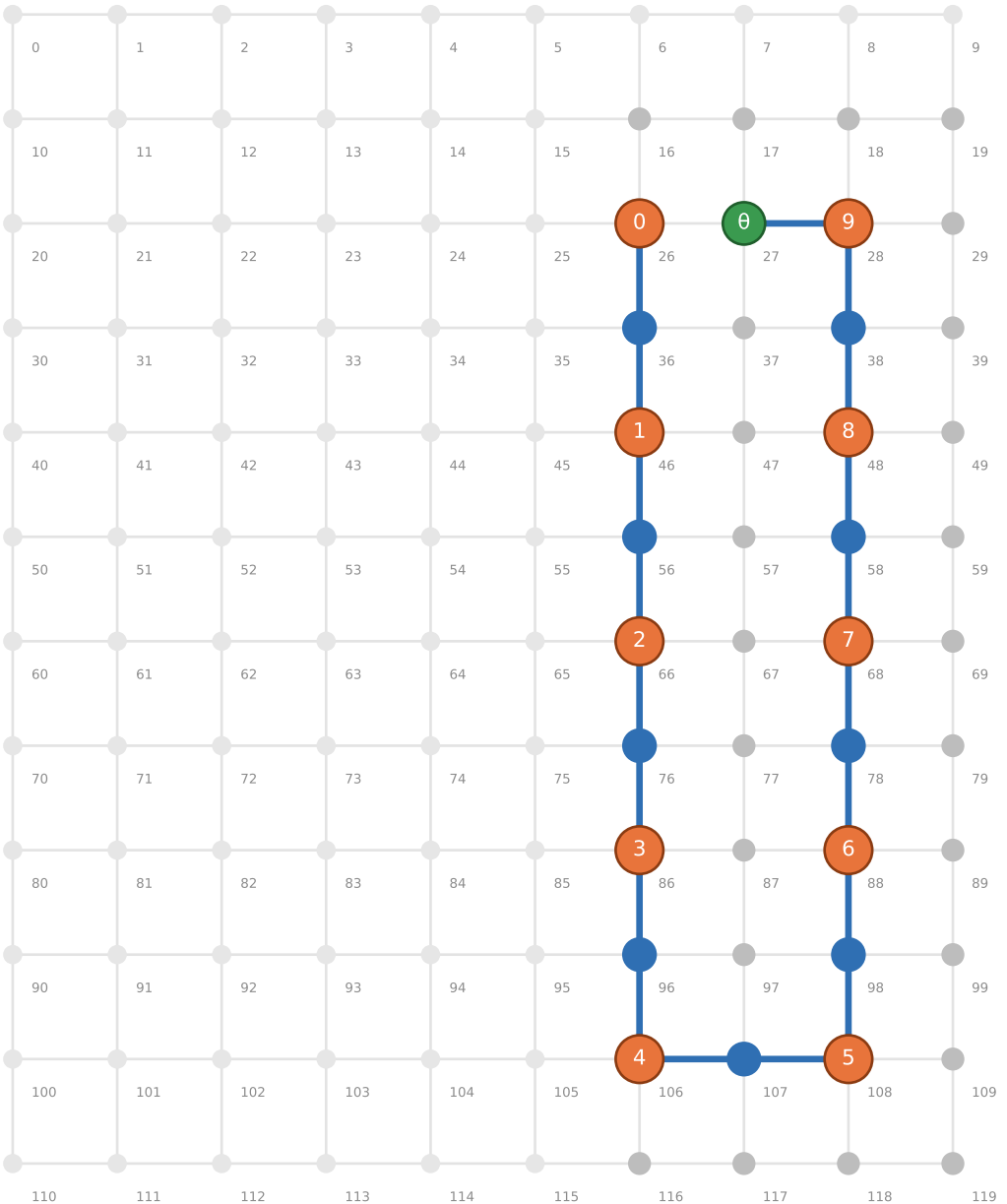

If
mid[7] ^ mid[8] ^ mid[19] ^ mi...
R_Z
π
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
R_Z
π
If
mid[6] ^ mid[7] ^ mid[8] ^ mid...
R_Z
π
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
R_Z
π
If
mid[5] ^ mid[6] ^ mid[7] ^ mid...
R_Z
π
xpr]
[expr]
[expr]
[expr]
[expr]

[Z10] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:45 sx:36 cz:18)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18, total depth 59

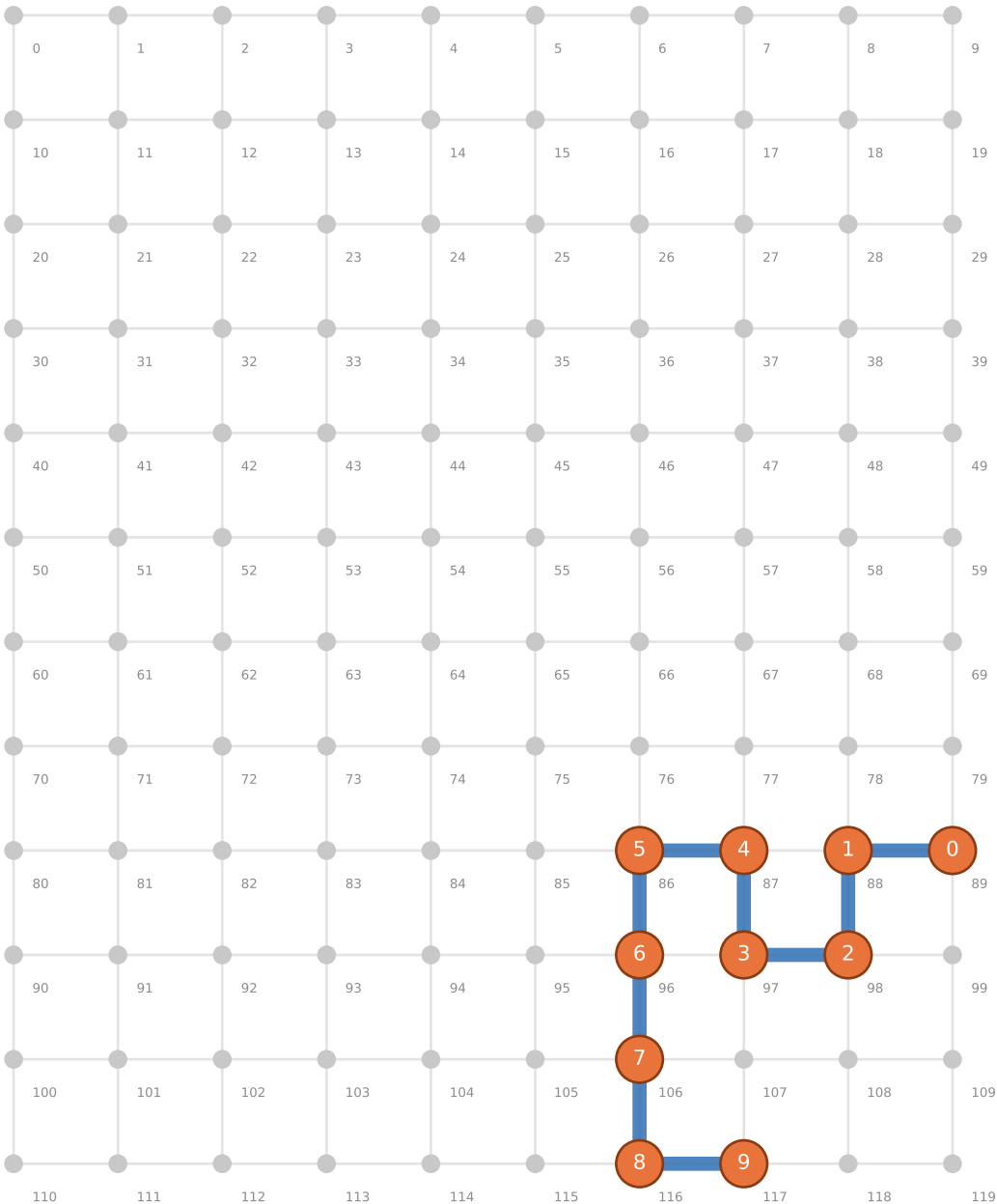


[Z10] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

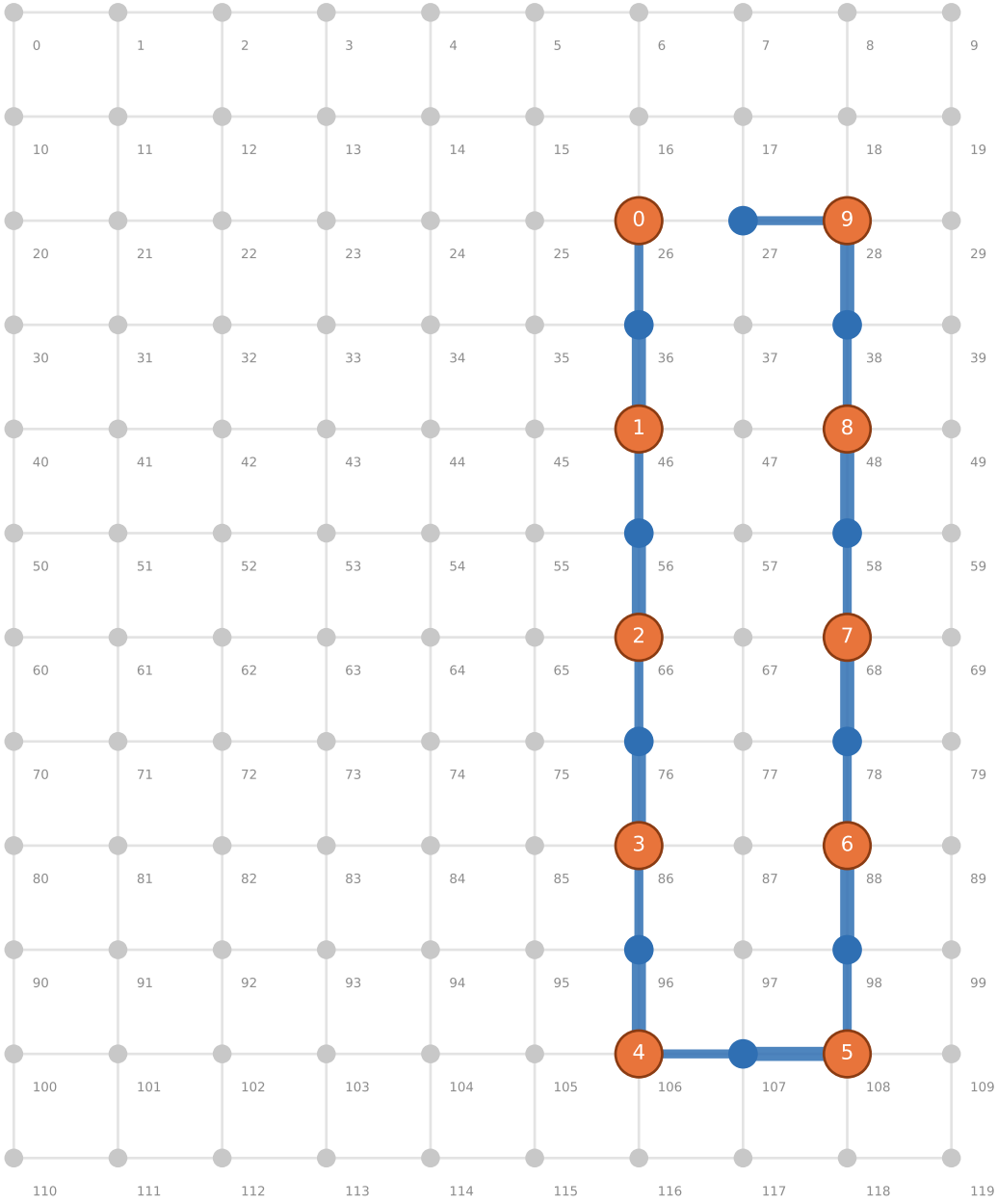
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9
chain length 19, 9 ancillas, 0 |0> fillers



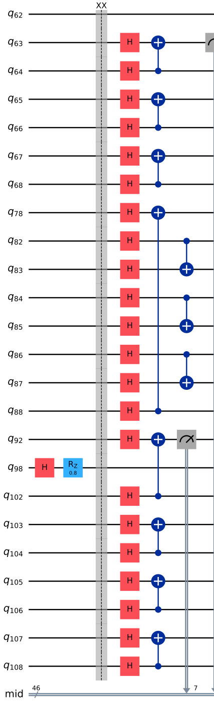
[Z10] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [89, 88, 98, 97, 87, 86, 96, 106, 116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 18, 2q gates 18 (+0 vs. untranspiled from routing)

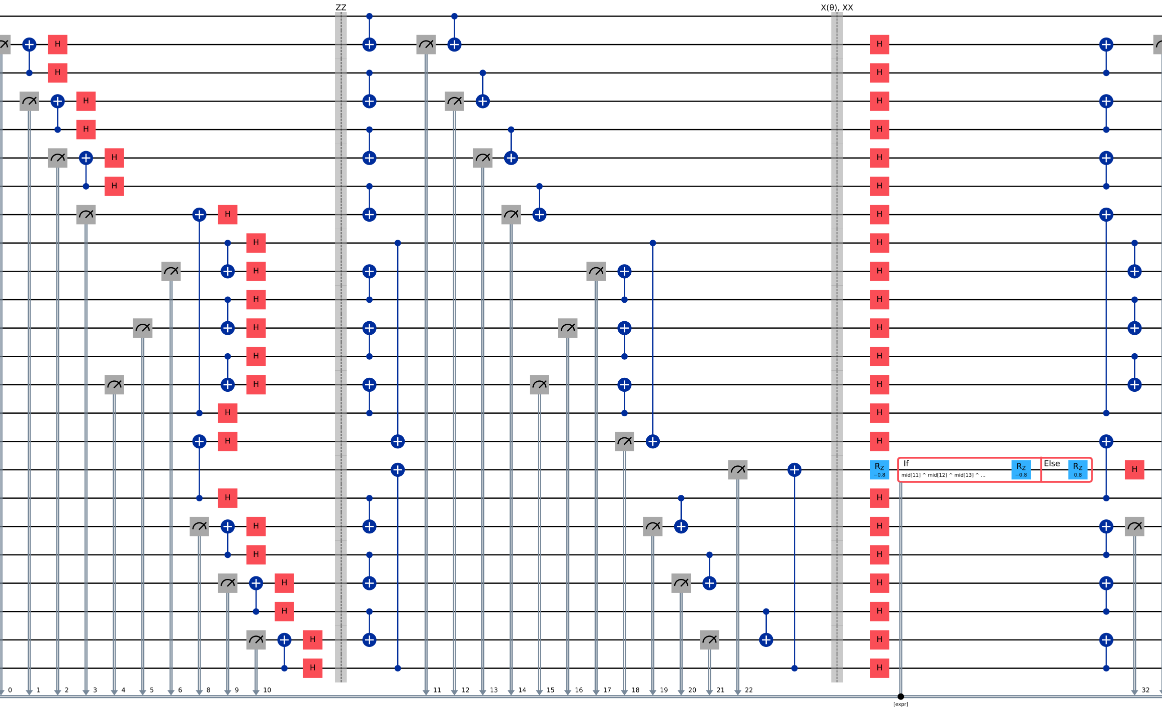


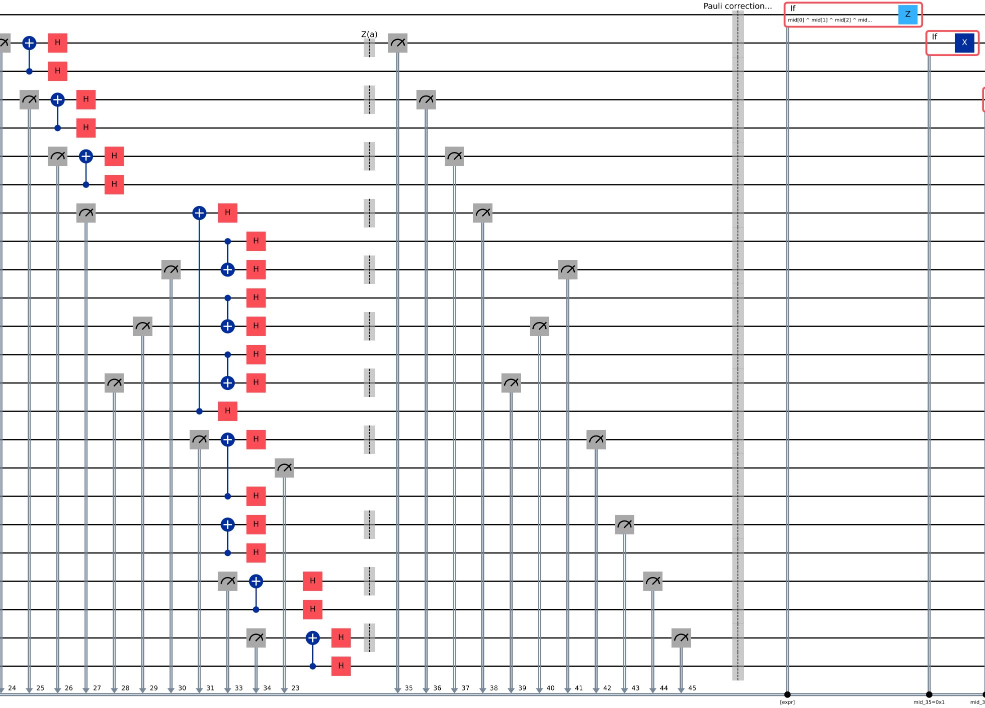
[Z10] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 56

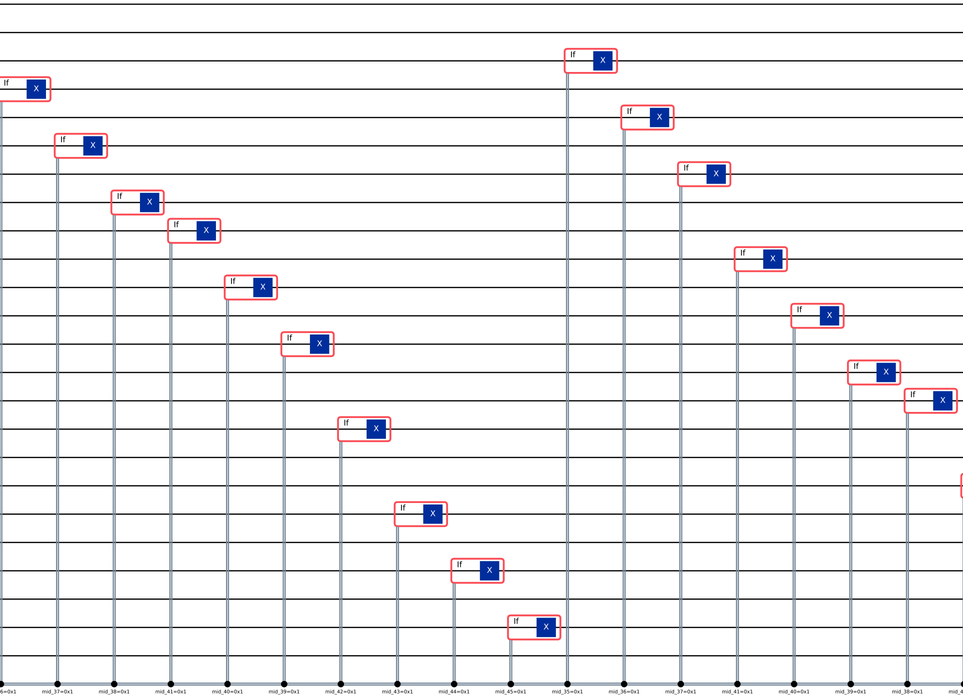


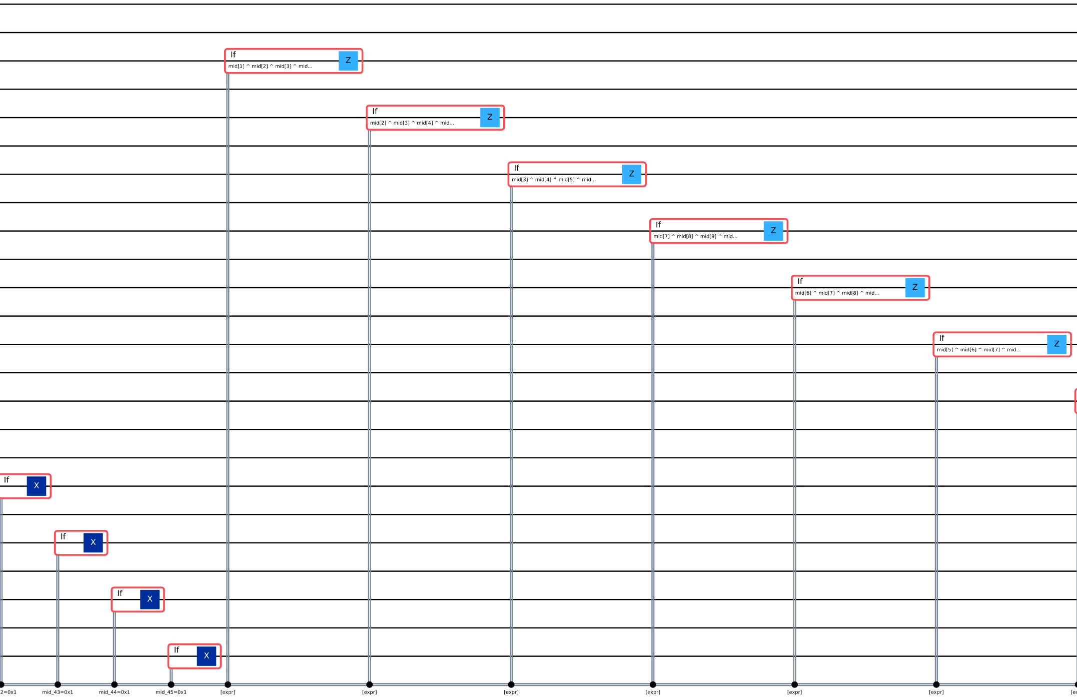
[Z12] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 6, 46 mid-circuit measurements, logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]

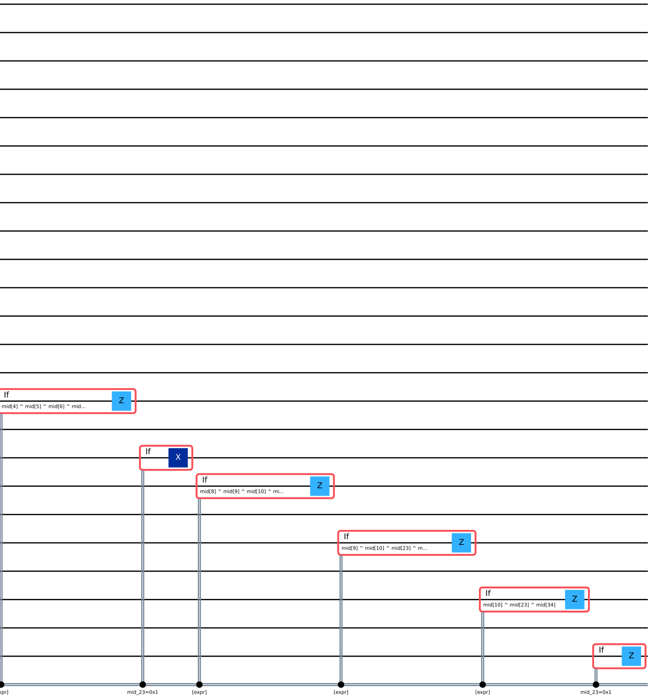

XX
q_{62}
q_{63}
q_{64}
q_{65}
q_{66}
q_{67}
q_{68}
q_{78}
q_{82}
q_{83}
q_{84}
q_{85}
q_{86}
q_{87}
q_{88}
q_{92}
q_{98}
q_{102}
q_{103}
q_{104}
q_{105}
q_{106}
q_{107}
q_{108}
mid
46
H
R$_Z$
0.8
7


ZZ
X(θ), XX
H
R_Z −0.8
If
mid[11] ^ mid[12] ^ mid[13] ^ ...
R_Z −0.8
Else
R_Z 0.8
0
1
2
3
4
5
6
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
32
[expr]

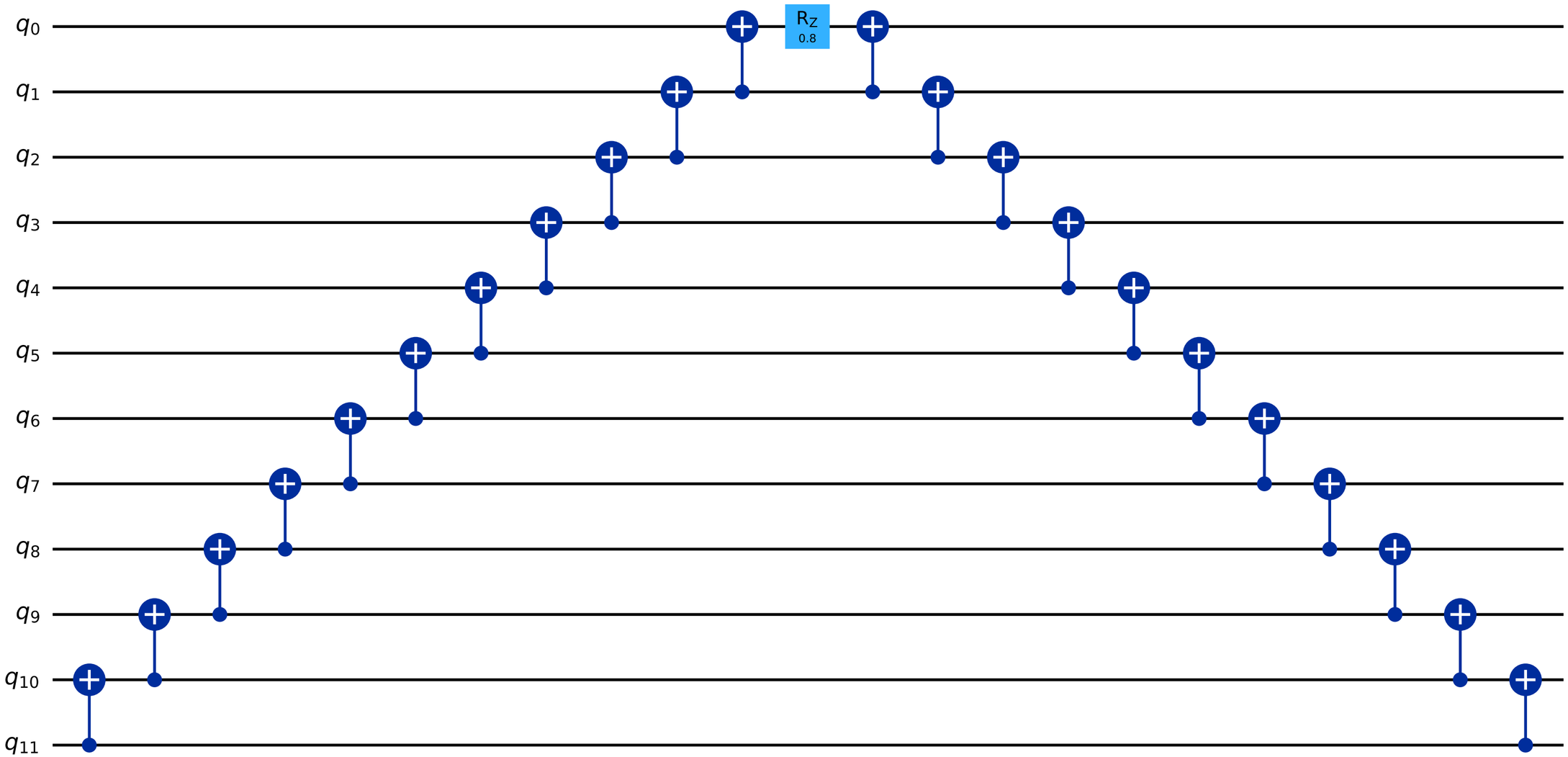

Pauli correction...
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
Z
If
X
Z(a)
H
24
25
26
27
28
29
30
31
33
34
23
35
36
37
38
39
40
41
42
43
44
45
[expr]
mid_35=0x1
mid_3


If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
6=0x1
mid_37=0x1
mid_38=0x1
mid_41=0x1
mid_40=0x1
mid_39=0x1
mid_42=0x1
mid_43=0x1
mid_44=0x1
mid_45=0x1
mid_35=0x1
mid_36=0x1
mid_37=0x1
mid_41=0x1
mid_40=0x1
mid_39=0x1
mid_38=0x1
mid_4

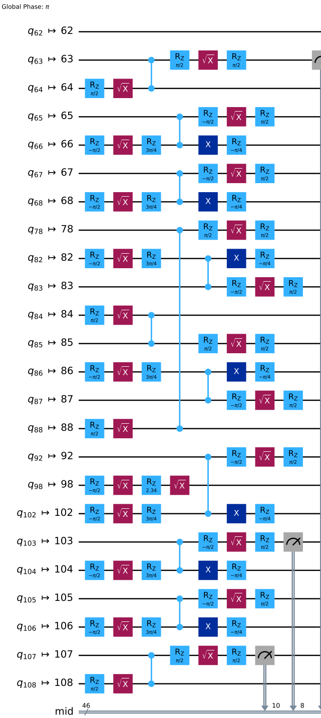

If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
Z
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
Z
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
Z
If
mid[7] ^ mid[8] ^ mid[9] ^ mid...
Z
If
mid[6] ^ mid[7] ^ mid[8] ^ mid...
Z
If
mid[5] ^ mid[6] ^ mid[7] ^ mid...
Z
If
X
If
X
If
X
If
X
2=0x1
mid_43=0x1
mid_44=0x1
mid_45=0x1
[expr]
[expr]
[expr]
[expr]
[expr]
[expr]
[ex

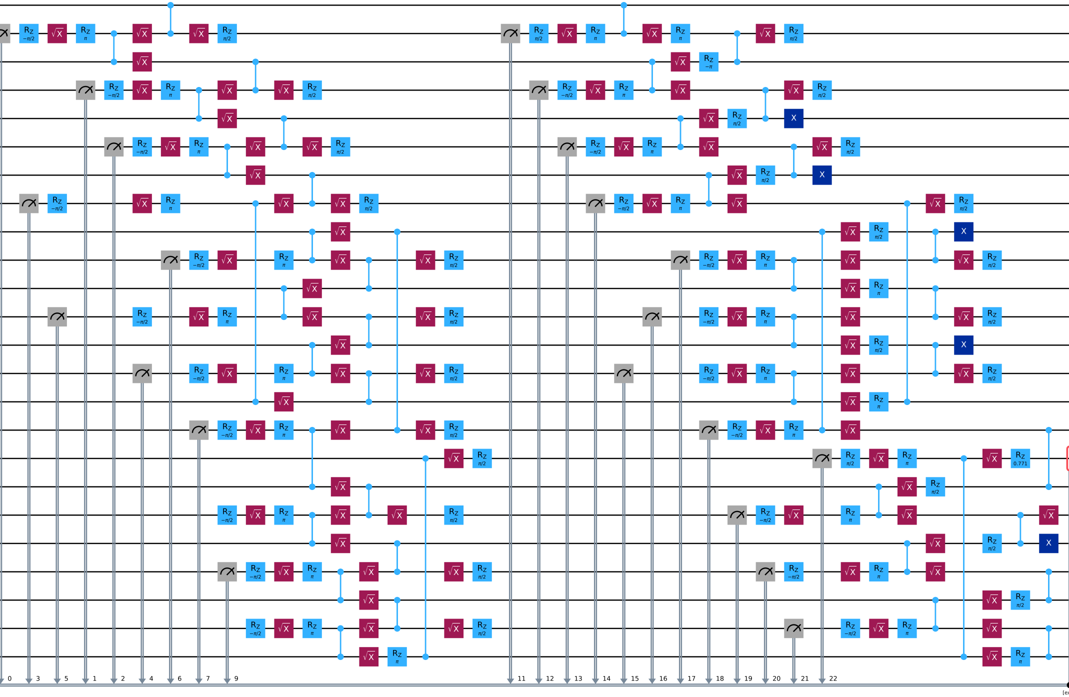

If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
Z
If
X
If
mid[8] ^ mid[9] ^ mid[10] ^ mi...
Z
If
mid[9] ^ mid[10] ^ mid[23] ^ m...
Z
If
mid[10] ^ mid[23] ^ mid[34]
Z
If
Z
xpr]
mid_23=0x1
[expr]
[expr]
[expr]
mid_23=0x1

[Z12] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 22

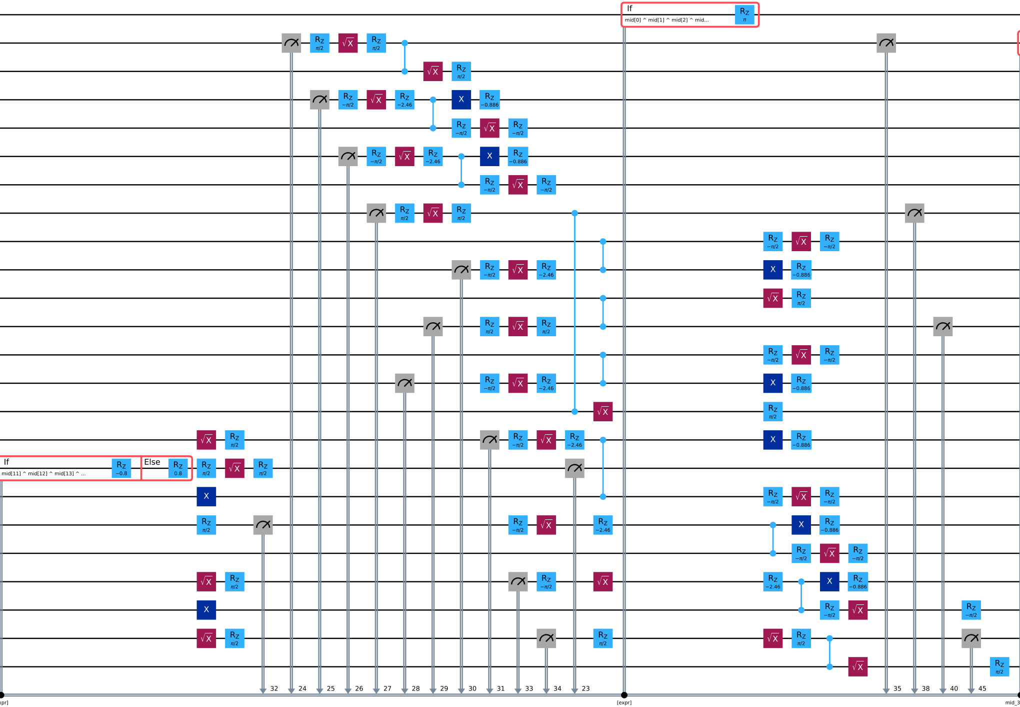


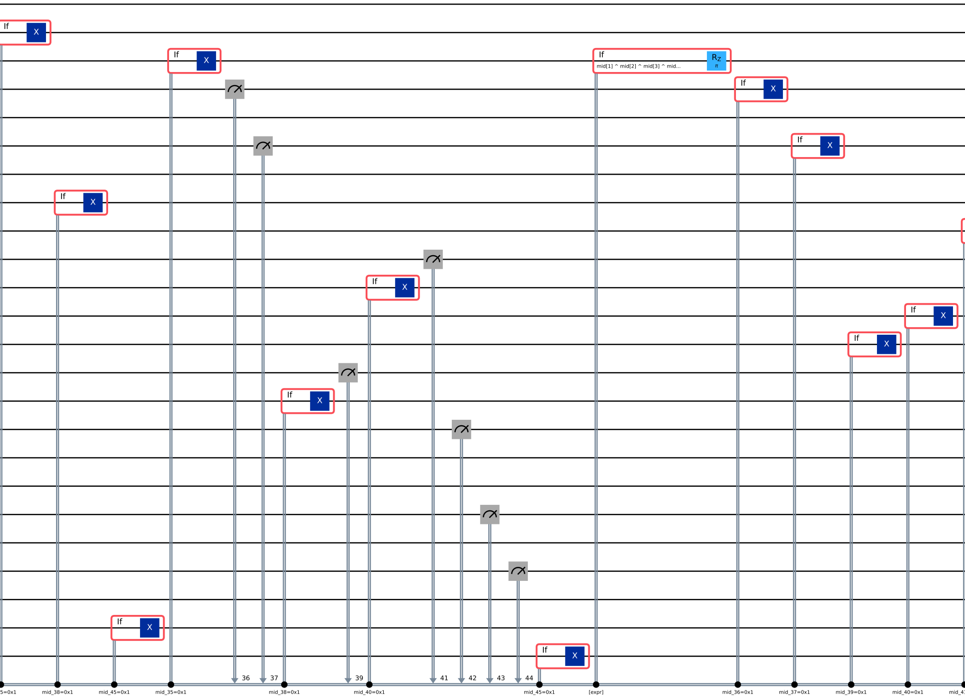
[Z12] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:181 sx:138 cz:68 measure:46 if_else:36 x:21)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 6, total depth 47

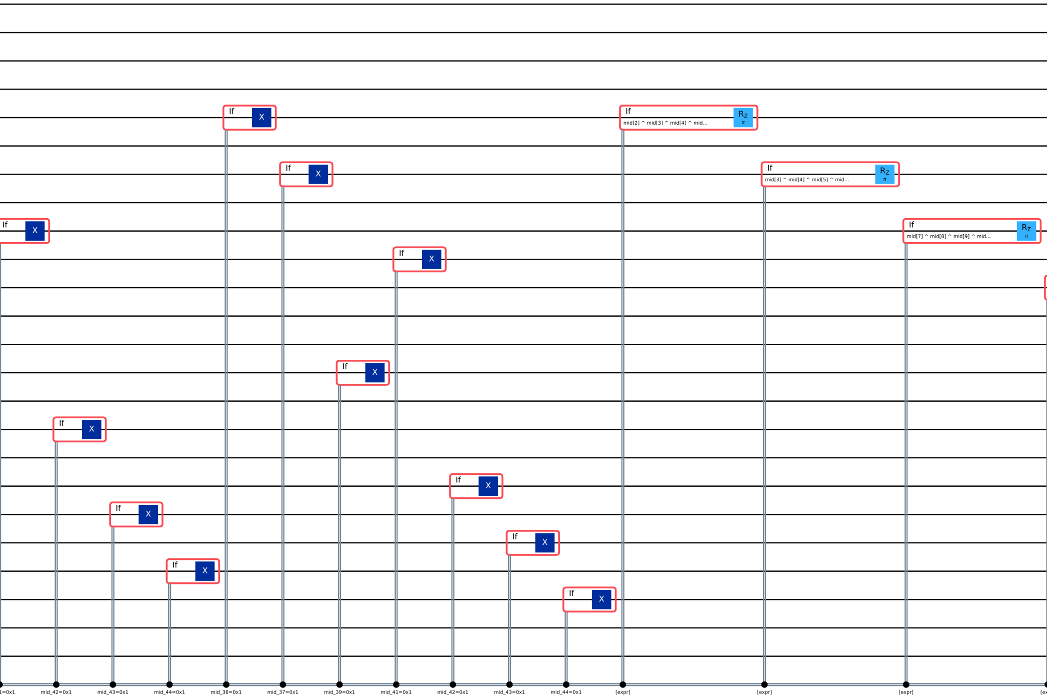

Global Phase: π
q_{62} ↦ 62
q_{63} ↦ 63
q_{64} ↦ 64
q_{65} ↦ 65
q_{66} ↦ 66
q_{67} ↦ 67
q_{68} ↦ 68
q_{78} ↦ 78
q_{82} ↦ 82
q_{83} ↦ 83
q_{84} ↦ 84
q_{85} ↦ 85
q_{86} ↦ 86
q_{87} ↦ 87
q_{88} ↦ 88
q_{92} ↦ 92
q_{98} ↦ 98
q_{102} ↦ 102
q_{103} ↦ 103
q_{104} ↦ 104
q_{105} ↦ 105
q_{106} ↦ 106
q_{107} ↦ 107
q_{108} ↦ 108
mid
46
10
8
R_Z π/2
R_Z −π/2
R_Z 3π/4
R_Z −π/4
R_Z 2.34
√X
X

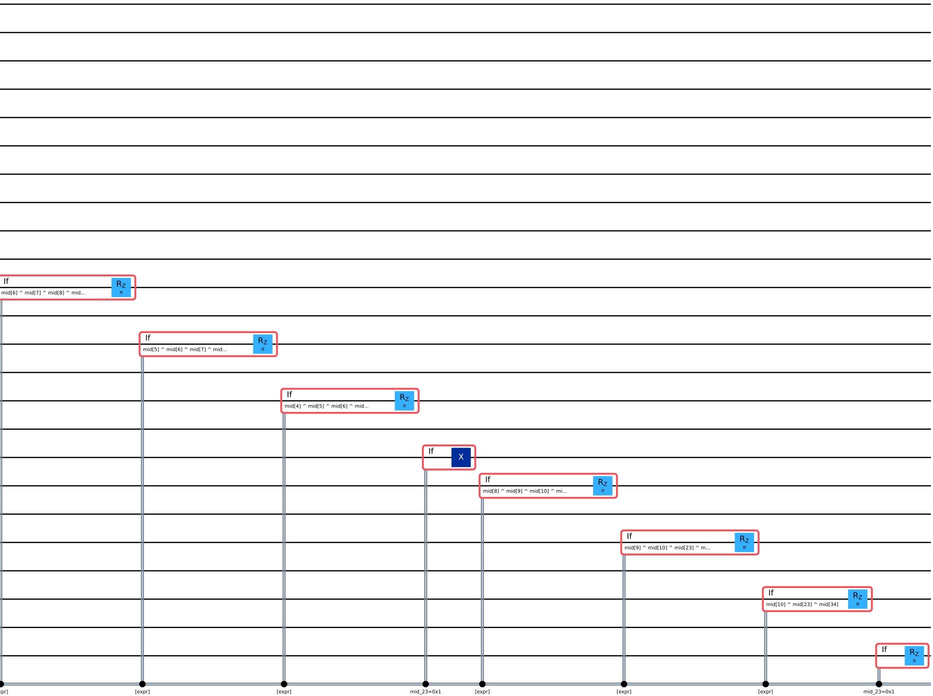


Rz
√X
X
−π/2
π
π/2
−π
0.771
0
3
5
1
2
4
6
7
9
11
12
13
14
15
16
17
18
19
20
21
22

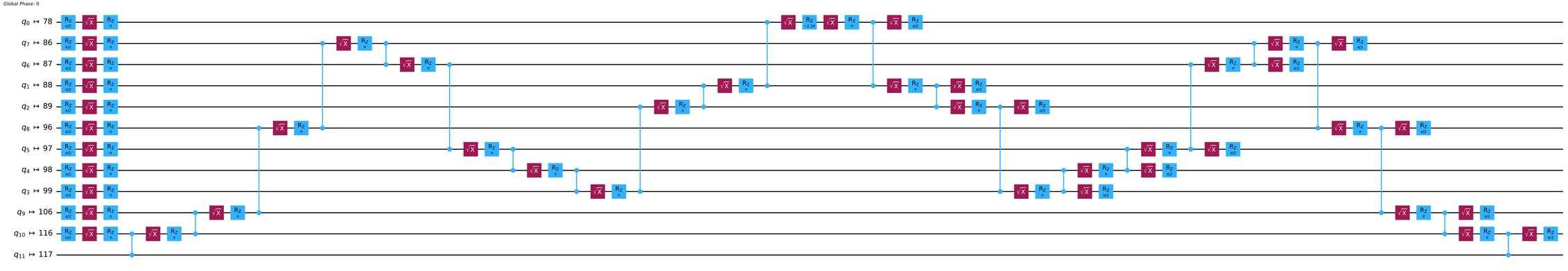

If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
If
mid[11] ^ mid[12] ^ mid[13] ^ ...
Else
RZ
π
−0.8
0.8
π/2
−π/2
−2.46
−0.886
√X
X
32
24
25
26
27
28
29
30
31
33
34
23
35
38
40
45
[expr]
[expr]
mid_3


If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
RZ
π
If
X
If
X
If
X
If
X
5=0x1
mid_38=0x1
mid_45=0x1
mid_35=0x1
36
37
mid_38=0x1
39
mid_40=0x1
41
42
43
44
mid_45=0x1
[expr]
mid_36=0x1
mid_37=0x1
mid_39=0x1
mid_40=0x1
mid_4


If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
R_Z
π
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
R_Z
π
If
mid[7] ^ mid[8] ^ mid[9] ^ mid...
R_Z
π
=0x1
mid_42=0x1
mid_43=0x1
mid_44=0x1
mid_36=0x1
mid_37=0x1
mid_39=0x1
mid_41=0x1
mid_42=0x1
mid_43=0x1
mid_44=0x1
[expr]
[expr]
[expr]

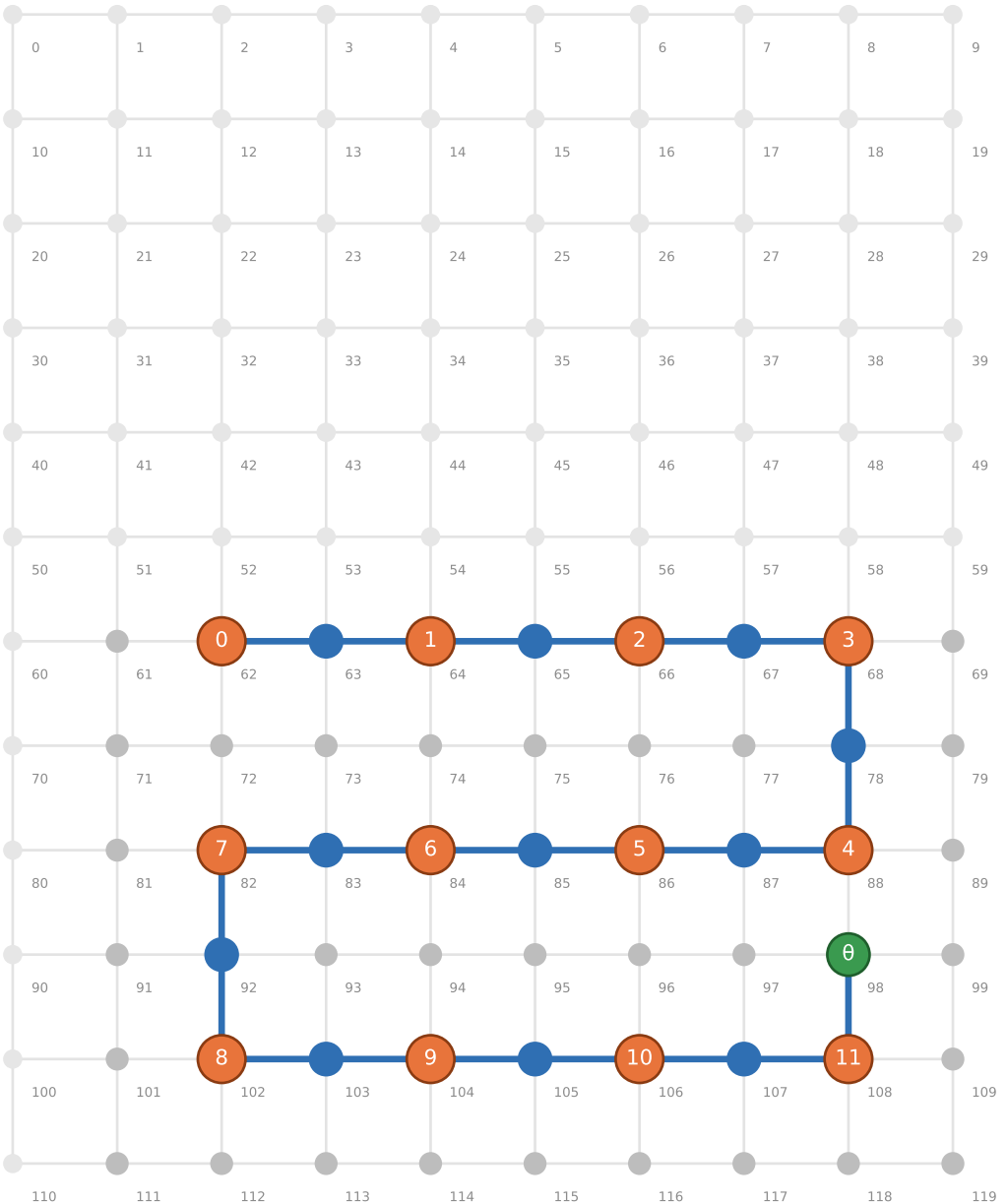

If
mid[6] ^ mid[7] ^ mid[8] ^ mid...
R_Z
π
If
mid[5] ^ mid[6] ^ mid[7] ^ mid...
R_Z
π
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
R_Z
π
If
X
If
mid[8] ^ mid[9] ^ mid[10] ^ mi...
R_Z
π
If
mid[9] ^ mid[10] ^ mid[23] ^ m...
R_Z
π
If
mid[10] ^ mid[23] ^ mid[34]
R_Z
π
If
R_Z
π
xpr]
[expr]
[expr]
mid_23=0x1
[expr]
[expr]
[expr]
mid_23=0x1

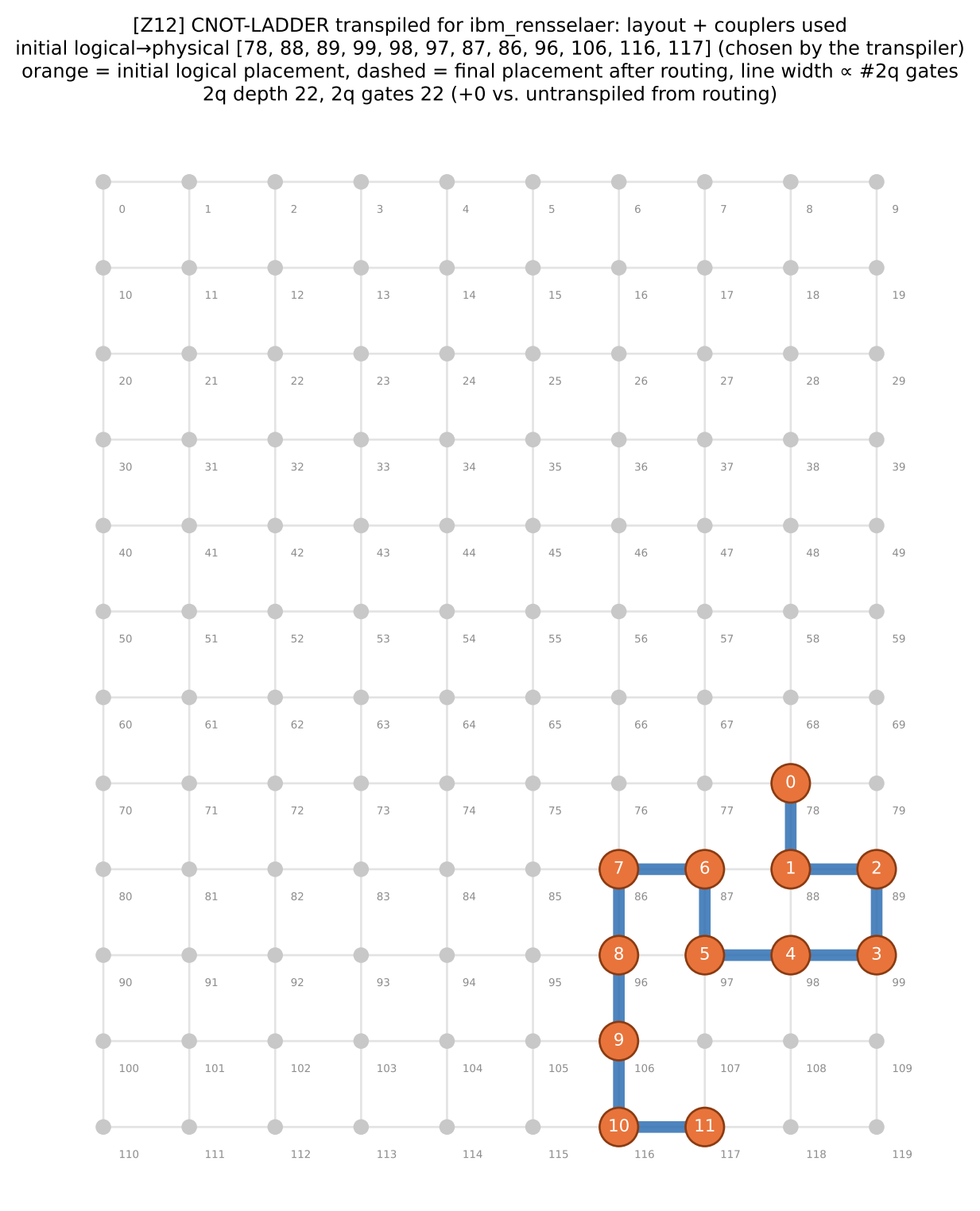
[Z12] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:55 sx:44 cz:22)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 (12 logical qubits)
2q depth 22, total depth 71



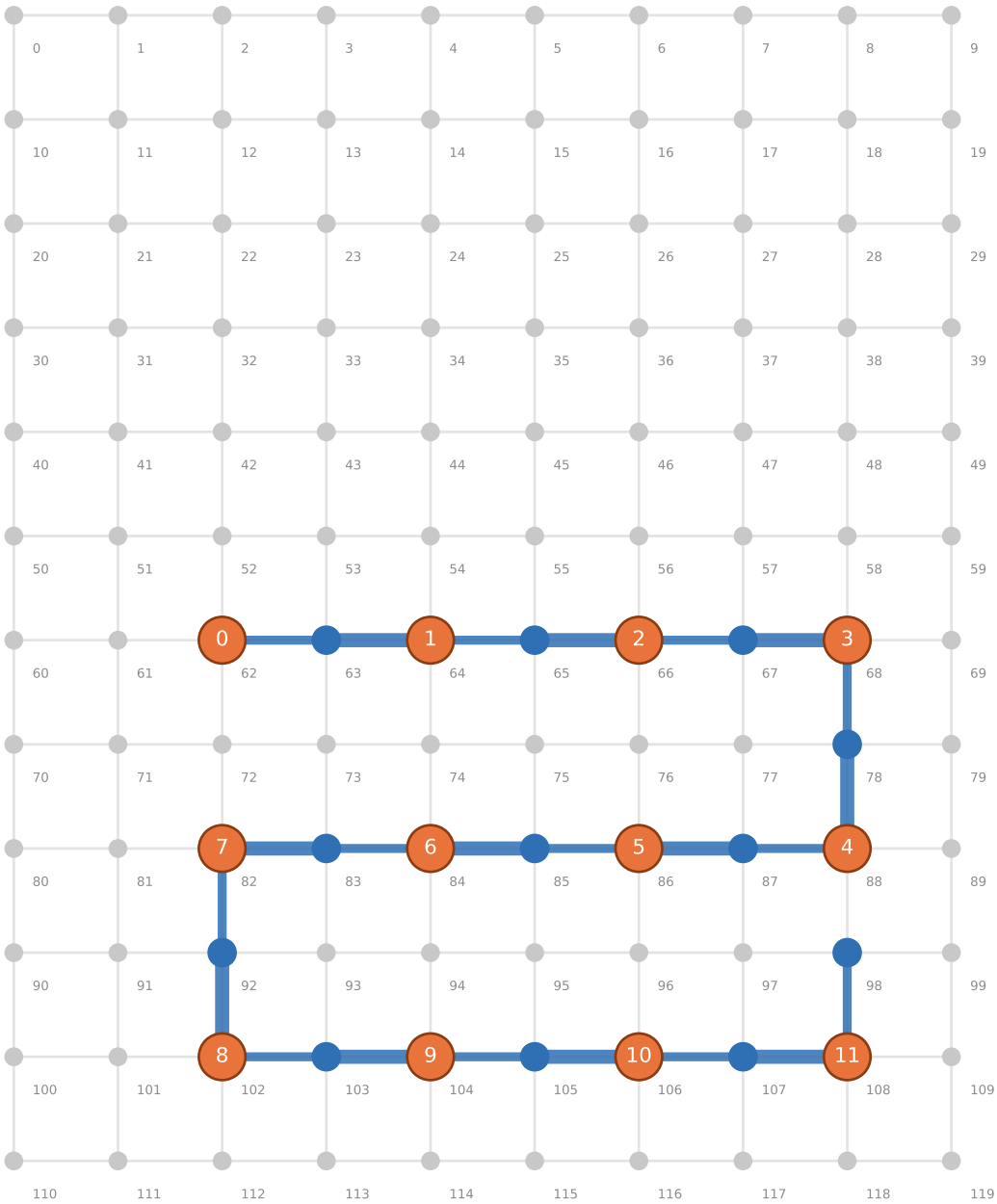
[Z12] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11
chain length 23, 11 ancillas, 0 |0> fillers

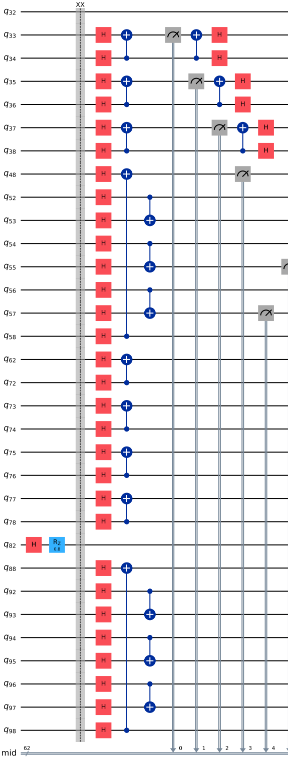


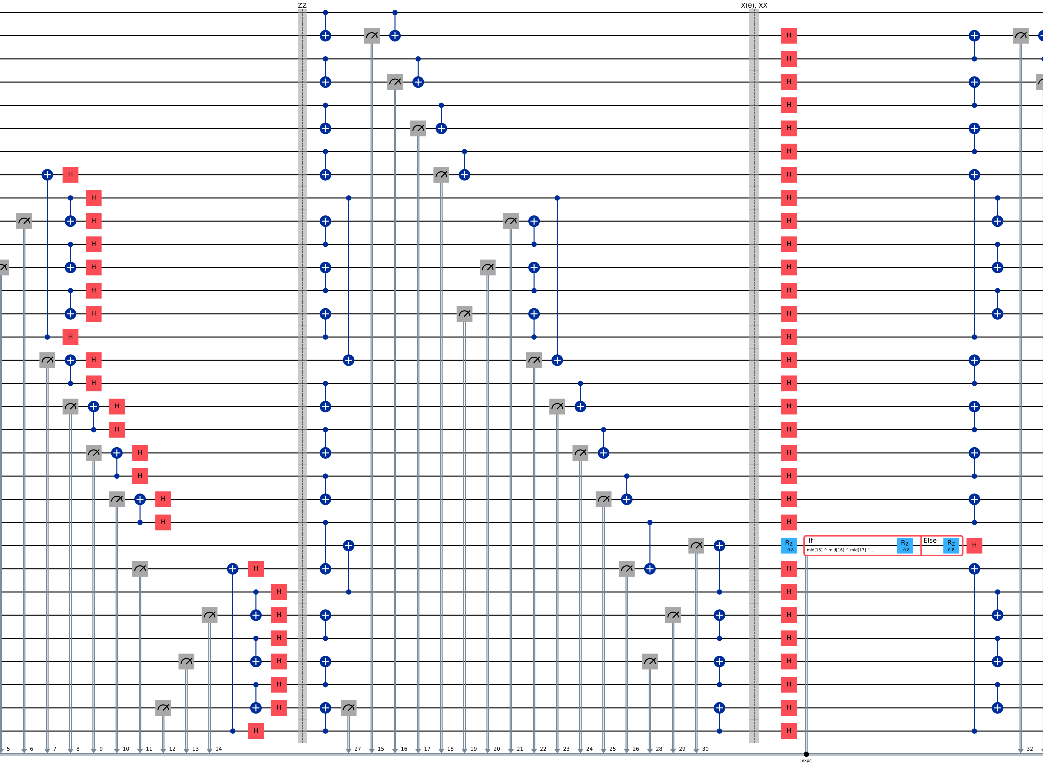

[Z12] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [78, 88, 89, 99, 98, 97, 87, 86, 96, 106, 116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 22, 2q gates 22 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
7 6 1 2
8 5 4 3
9
10 11

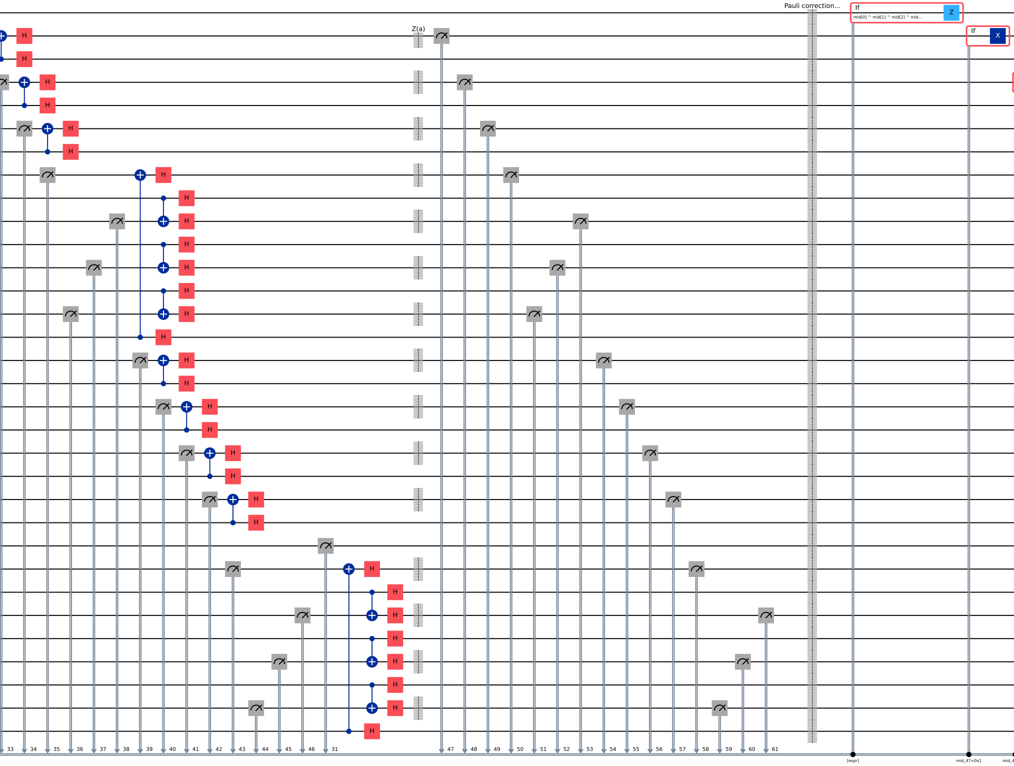
[Z12] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 68

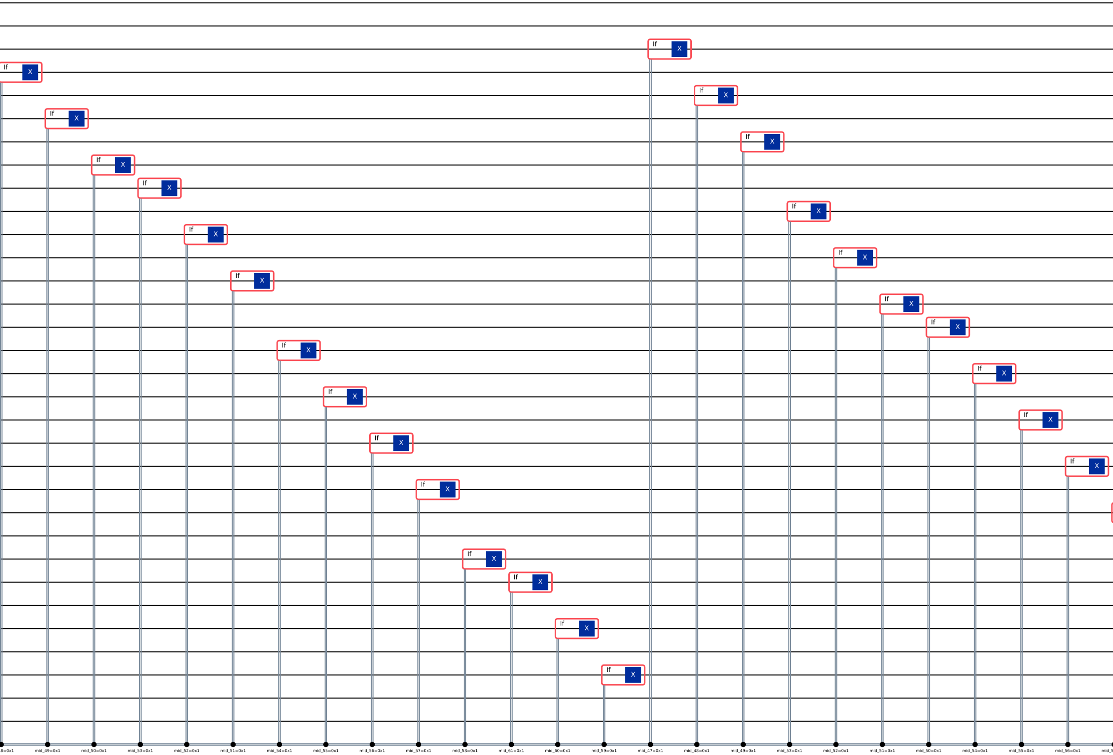


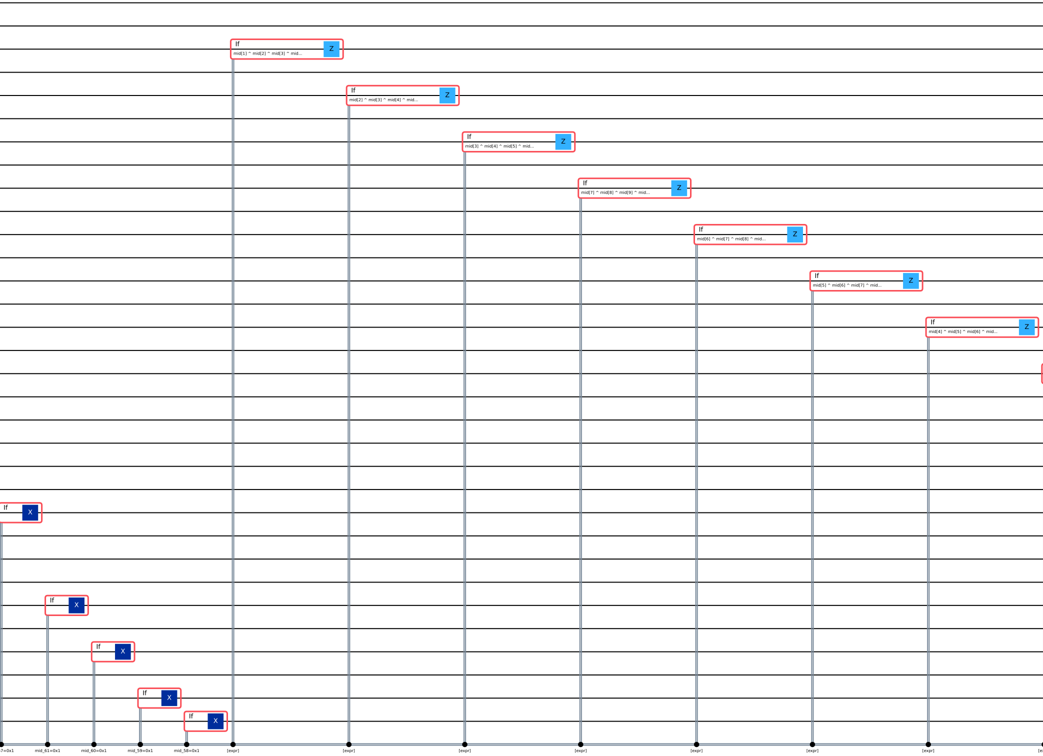
[Z16] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 6, 62 mid-circuit measurements, logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]

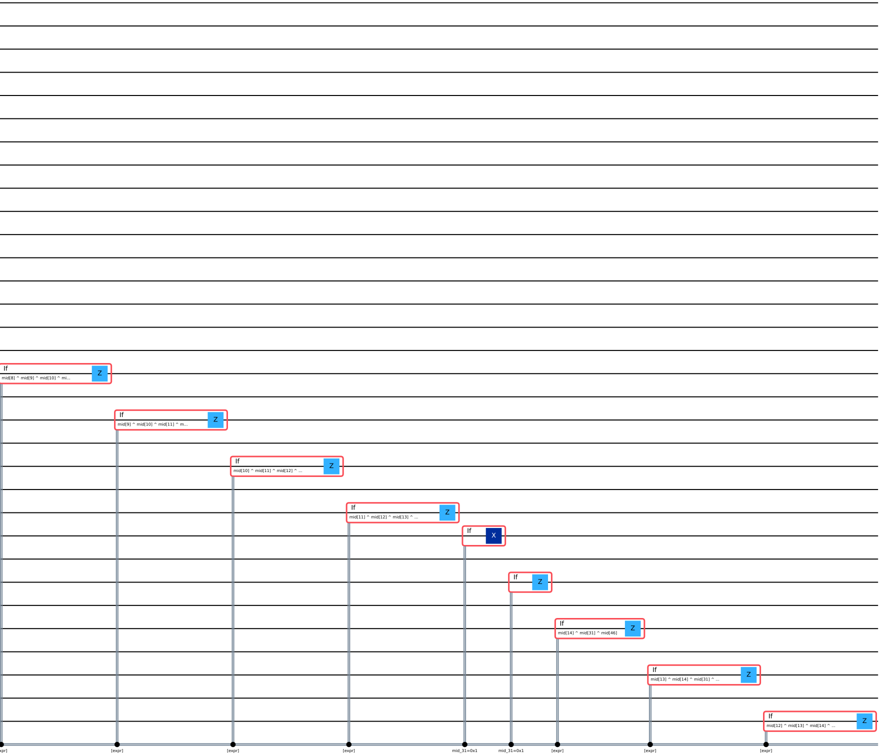

XX
q_{32}
q_{33}
q_{34}
q_{35}
q_{36}
q_{37}
q_{38}
q_{48}
q_{52}
q_{53}
q_{54}
q_{55}
q_{56}
q_{57}
q_{58}
q_{62}
q_{72}
q_{73}
q_{74}
q_{75}
q_{76}
q_{77}
q_{78}
q_{82}
q_{88}
q_{92}
q_{93}
q_{94}
q_{95}
q_{96}
q_{97}
q_{98}
mid
62
H
R_Z
0.8
0
1
2
3
4


ZZ
X(θ), XX
H
R_Z −0.8
If
mid[15] ^ mid[16] ^ mid[17] ^ ...
R_Z −0.8
Else
R_Z 0.8
5
6
7
8
9
10
11
12
13
14
27
15
16
17
18
19
20
21
22
23
24
25
26
28
29
30
32
[expr]

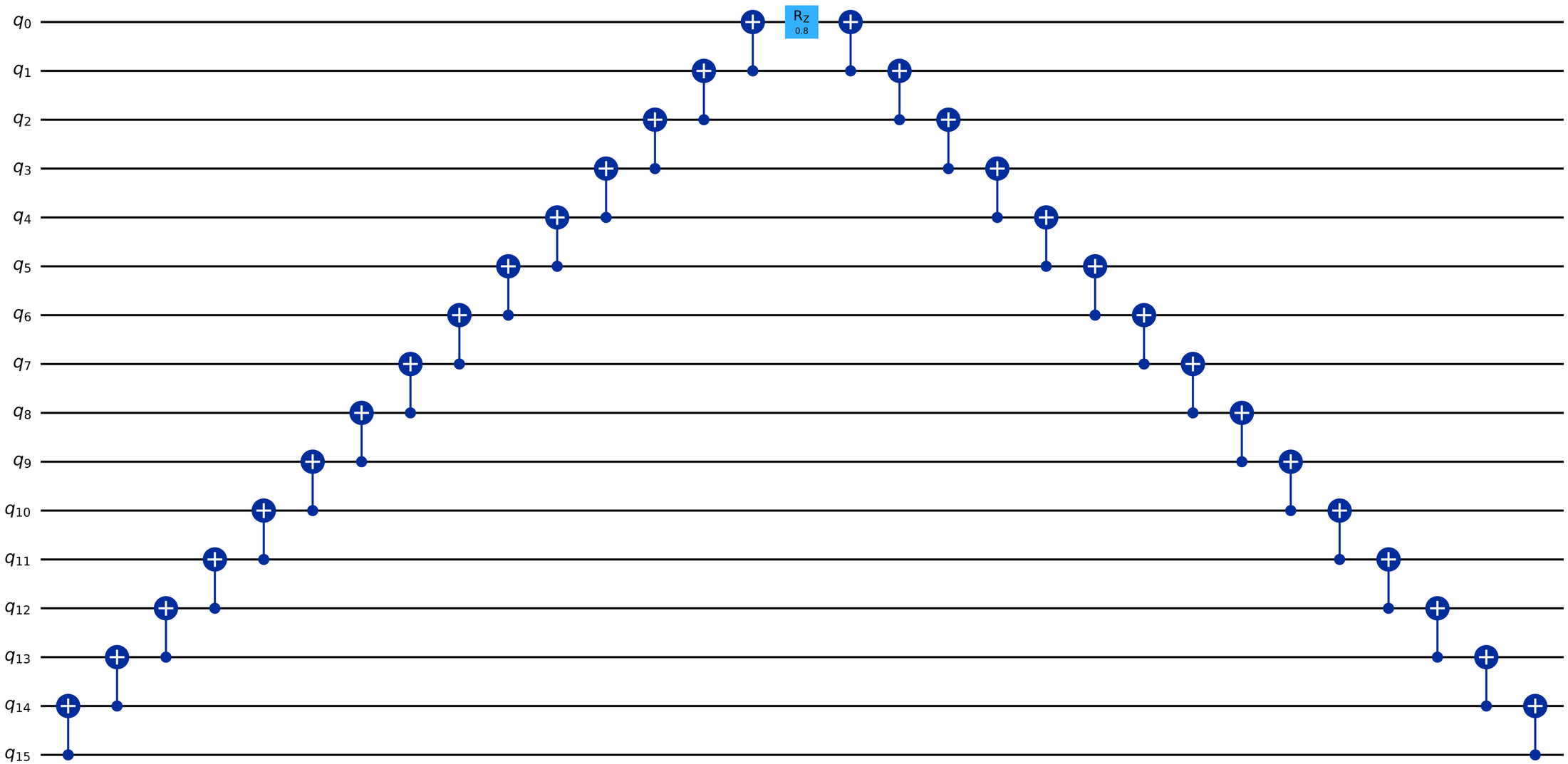

Pauli correction...
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
Z
If
X
Z(a)
H
33
34
35
36
37
38
39
40
41
42
43
44
45
46
31
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
[expr]
mid_47=0x1


If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
-8=0x1
mid_49=0x1
mid_50=0x1
mid_53=0x1
mid_52=0x1
mid_51=0x1
mid_54=0x1
mid_55=0x1
mid_56=0x1
mid_57=0x1
mid_58=0x1
mid_61=0x1
mid_60=0x1
mid_59=0x1
mid_47=0x1
mid_48=0x1
mid_49=0x1
mid_53=0x1
mid_52=0x1
mid_51=0x1
mid_50=0x1
mid_54=0x1
mid_55=0x1
mid_56=0x1
mid_5

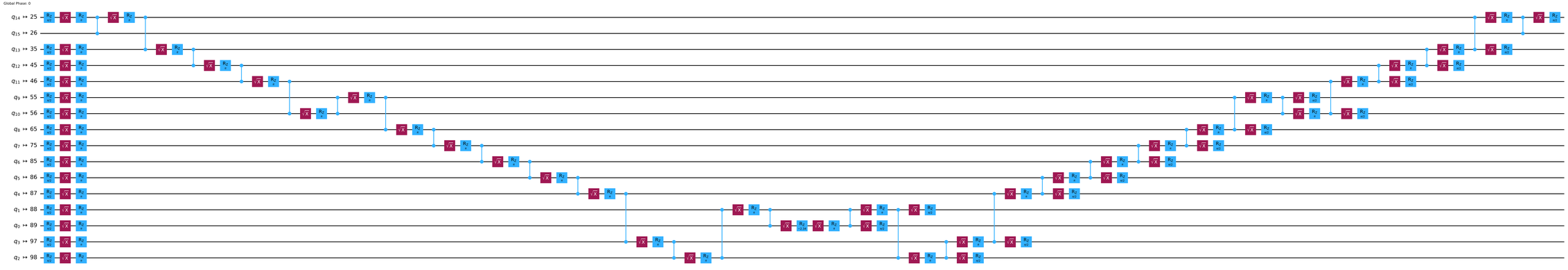

If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
Z
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
Z
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
Z
If
mid[7] ^ mid[8] ^ mid[9] ^ mid...
Z
If
mid[6] ^ mid[7] ^ mid[8] ^ mid...
Z
If
mid[5] ^ mid[6] ^ mid[7] ^ mid...
Z
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
Z
If
X
If
X
If
X
If
X
If
X
7=0x1
mid_61=0x1
mid_60=0x1
mid_59=0x1
mid_58=0x1
[expr]
[expr]
[expr]
[expr]
[expr]
[expr]
[expr]


If
mid[8] ^ mid[9] ^ mid[10] ^ mi...
Z
If
mid[9] ^ mid[10] ^ mid[11] ^ m...
Z
If
mid[10] ^ mid[11] ^ mid[12] ^ ...
Z
If
mid[11] ^ mid[12] ^ mid[13] ^ ...
Z
If
X
If
Z
If
mid[14] ^ mid[31] ^ mid[46]
Z
If
mid[13] ^ mid[14] ^ mid[31] ^ ...
Z
If
mid[12] ^ mid[13] ^ mid[14] ^ ...
Z
xpr]
[expr]
[expr]
[expr]
mid_31=0x1
mid_31=0x1
[expr]
[expr]
[expr]

[Z16] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 30

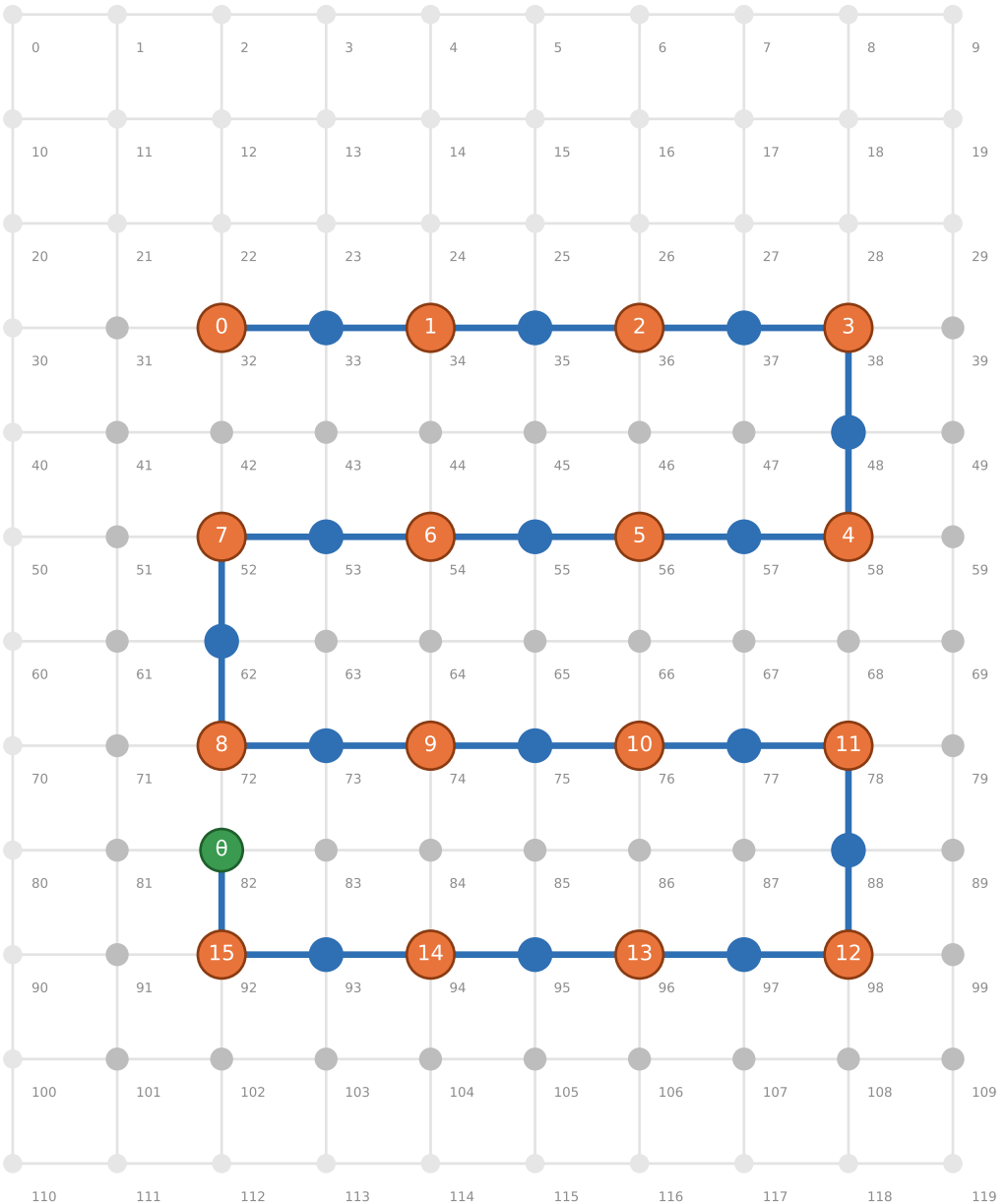


[Z16] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:75 sx:60 cz:30)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 (16 logical qubits)
2q depth 30, total depth 95

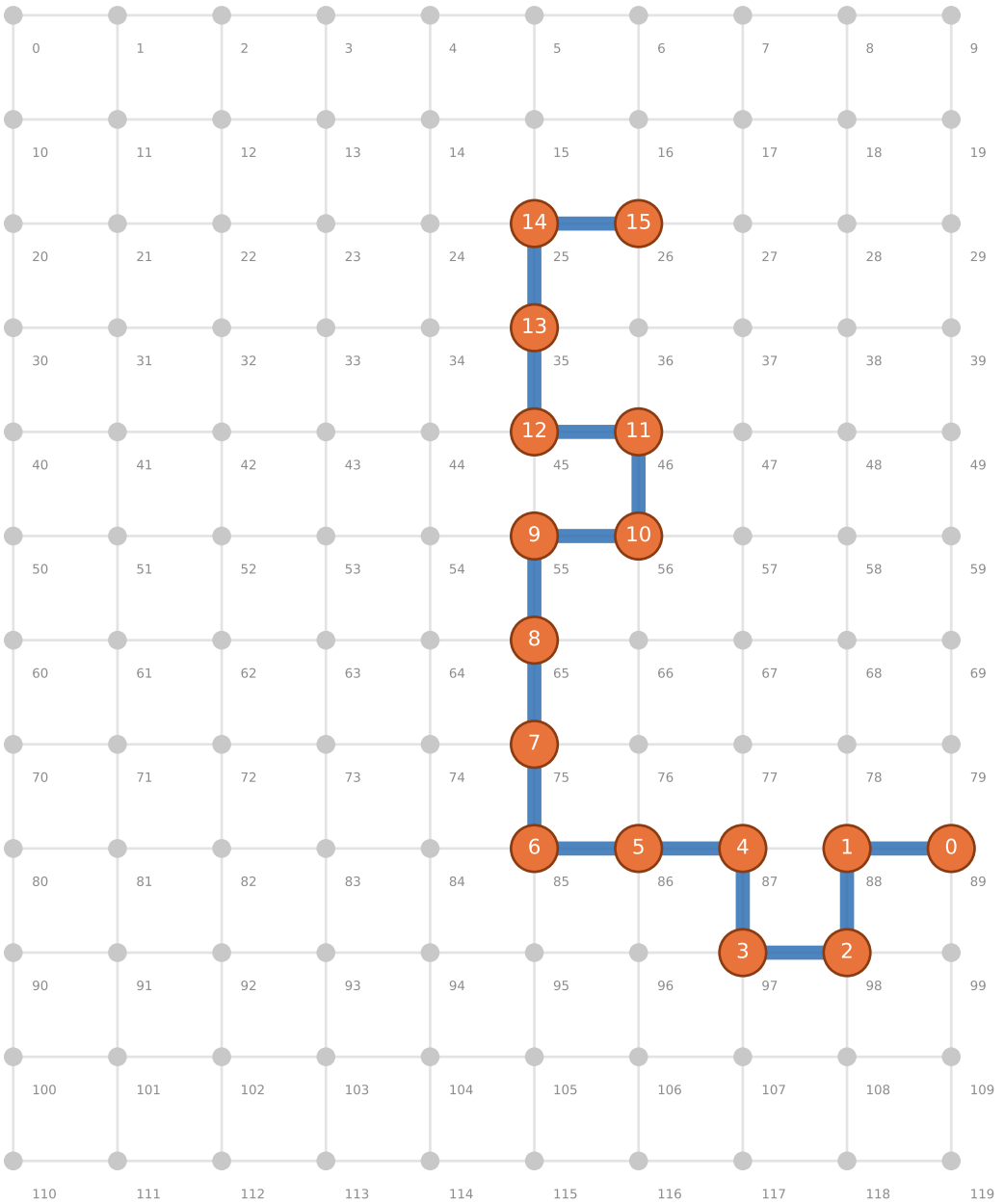


[Z16] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

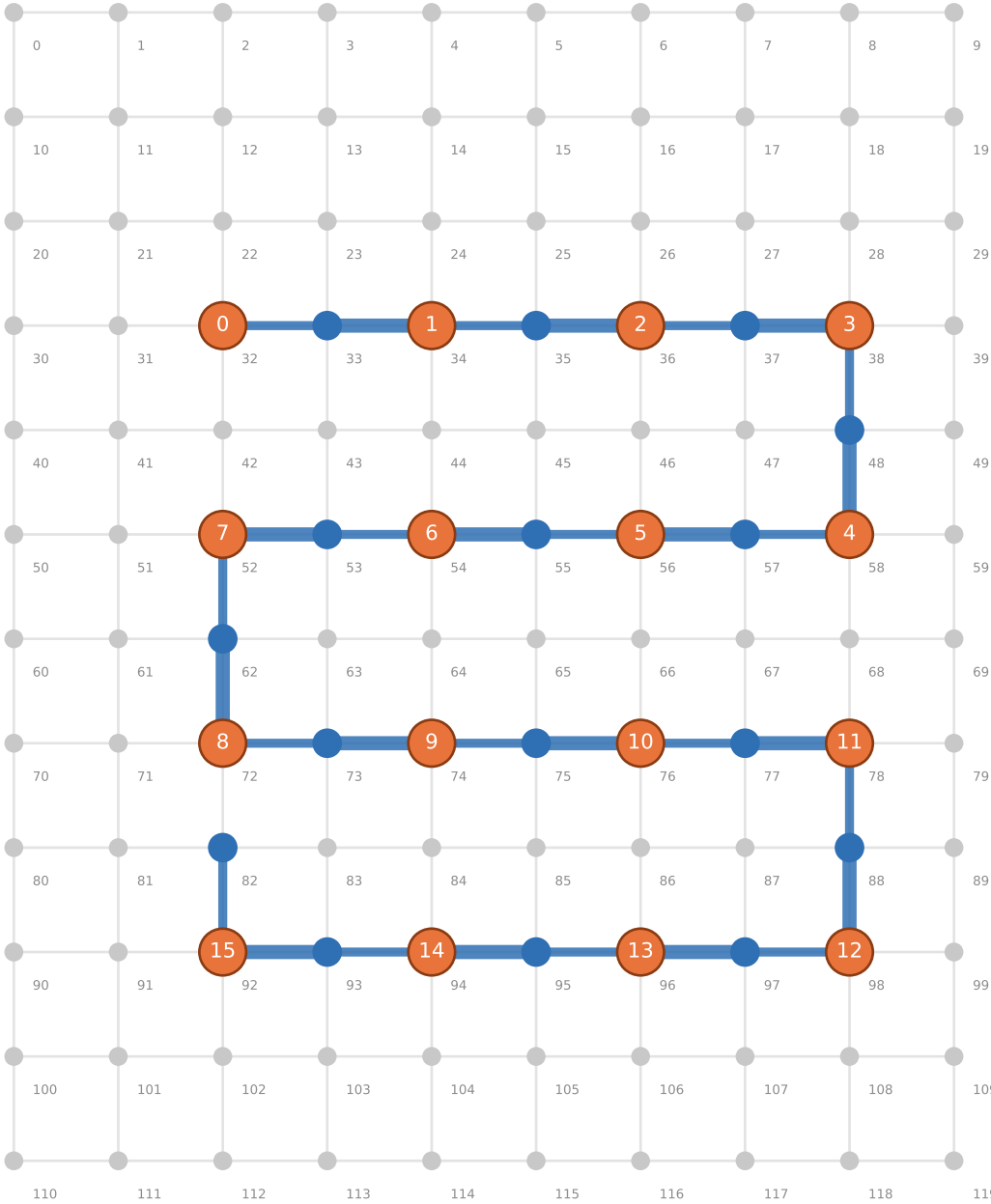
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15
chain length 31, 15 ancillas, 0 |0> fillers



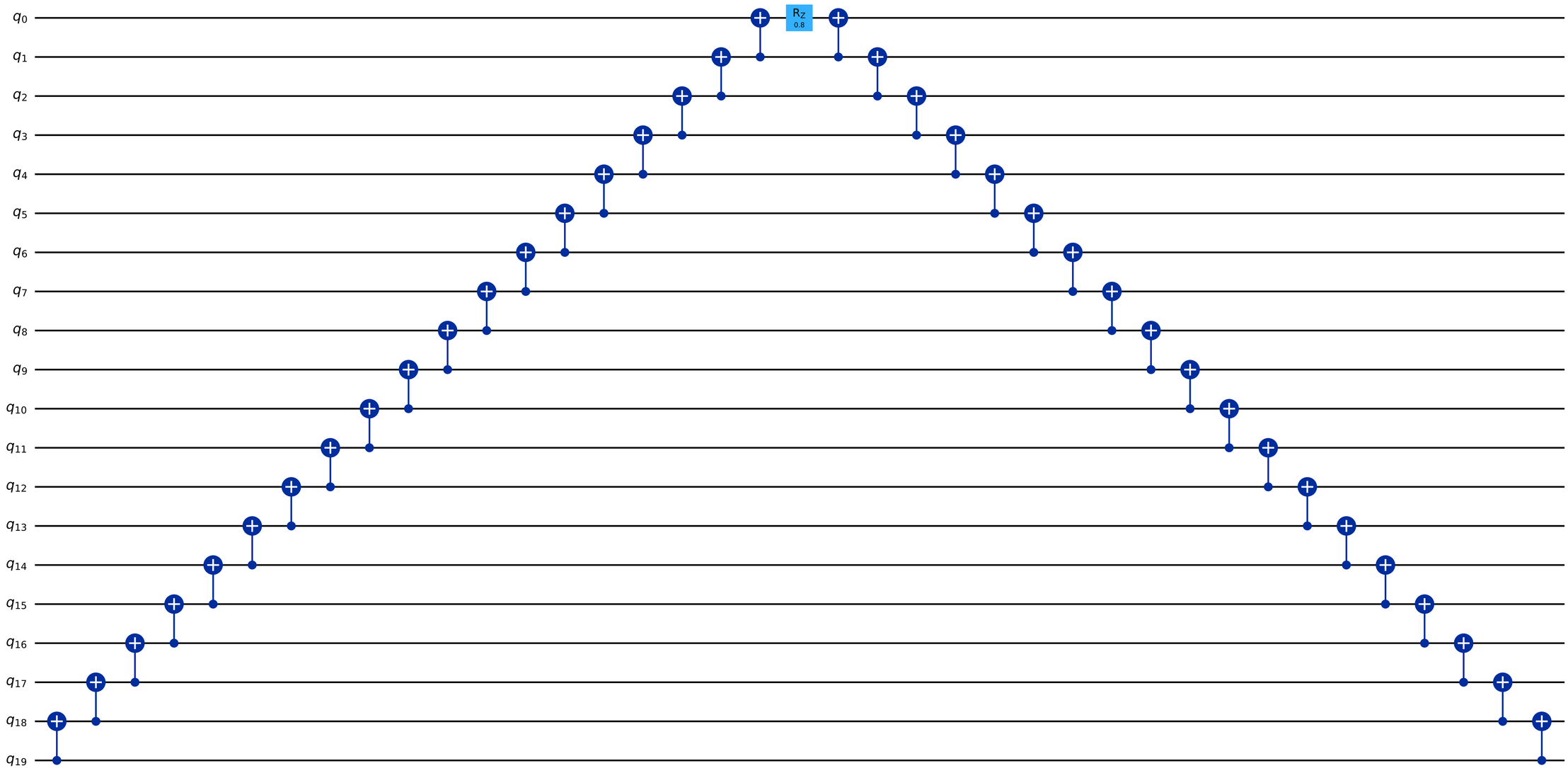
[Z16] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [89, 88, 98, 97, 87, 86, 85, 75, 65, 55, 56, 46, 45, 35, 25, 26] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 30, 2q gates 30 (+0 vs. untranspiled from routing)



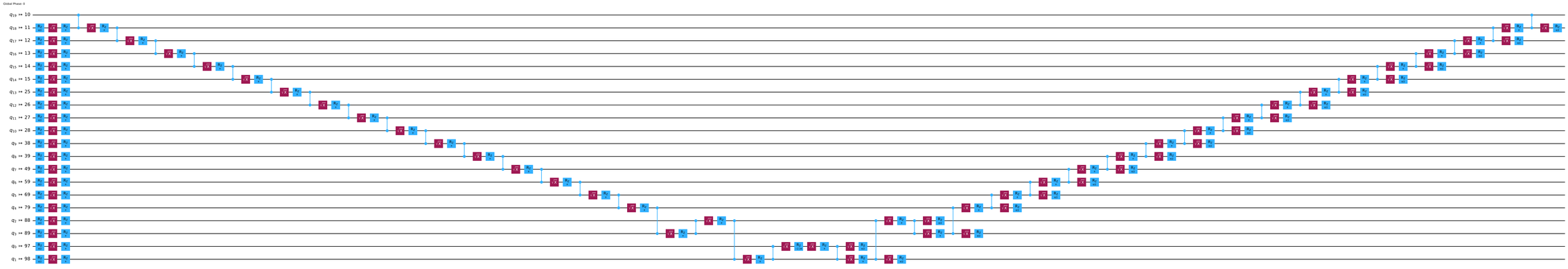
[Z16] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 92



[Z20] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 38

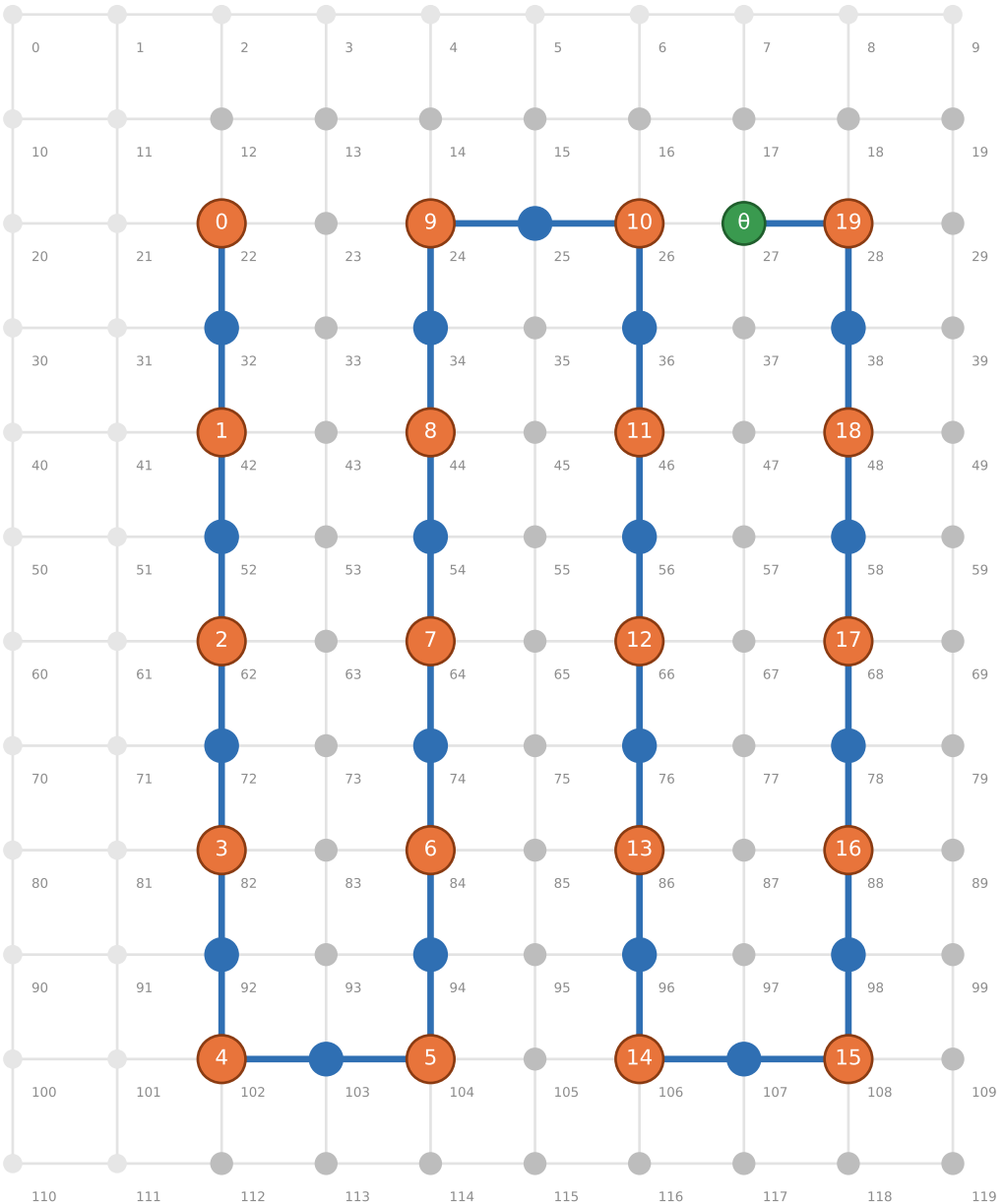


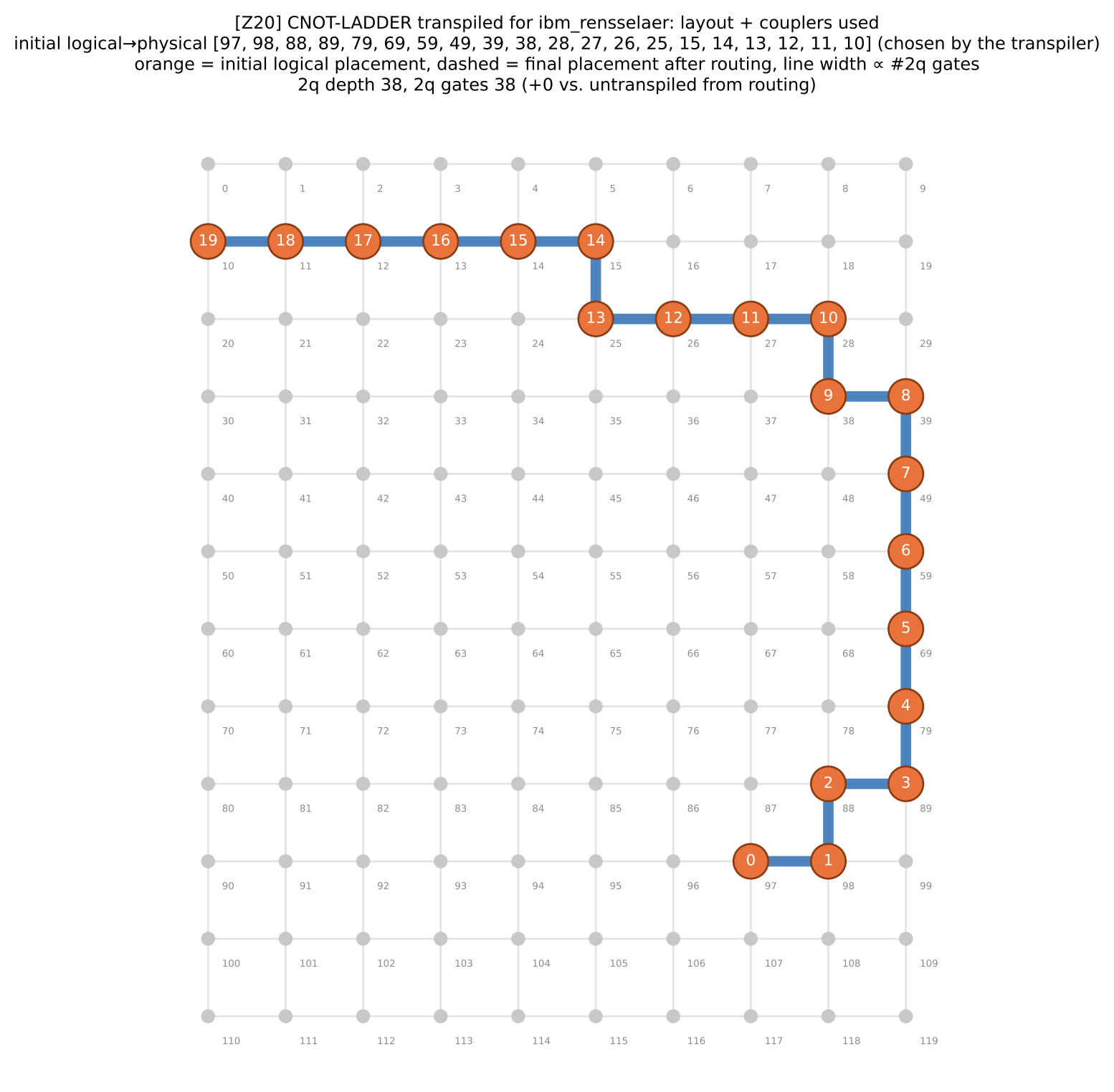
[Z20] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:95 sx:76 cz:38)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 (20 logical qubits)
2q depth 38, total depth 119



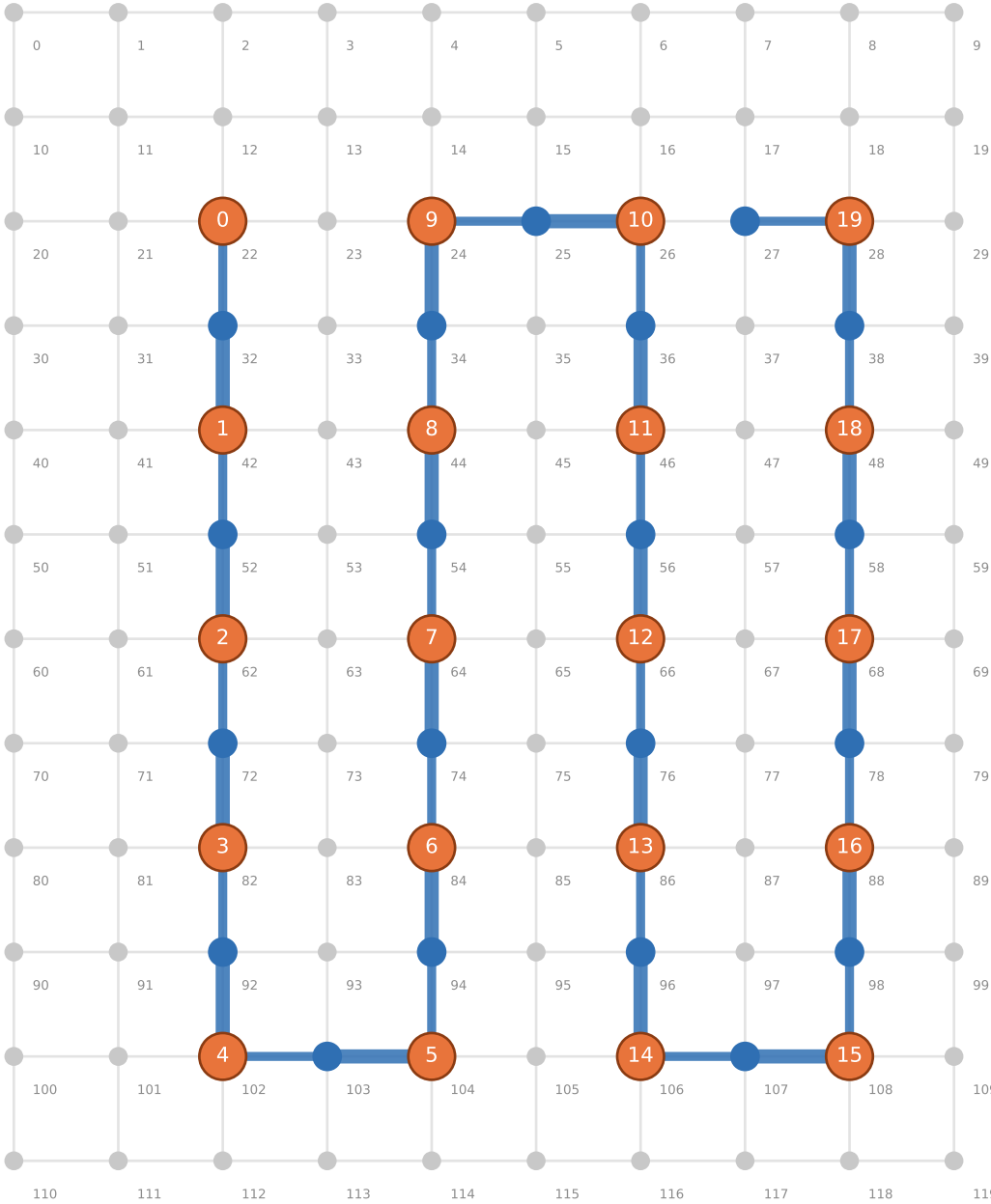
[Z20] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [22, 42, 62, 82, 102, 104, 84, 64, 44, 24, 26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19
chain length 39, 19 ancillas, 0 |0> fillers

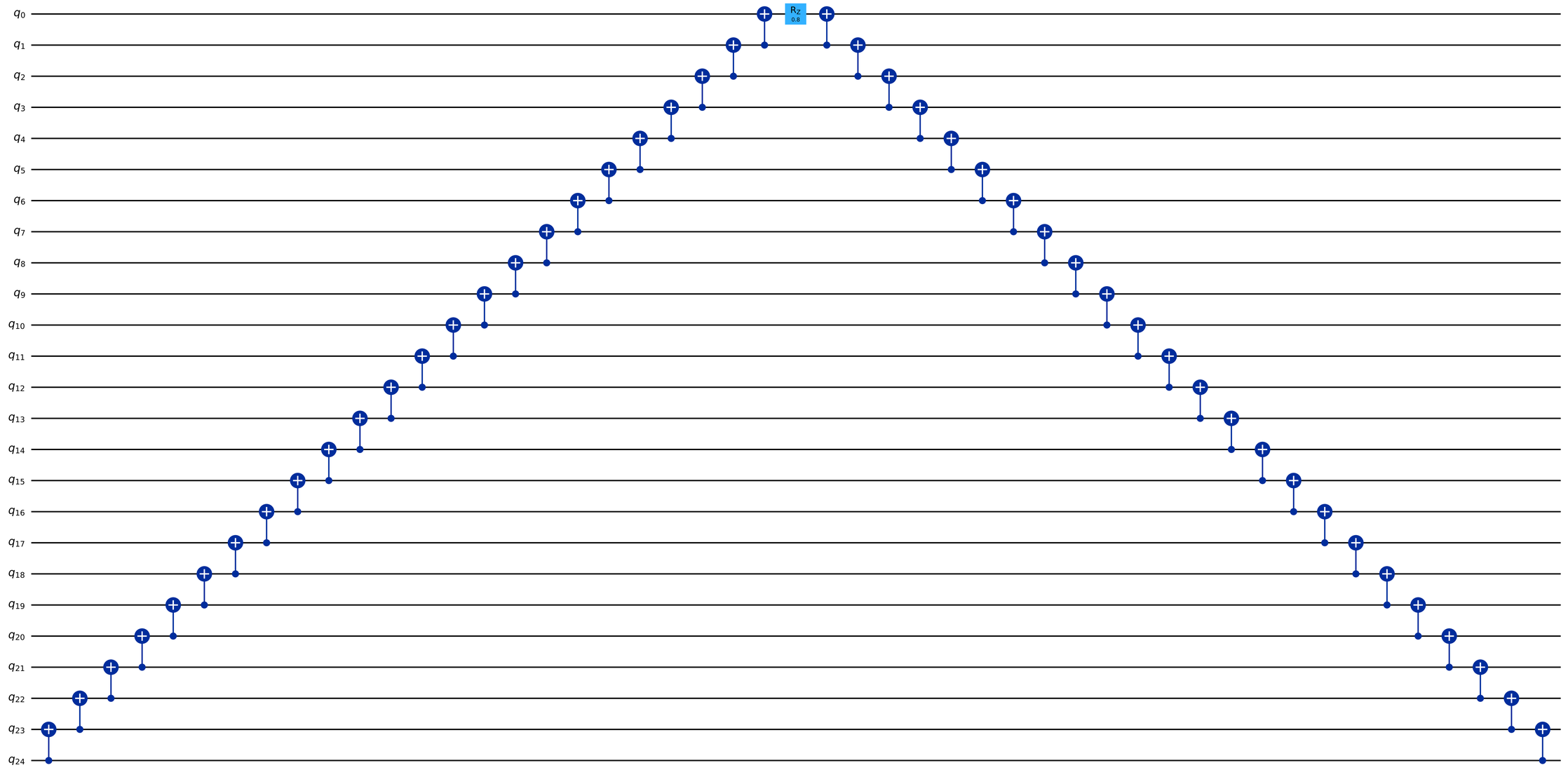


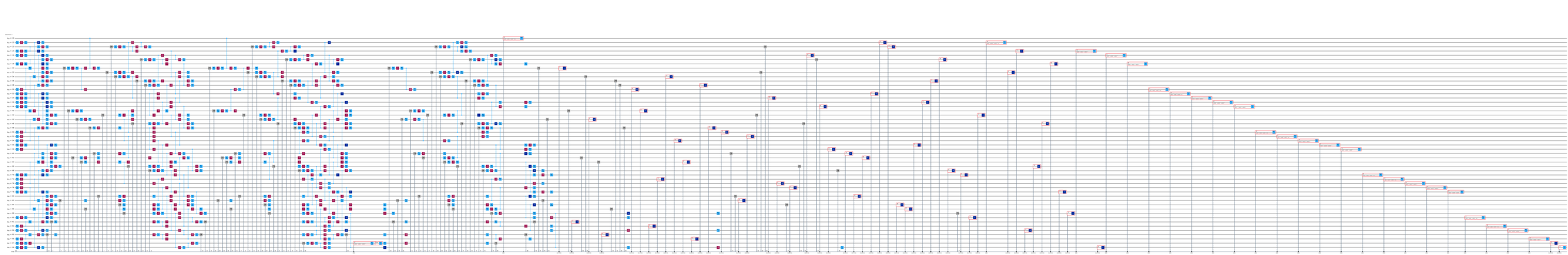

[Z20] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [97, 98, 88, 89, 79, 69, 59, 49, 39, 38, 28, 27, 26, 25, 15, 14, 13, 12, 11, 10] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 38, 2q gates 38 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
19 18 17 16 15 14
10 11 12 13 14 15 16 17 18 19
13 12 11 10
20 21 22 23 24 25 26 27 28 29
9 8
30 31 32 33 34 35 36 37 38 39
7
40 41 42 43 44 45 46 47 48 49
6
50 51 52 53 54 55 56 57 58 59
5
60 61 62 63 64 65 66 67 68 69
4
70 71 72 73 74 75 76 77 78 79
2 3
80 81 82 83 84 85 86 87 88 89
0 1
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

[Z20] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [22, 42, 62, 82, 102, 104, 84, 64, 44, 24, 26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 116

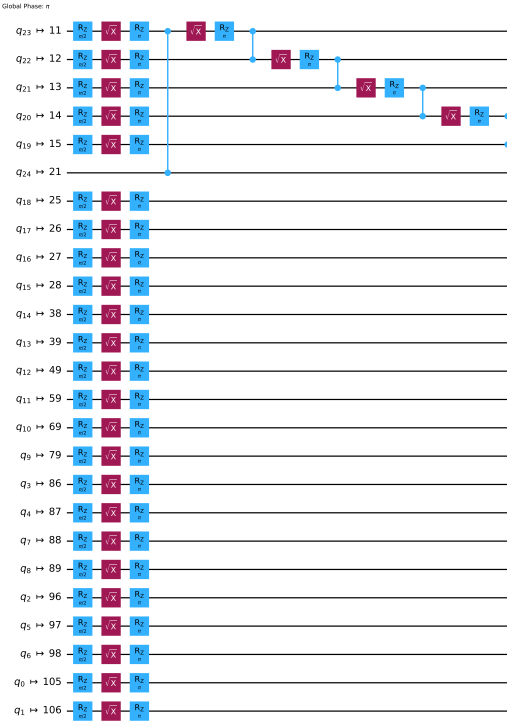


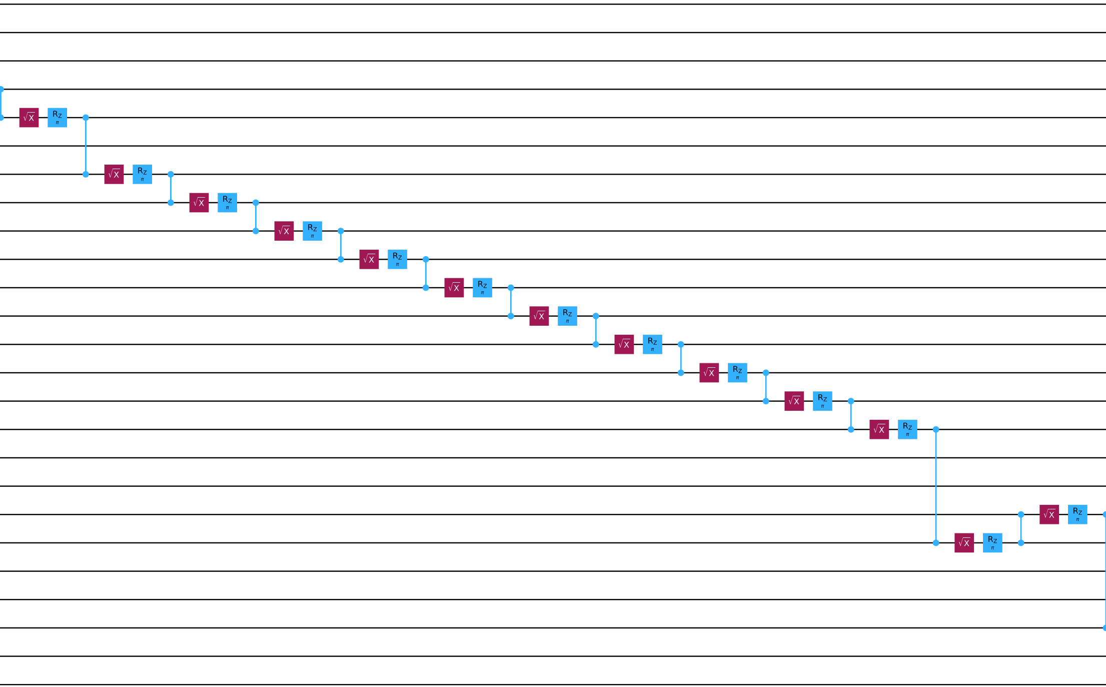
[Z25] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48

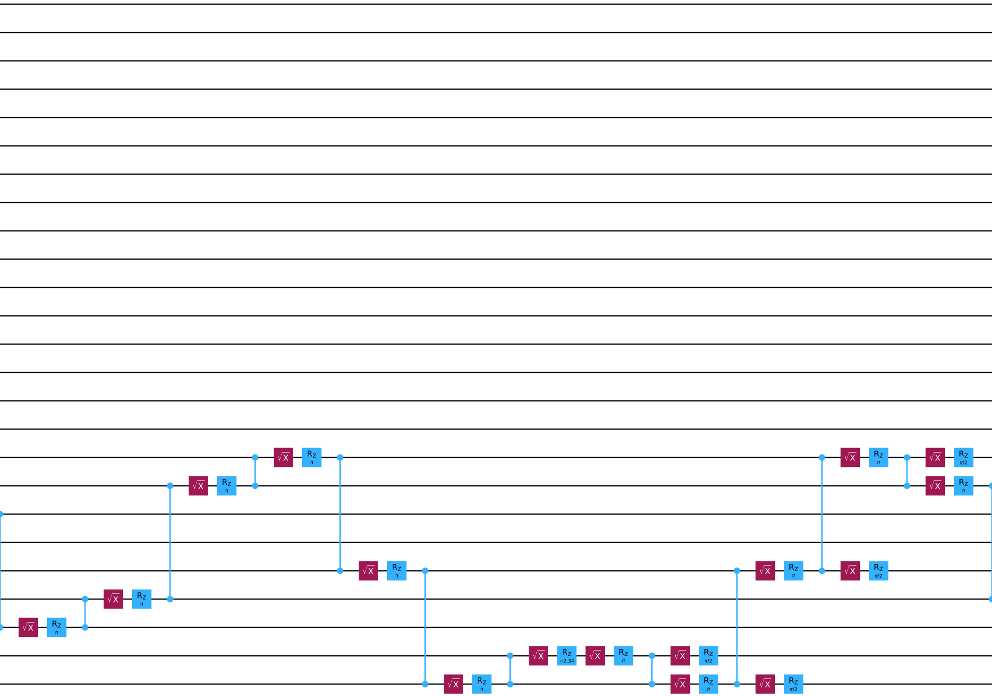


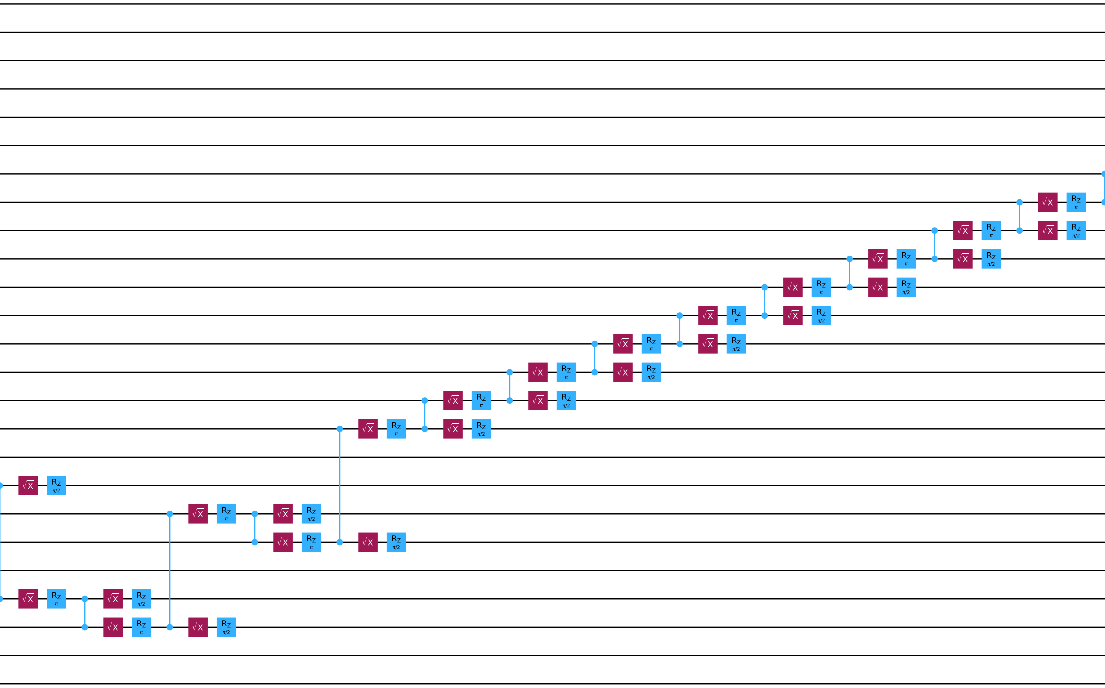


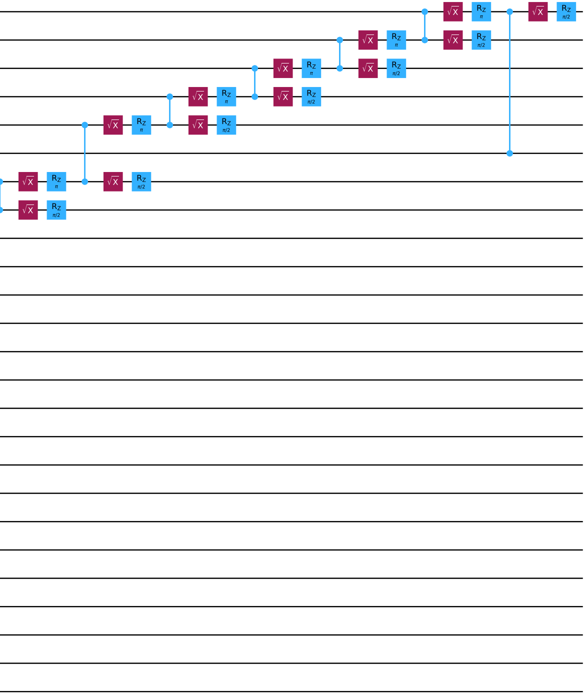
[Z25] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:120 sx:96 cz:48)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48, total depth 149


Global Phase: π
$q_{23} \mapsto 11$
$q_{22} \mapsto 12$
$q_{21} \mapsto 13$
$q_{20} \mapsto 14$
$q_{19} \mapsto 15$
$q_{24} \mapsto 21$
$q_{18} \mapsto 25$
$q_{17} \mapsto 26$
$q_{16} \mapsto 27$
$q_{15} \mapsto 28$
$q_{14} \mapsto 38$
$q_{13} \mapsto 39$
$q_{12} \mapsto 49$
$q_{11} \mapsto 59$
$q_{10} \mapsto 69$
$q_9 \mapsto 79$
$q_3 \mapsto 86$
$q_4 \mapsto 87$
$q_7 \mapsto 88$
$q_8 \mapsto 89$
$q_2 \mapsto 96$
$q_5 \mapsto 97$
$q_6 \mapsto 98$
$q_0 \mapsto 105$
$q_1 \mapsto 106$
R_Z π/2
$\sqrt{X}$
R_Z π



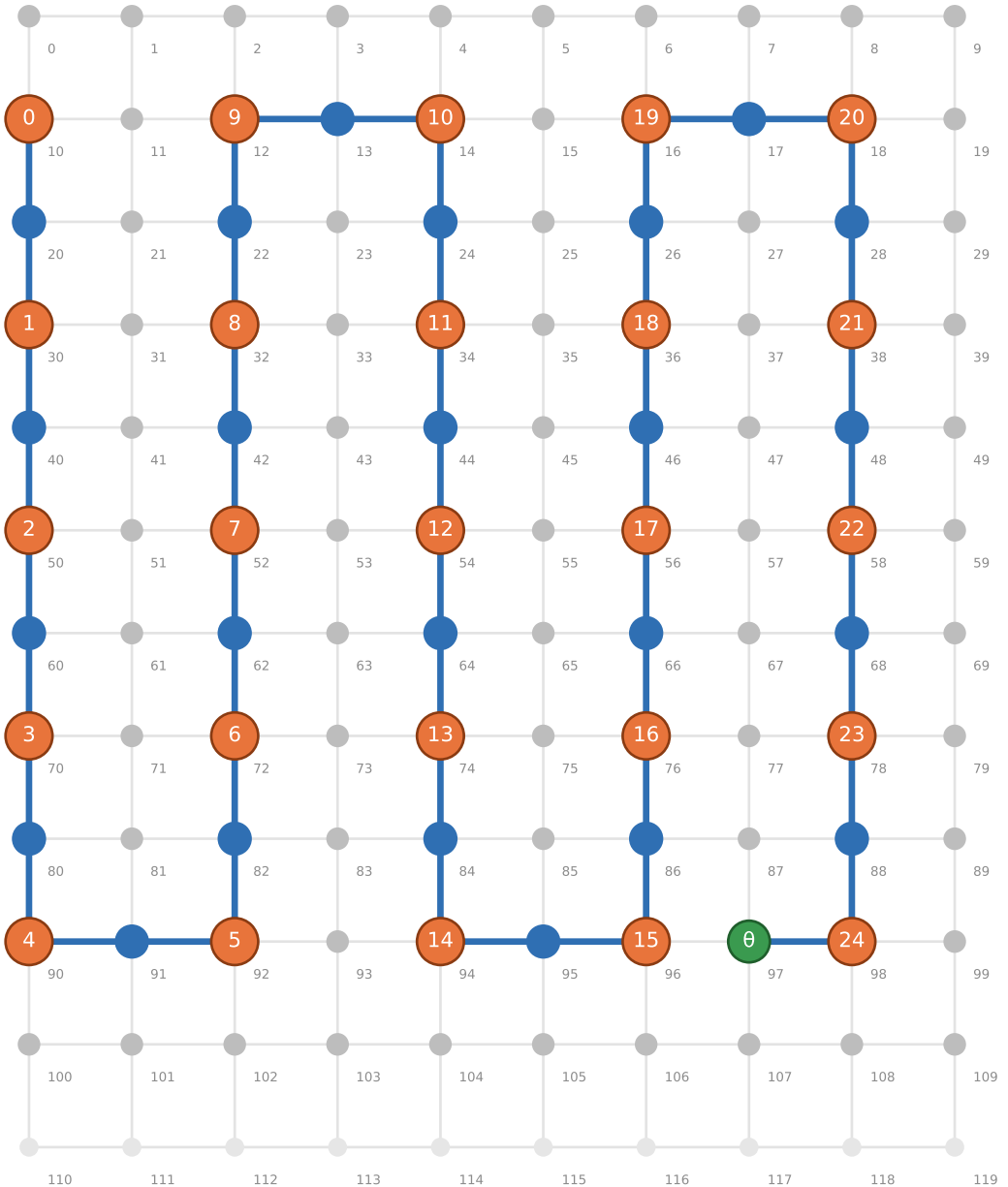

√X
Rz
π
√X
Rz
π
√X
Rz
π
√X
Rz
π
√X
Rz
π
√X
Rz
π
√X
Rz
−2.34
√X
Rz
π
√X
Rz
π/2
√X
Rz
π
√X
Rz
π
√X
Rz
π/2
√X
Rz
π
√X
Rz
π/2
√X
Rz
π
√X
Rz
π/2
√X
Rz
π



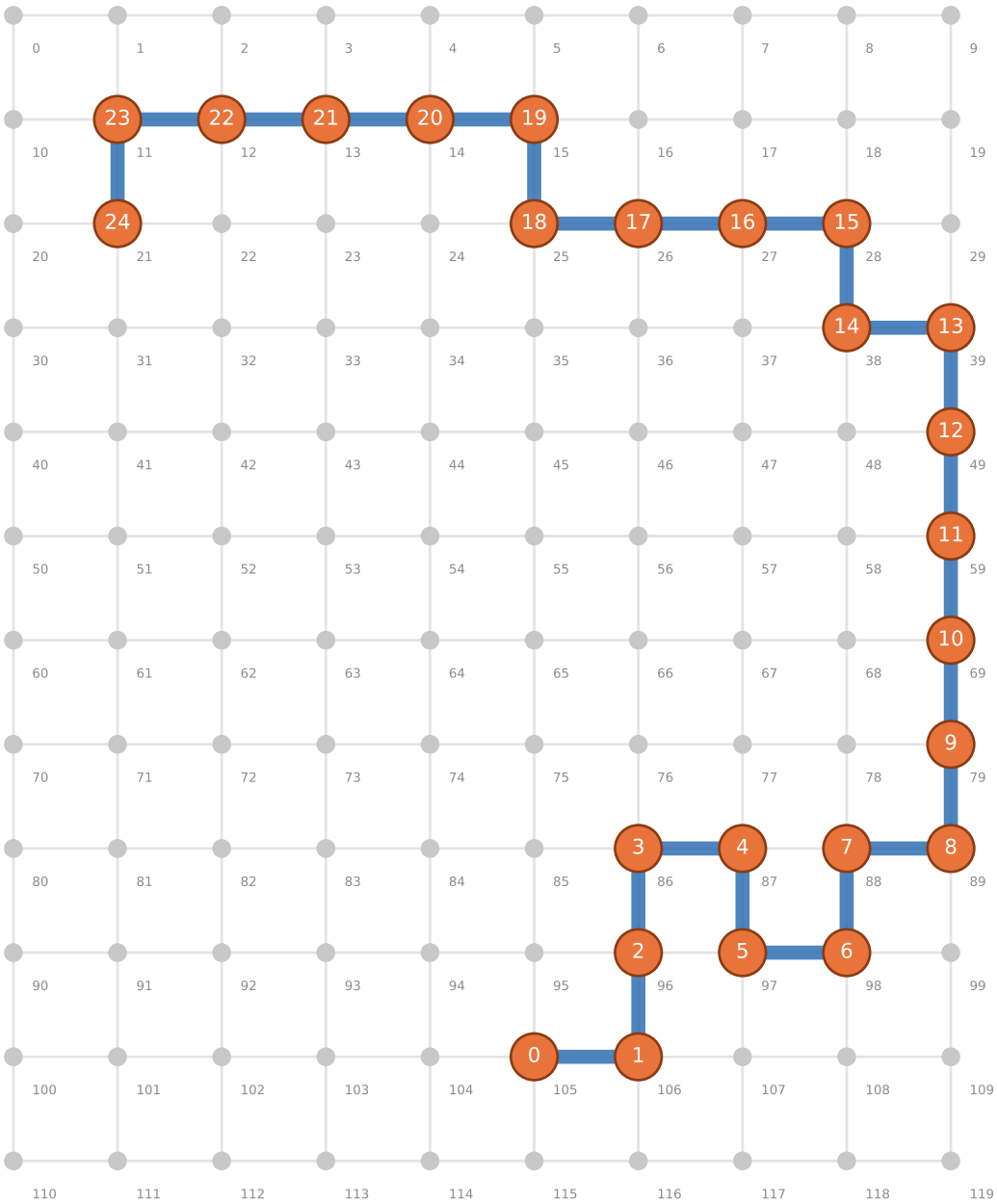


[Z25] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [10, 30, 50, 70, 90, 92, 72, 52, 32, 12, 14, 34, 54, 74, 94, 96, 76, 56, 36, 16, 18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

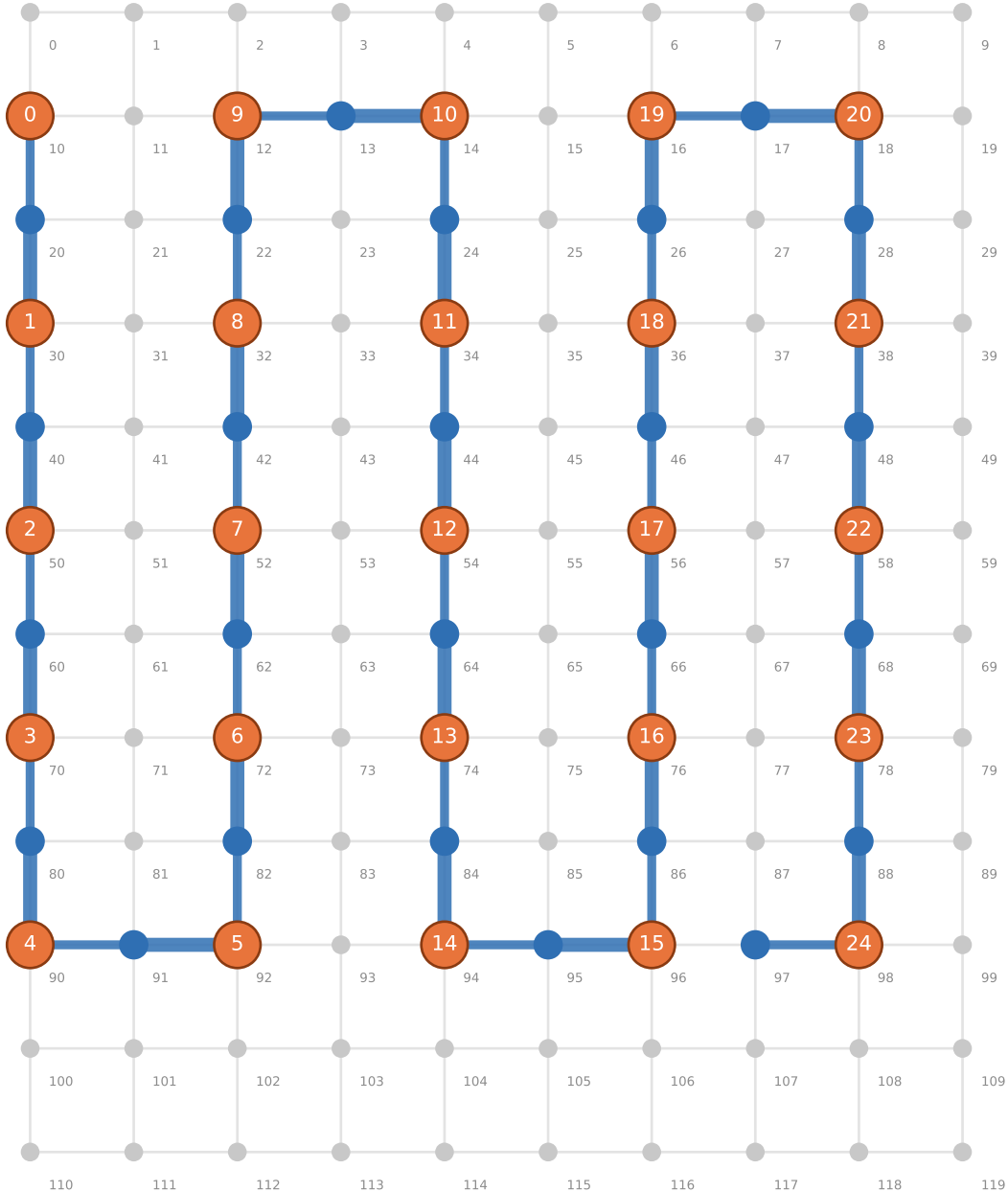
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24
chain length 49, 24 ancillas, 0 |0> fillers



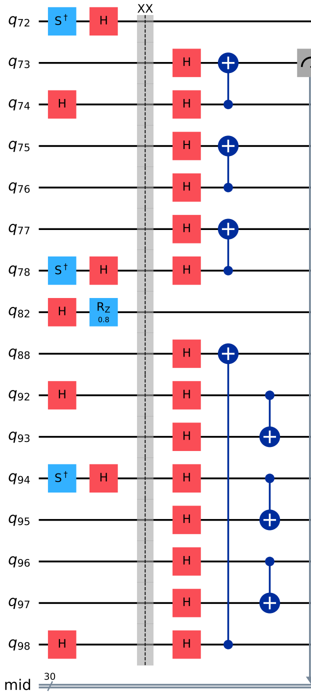
[Z25] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [105, 106, 96, 86, 87, 97, 98, 88, 89, 79, 69, 59, 49, 39, 38, 28, 27, 26, 25, 15, 14, 13, 12, 11, 21] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 48, 2q gates 48 (+0 vs. untranspiled from routing)

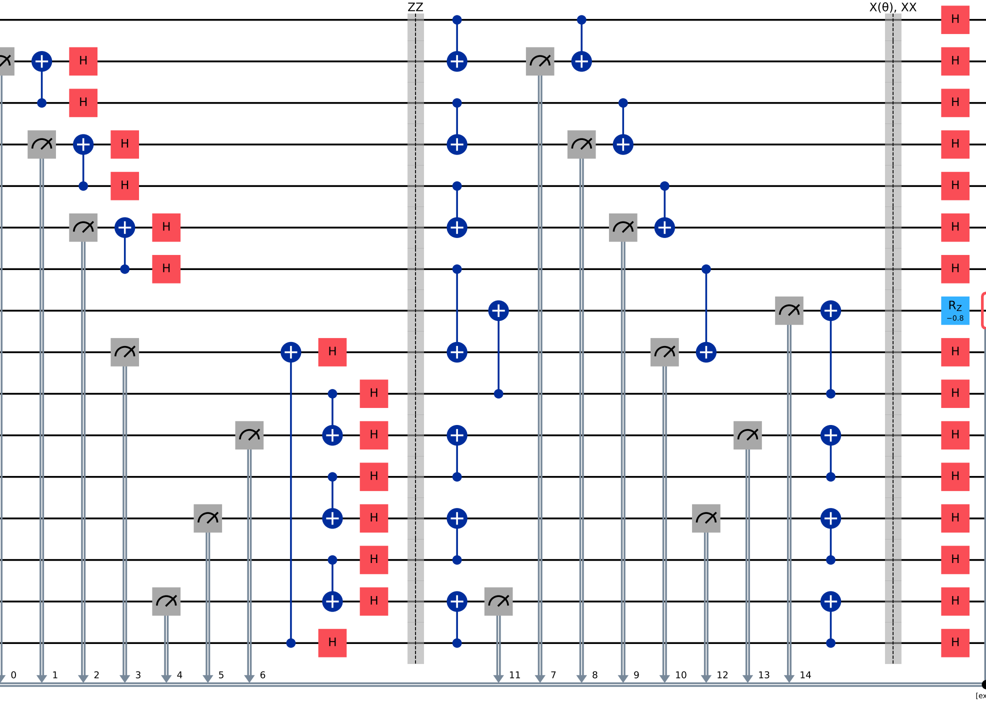


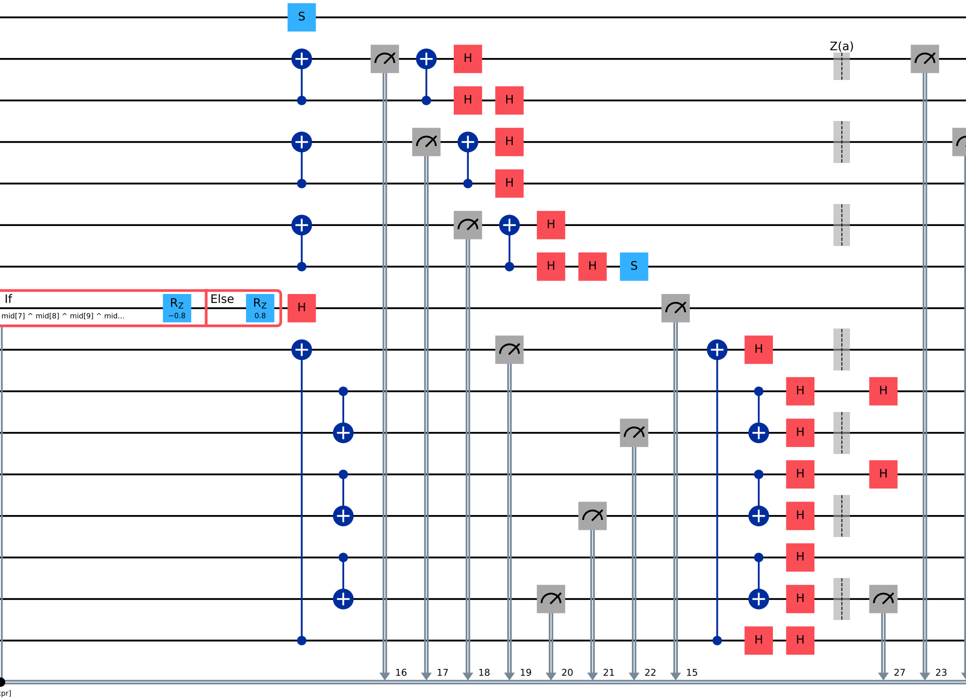
[Z25] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [10, 30, 50, 70, 90, 92, 72, 52, 32, 12, 14, 34, 54, 74, 94, 96, 76, 56, 36, 16, 18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 146

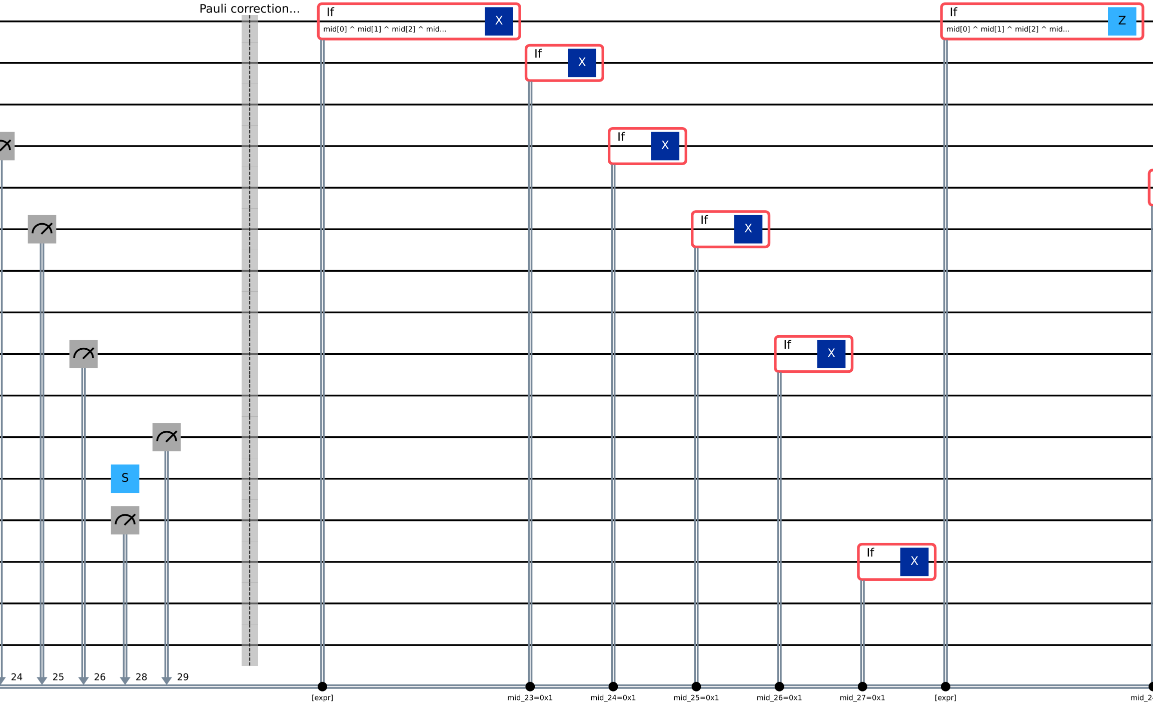


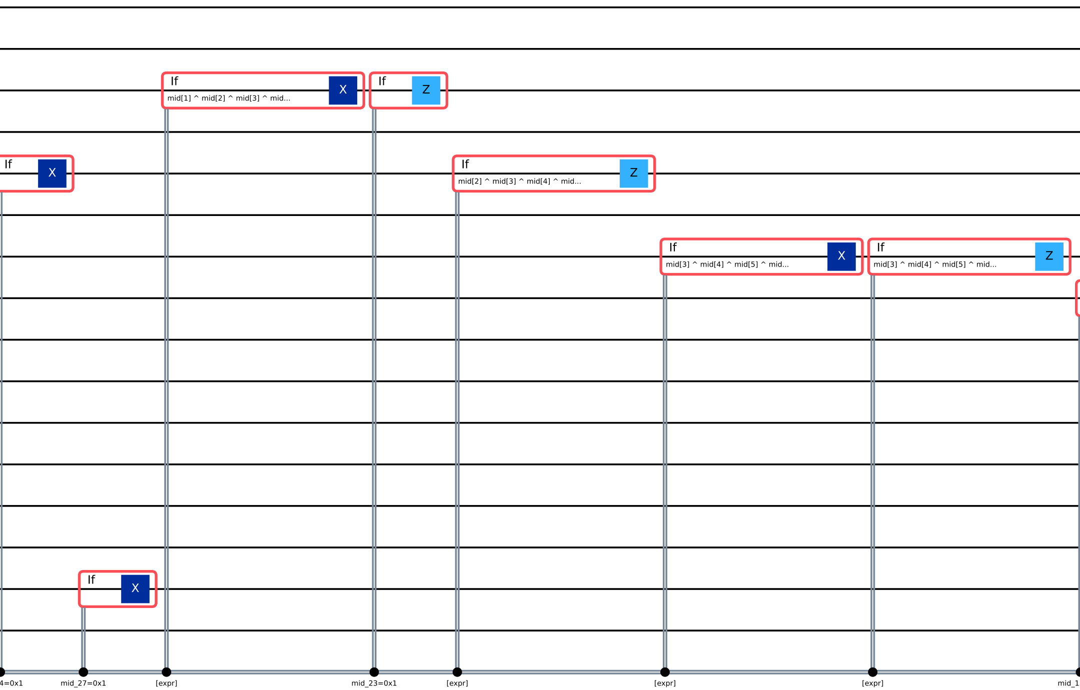
[XYZXYZXY] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 6, 30 mid-circuit measurements, logical→physical [72, 74, 76, 78, 98, 96, 94, 92]

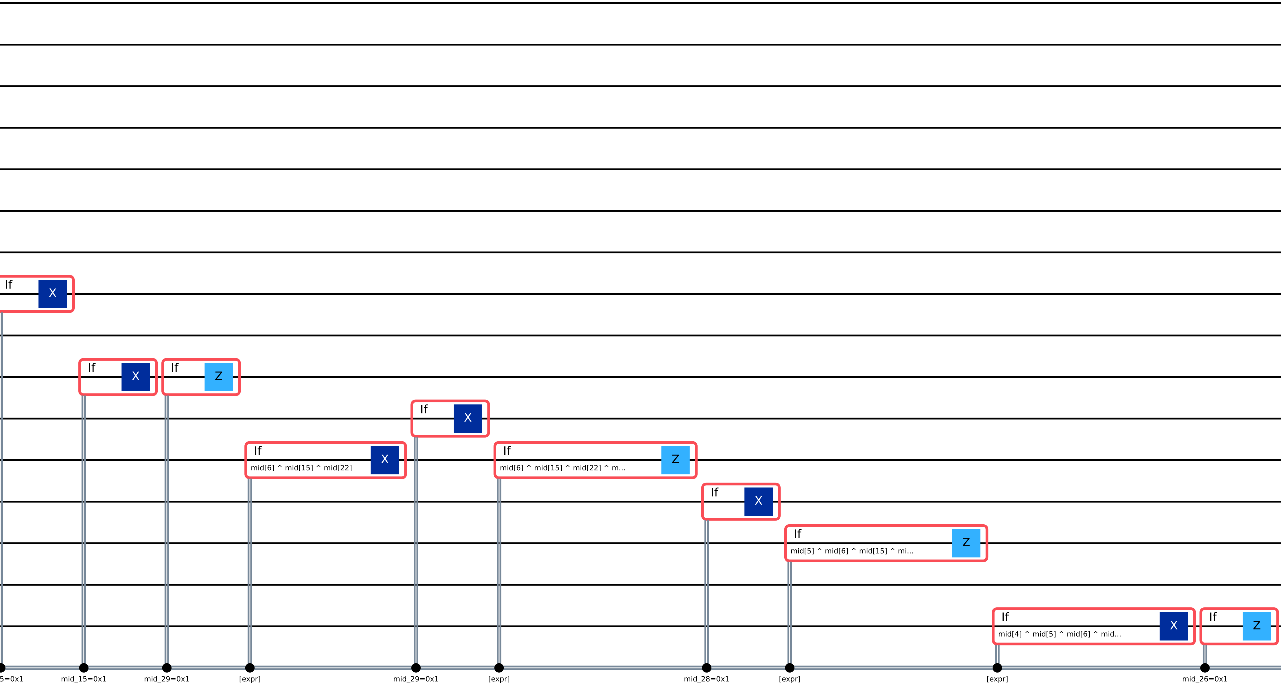

XX
q_{72}
q_{73}
q_{74}
q_{75}
q_{76}
q_{77}
q_{78}
q_{82}
q_{88}
q_{92}
q_{93}
q_{94}
q_{95}
q_{96}
q_{97}
q_{98}
mid
30
S†
H
R_Z
0.8


ZZ
X(θ), XX
H
R_Z
−0.8
0
1
2
3
4
5
6
11
7
8
9
10
12
13
14
[ex

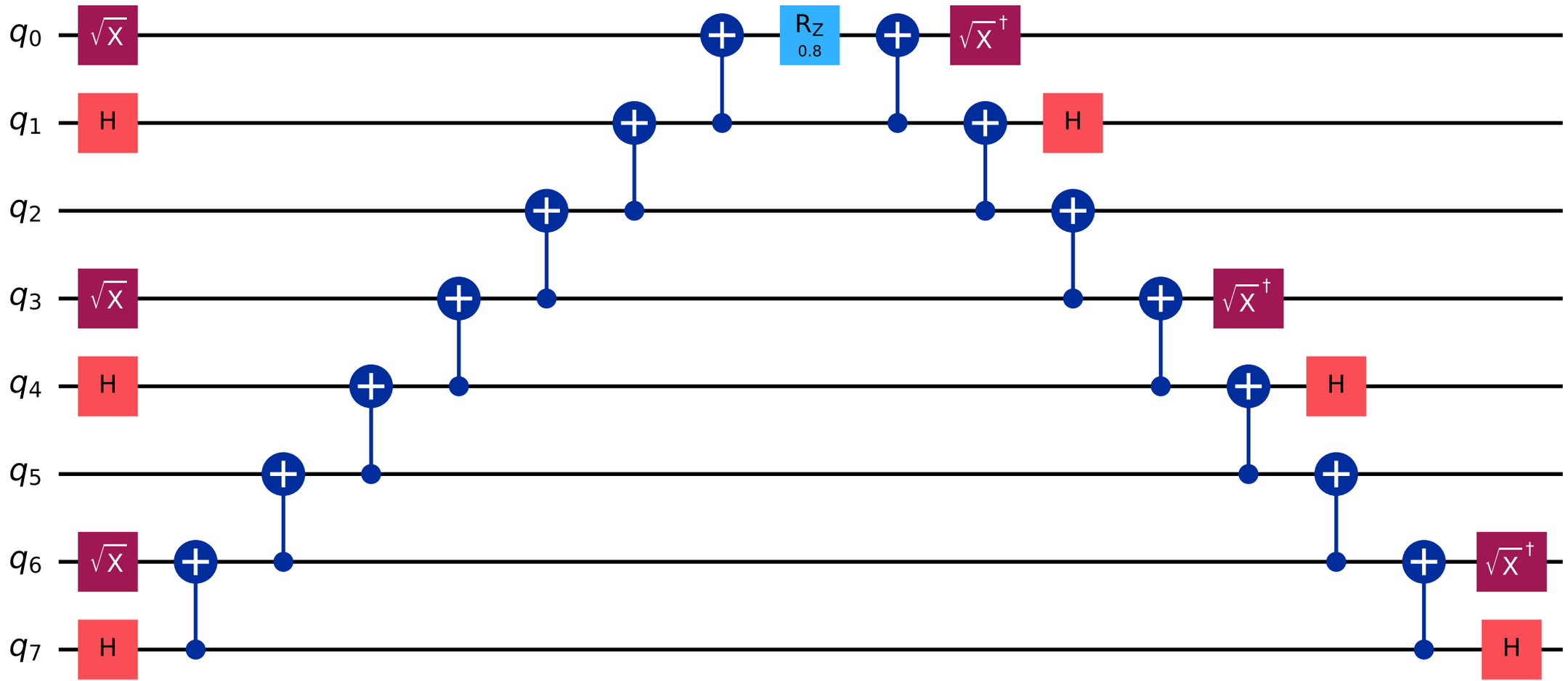

S
Z(a)
H
If
mid[7] ^ mid[8] ^ mid[9] ^ mid...
RZ
−0.8
Else
RZ
0.8
S
16
17
18
19
20
21
22
15
27
23
pr]


Pauli correction...
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
X
If
X
If
X
If
X
If
X
If
X
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
Z
S
24
25
26
28
29
[expr]
mid_23=0x1
mid_24=0x1
mid_25=0x1
mid_26=0x1
mid_27=0x1
[expr]
mid_2

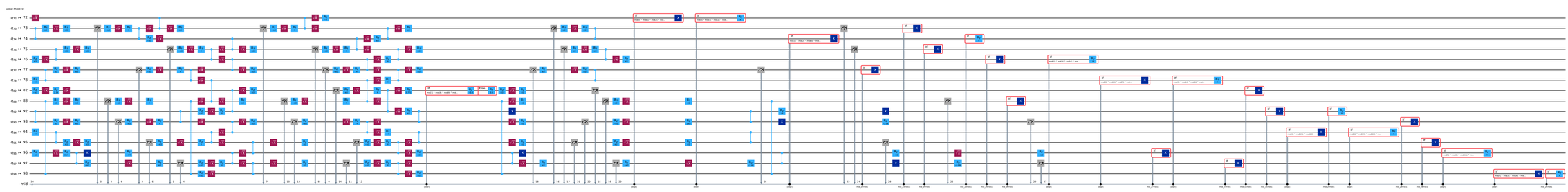

If
X
4=0x1
If
X
mid_27=0x1
If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
X
[expr]
If
Z
mid_23=0x1
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
Z
[expr]
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
X
[expr]
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
Z
[expr]
mid_1


If
X
5=0x1
If
X
mid_15=0x1
If
Z
mid_29=0x1
If
mid[6] ^ mid[15] ^ mid[22]
X
[expr]
If
X
mid_29=0x1
If
mid[6] ^ mid[15] ^ mid[22] ^ m...
Z
[expr]
If
X
mid_28=0x1
If
mid[5] ^ mid[6] ^ mid[15] ^ mi...
Z
[expr]
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
X
[expr]
If
Z
mid_26=0x1

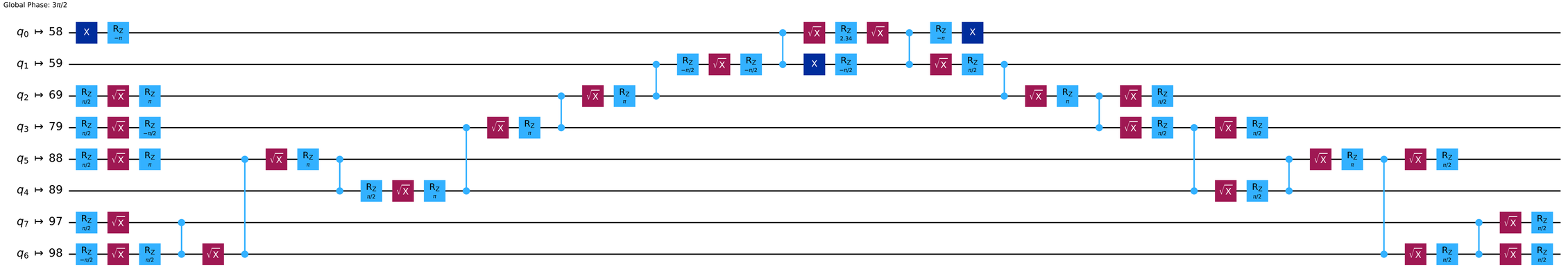
[XYZXYZXY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14



[XYZXYZXY] DYNAMIC circuit, fault-tolerant (dfs- ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:102 sx:82 cz:44 measure:30 if_else:25 x:6)
exp(-i P/ 8=0.4 P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7 (8 logical qubits)
2q depth 6, total depth 43

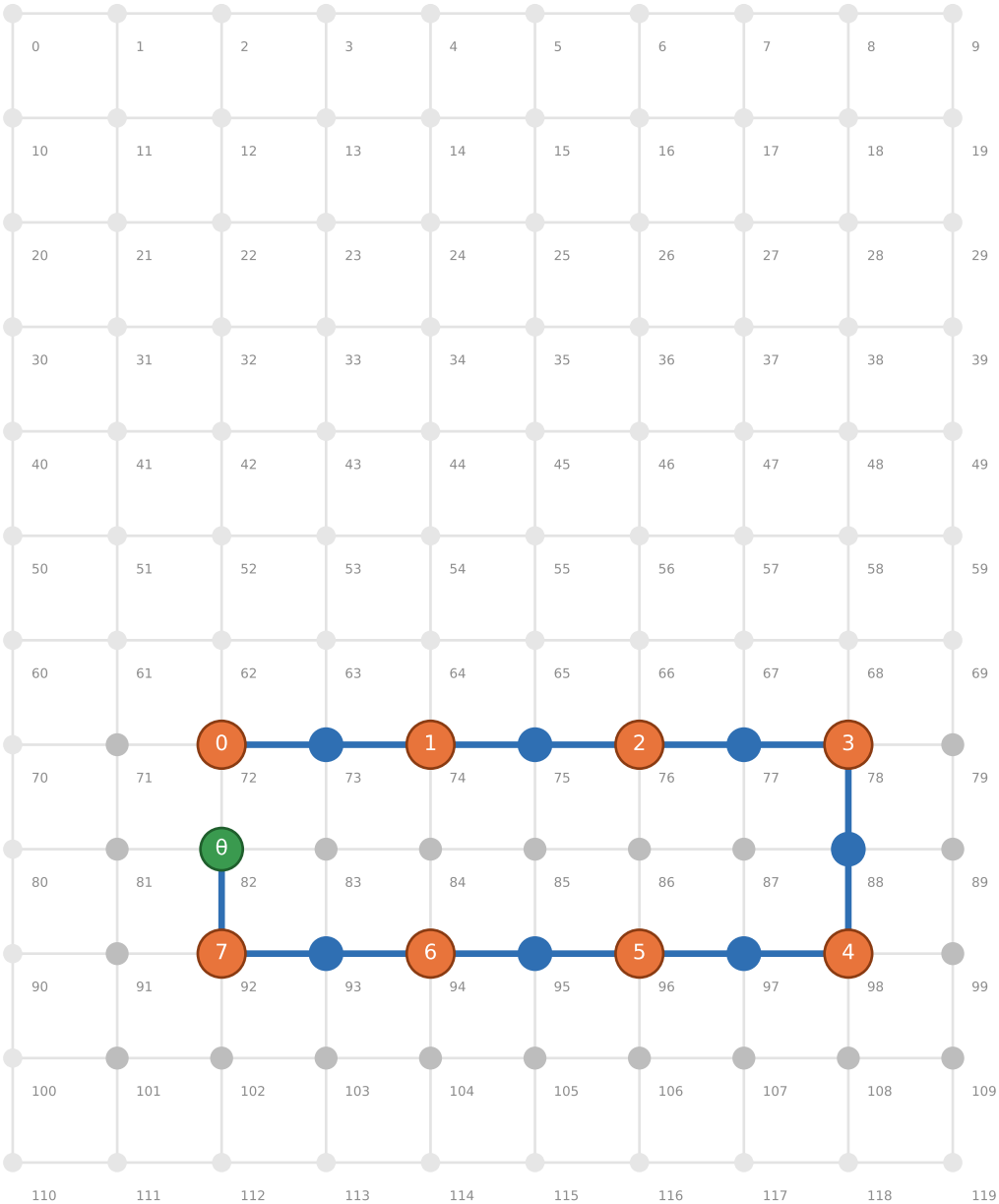


[XYZXYZXY] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:31 sx:24 cz:14 x:3)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14, total depth 47

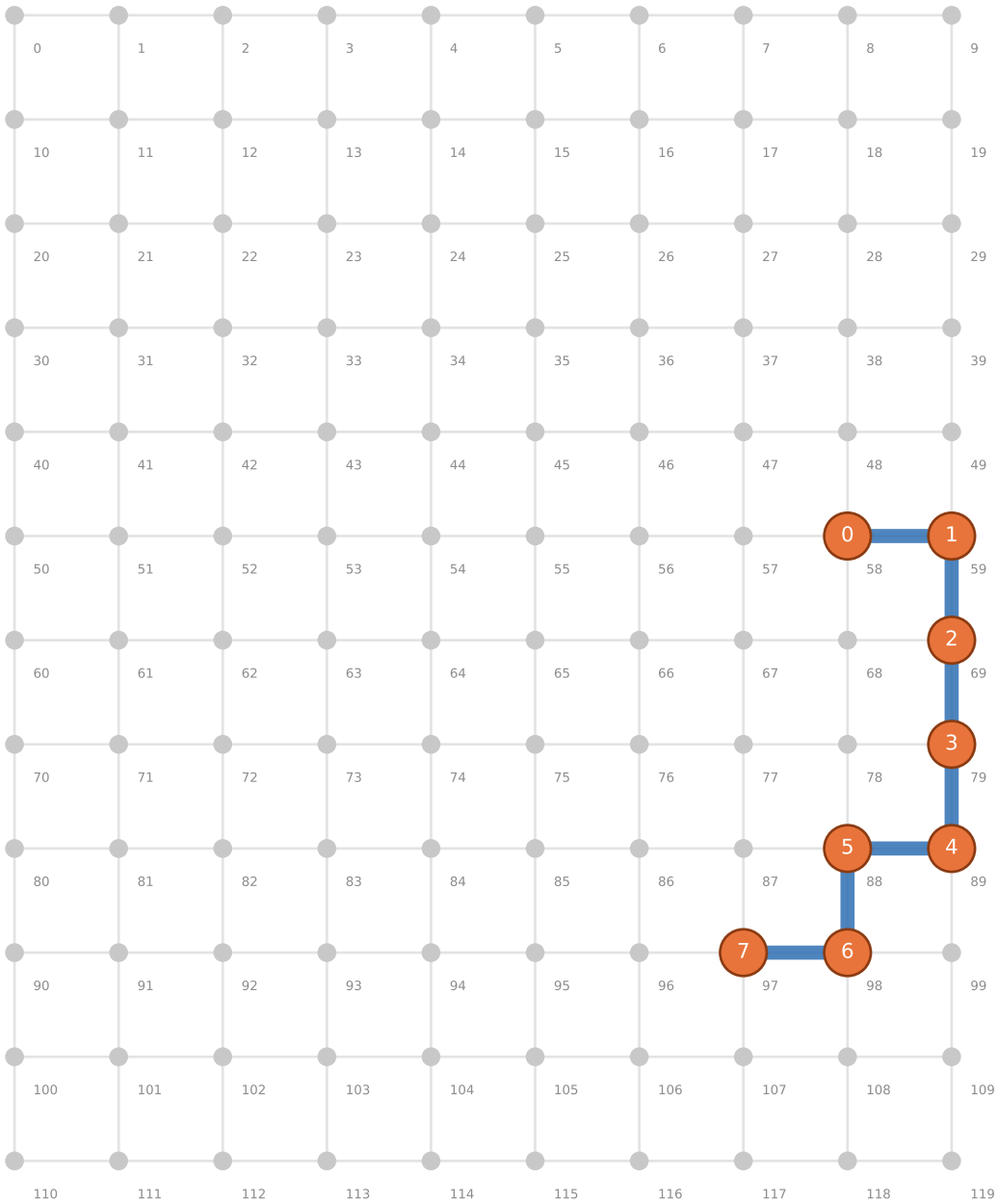


[XYZXYZXY] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

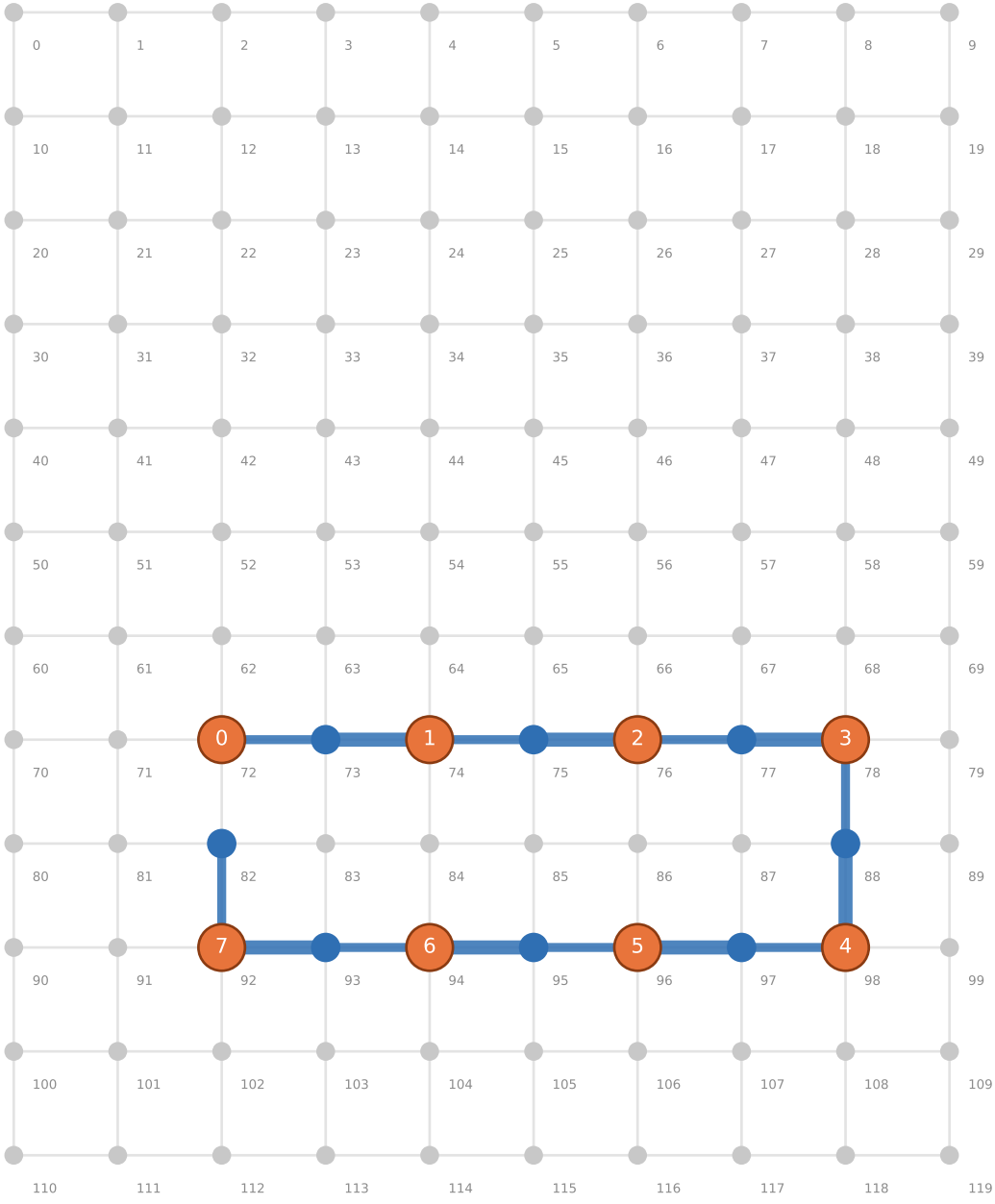
term 0: Y0 X1 Z2 Y3 X4 Z5 Y6 X7
chain length 15, 7 ancillas, 0 |0> fillers



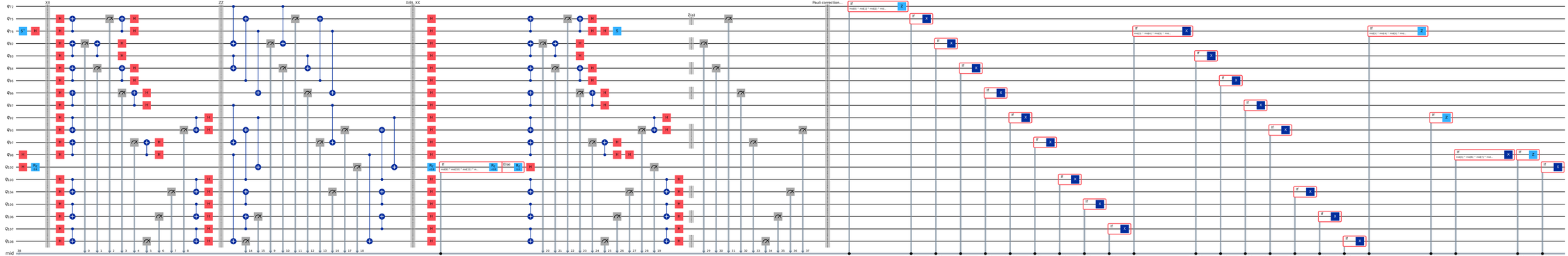
[XYZXYZXY] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [58, 59, 69, 79, 89, 88, 98, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



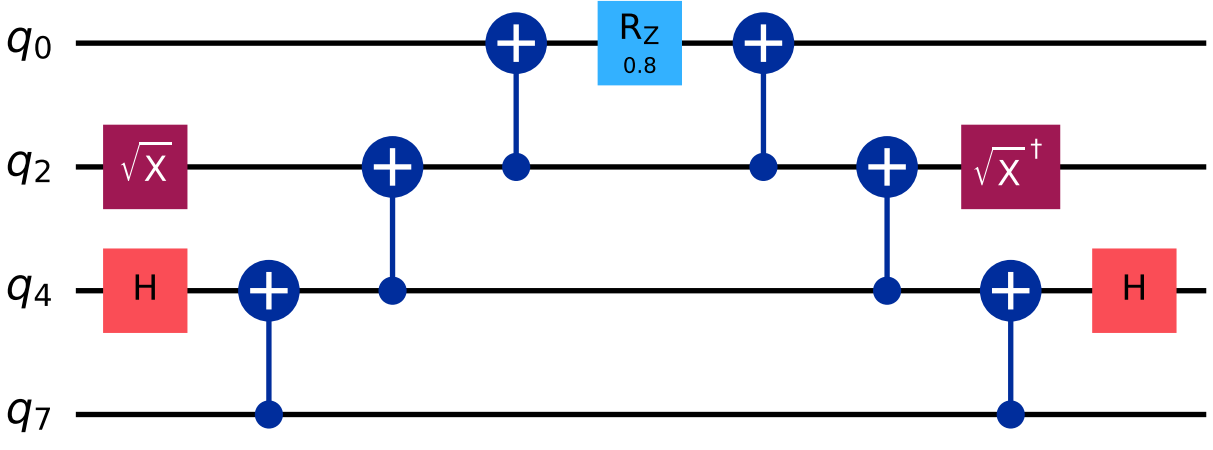
[XYZXYZXY] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 44



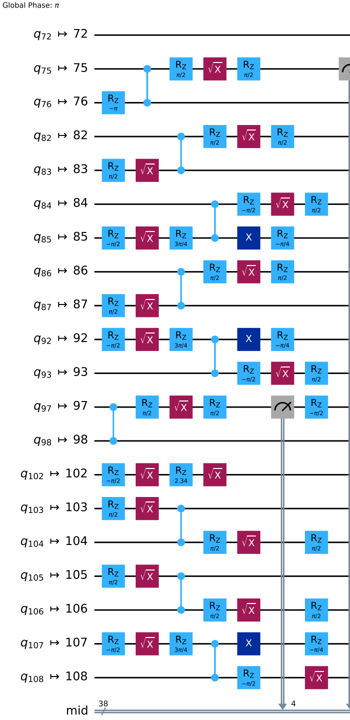
[ZIXIYIZ_gapped] DYNAMIC constant-depth circuit, fault-tolerant (|0> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P)_θ=0.4: P = ZQ Y2 X4 Z7 (8 logical qubits)
2q depth 6, 38 mid-circuit measurements, logical~physical [72, 74, 76, 78, 98, 96, 94, 92]

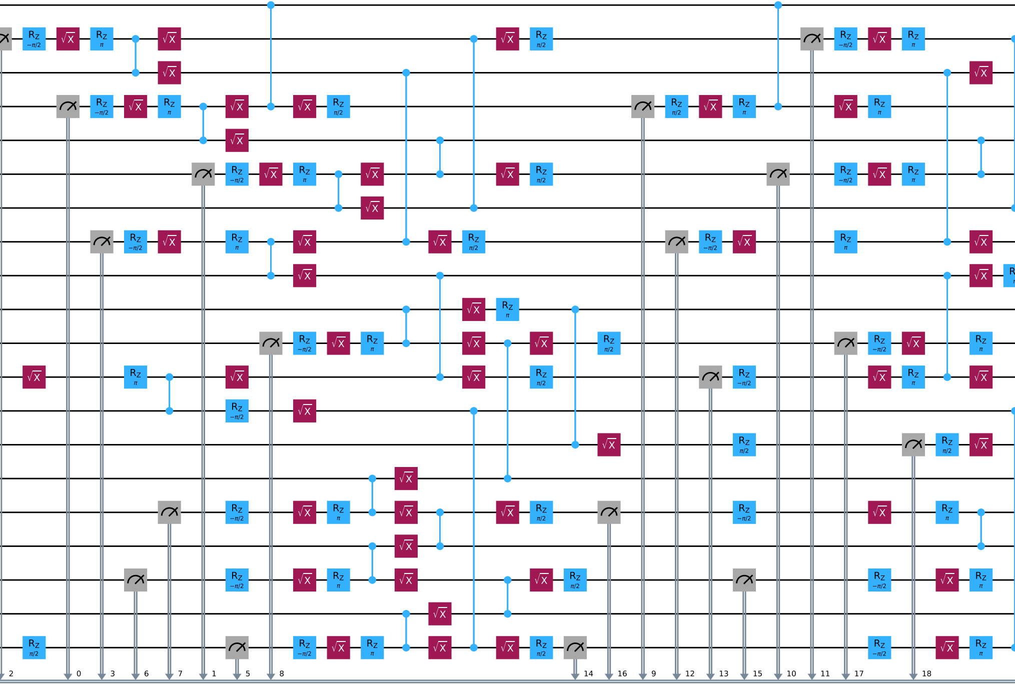


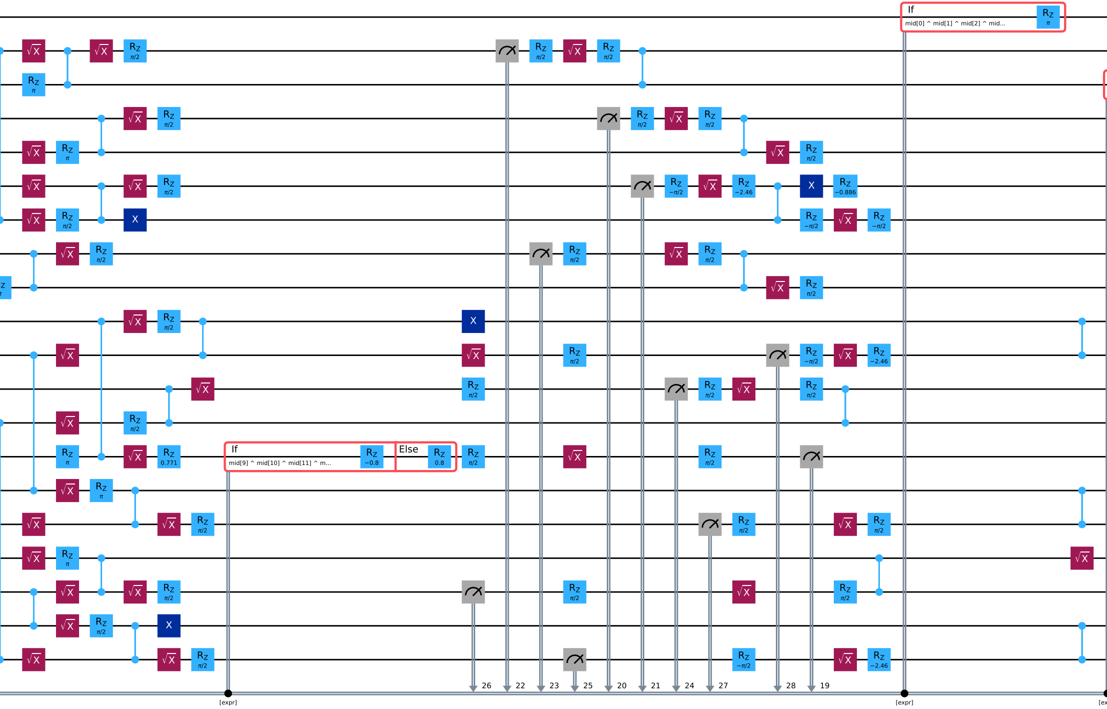
[ZIIXIYIZ_gapped] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 6

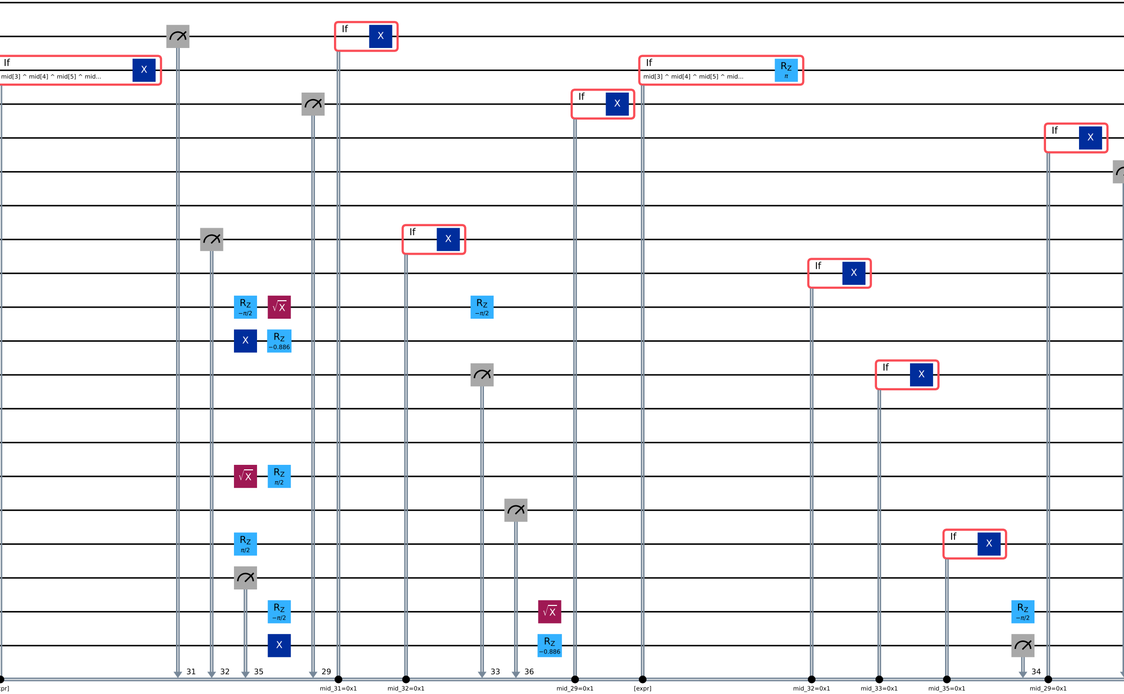


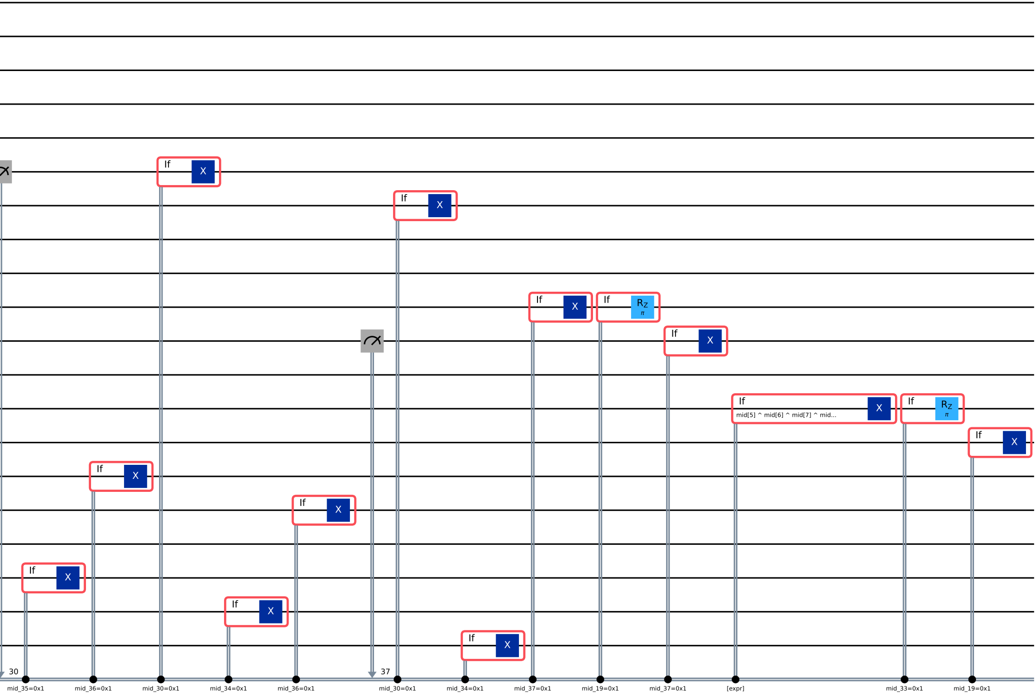
[ZIIXIYIZ_gapped] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:137 sx:110 cz:56 measure:38 if_else:24 x:9)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 6, total depth 39


Global Phase: π
q72 ↦ 72
q75 ↦ 75
q76 ↦ 76
q82 ↦ 82
q83 ↦ 83
q84 ↦ 84
q85 ↦ 85
q86 ↦ 86
q87 ↦ 87
q92 ↦ 92
q93 ↦ 93
q97 ↦ 97
q98 ↦ 98
q102 ↦ 102
q103 ↦ 103
q104 ↦ 104
q105 ↦ 105
q106 ↦ 106
q107 ↦ 107
q108 ↦ 108
mid
38
4
RZ π/2
RZ −π
RZ −π/2
RZ 3π/4
RZ −π/4
RZ 2.34
√X
X

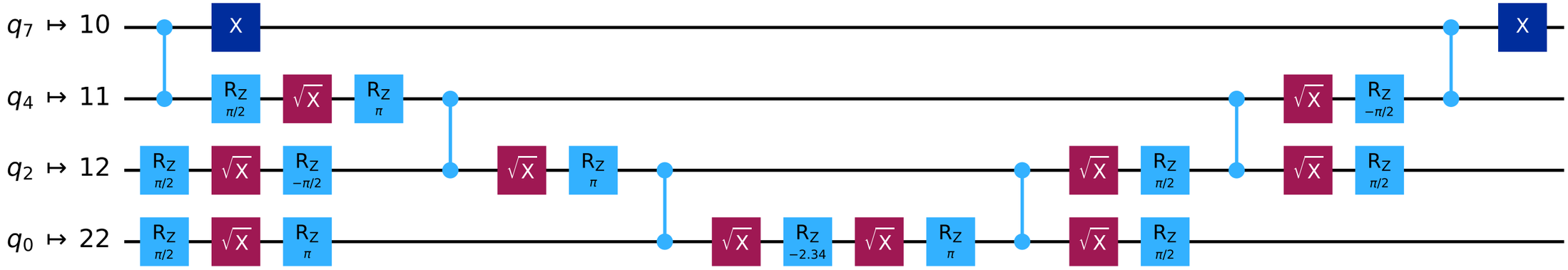

RZ
√X
π
π/2
−π/2
2
0
3
6
7
1
5
8
14
16
9
12
13
15
10
11
17
18


If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
If
mid[9] ^ mid[10] ^ mid[11] ^ m...
Else
√X
X
RZ π/2
RZ −π/2
RZ π
RZ 0.771
RZ −0.8
RZ 0.8
RZ −2.46
RZ −0.886
[expr]
26
22
23
25
20
21
24
27
28
19


If
X
mid[3] ^ mid[4] ^ mid[5] ^ mid...
If
X
If
X
If
R$_Z$
π
mid[3] ^ mid[4] ^ mid[5] ^ mid...
If
X
If
X
If
X
R$_Z$
$-\pi/2$
$\sqrt{X}$
X
R$_Z$
-0.886
R$_Z$
$-\pi/2$
If
X
$\sqrt{X}$
R$_Z$
$\pi/2$
R$_Z$
$\pi/2$
If
X
R$_Z$
$-\pi/2$
X
$\sqrt{X}$
R$_Z$
-0.886
R$_Z$
$-\pi/2$
31
32
35
29
33
36
34
pr]
mid_31=0x1
mid_32=0x1
mid_29=0x1
[expr]
mid_32=0x1
mid_33=0x1
mid_35=0x1
mid_29=0x1

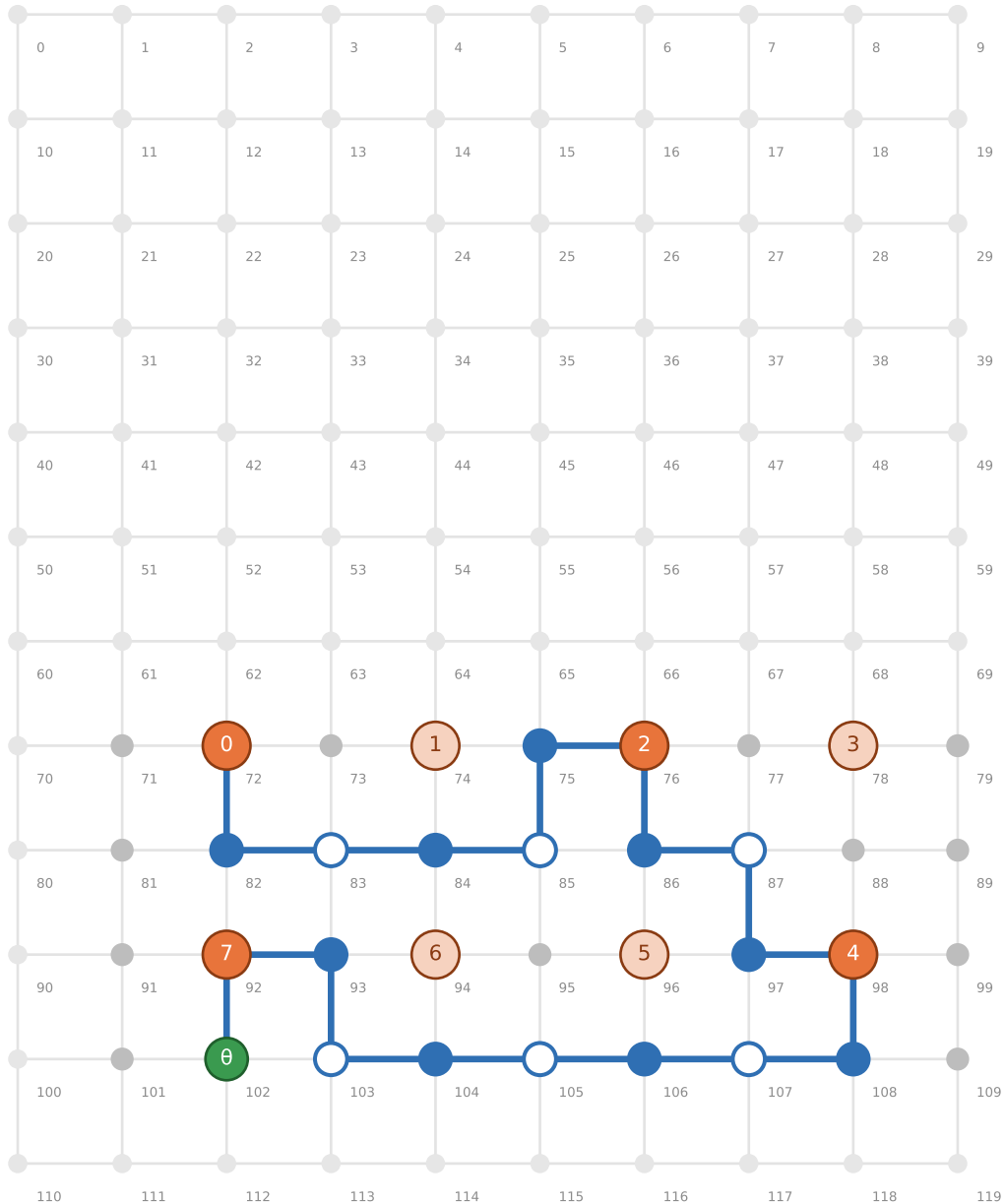

If
X
30
mid_35=0x1
mid_36=0x1
mid_30=0x1
mid_34=0x1
mid_36=0x1
37
mid_30=0x1
mid_34=0x1
mid_37=0x1
R_Z
π
mid_19=0x1
mid_37=0x1
mid[5] ^ mid[6] ^ mid[7] ^ mid...
[expr]
mid_33=0x1
mid_19=0x1

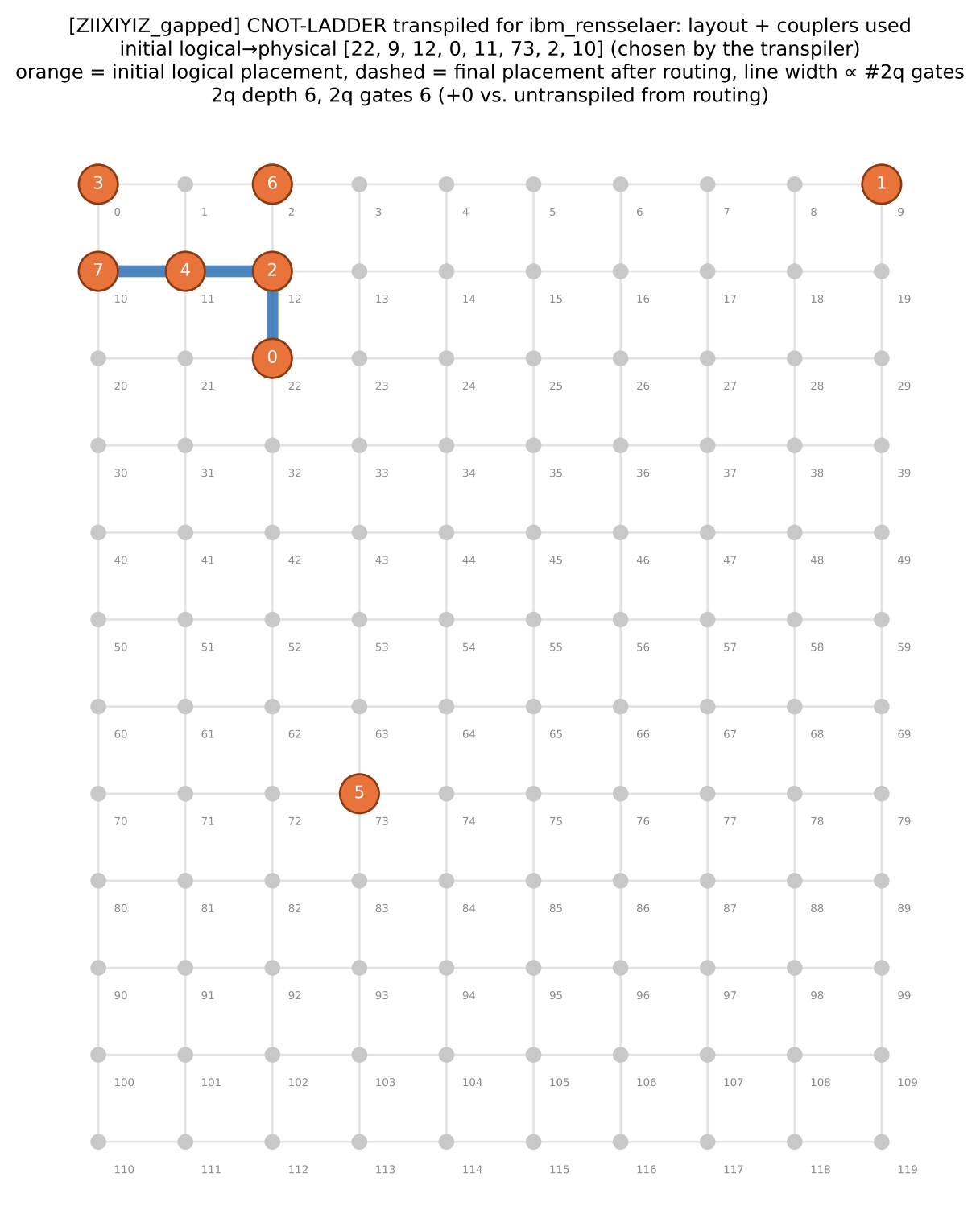
[ZIIXIYIZ_gapped] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:13 sx:10 cz:6 x:2)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
Global Phase: 0
2q depth 6, total depth 20



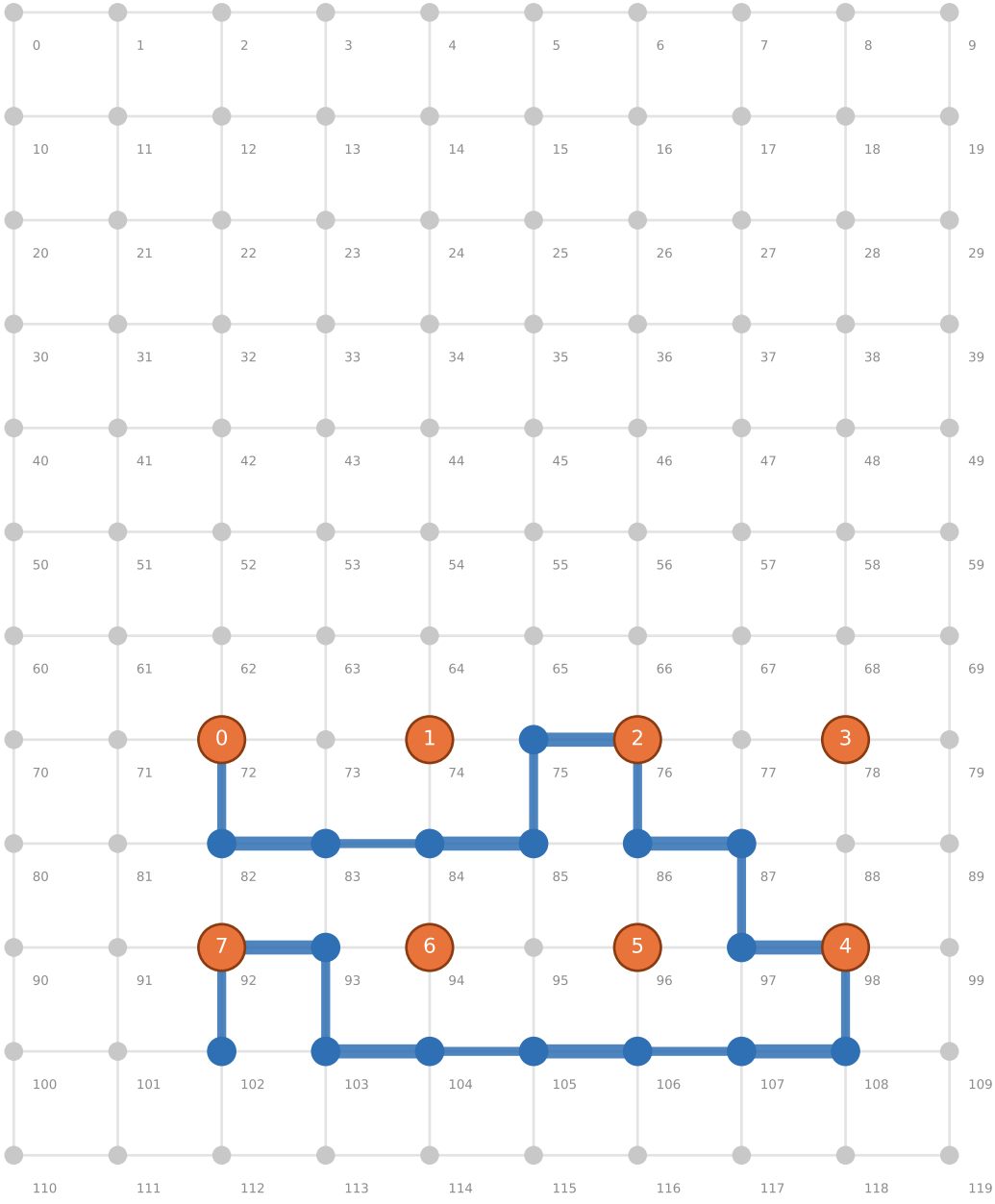
[ZIIXIYIZ_gapped] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Y2 X4 Z7
chain length 19, 9 ancillas, 6 |0> fillers

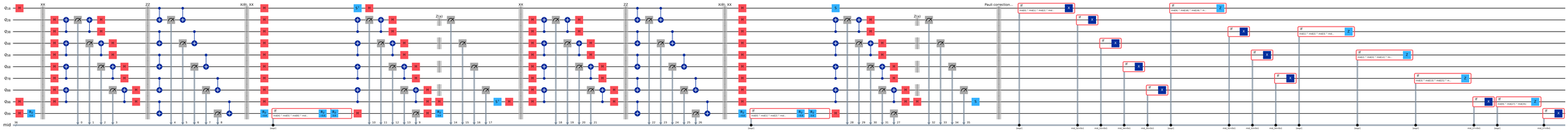



[ZIIXIYIZ_gapped] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 9, 12, 0, 11, 73, 2, 10] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
3
6
1
7
4
2
0
5
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

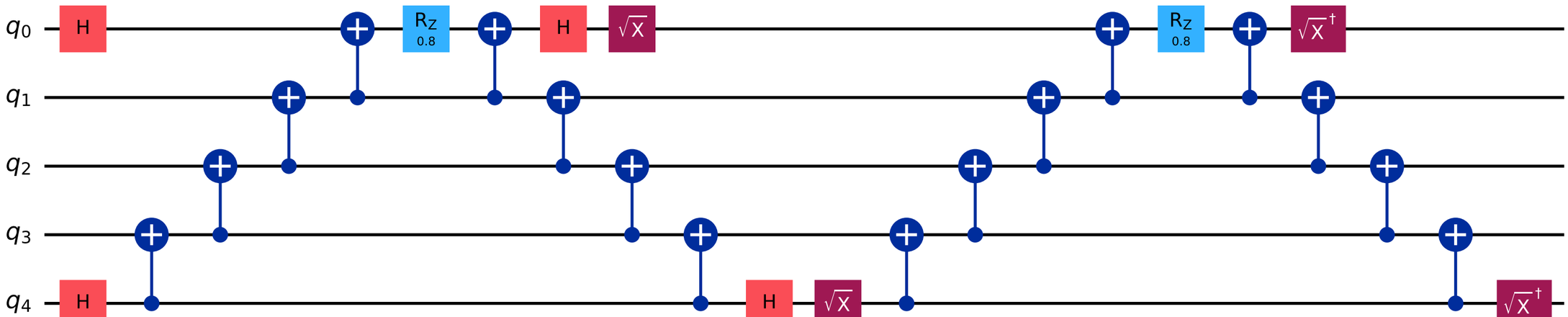
[ZIIXIYIZ_gapped] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 56



[XZZZX+YZZZY] DYNAMIC constant-depth circuit, fault-tolerant (@> ancilla) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 ·Y0 Z1 Z2 Z3 Y4 · (5 logical qubits)
2q depth 12, 36 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]

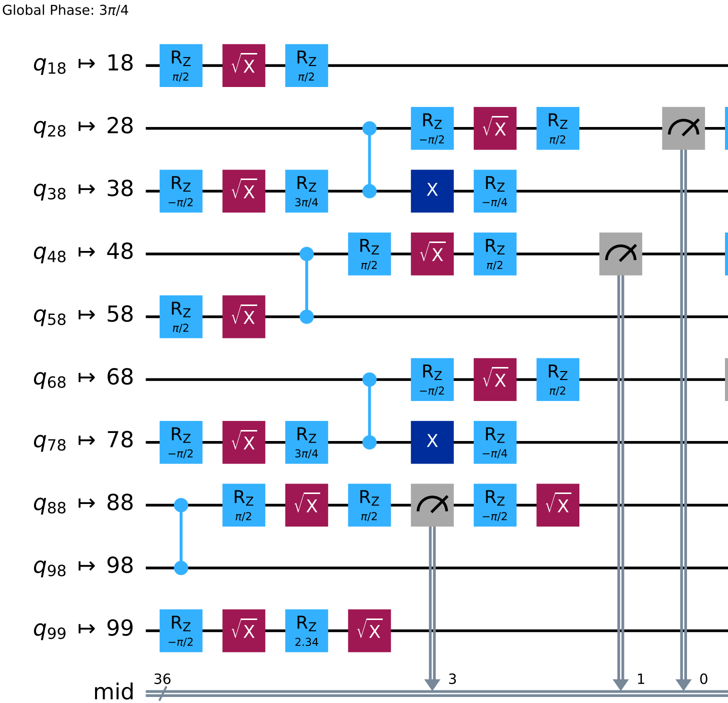


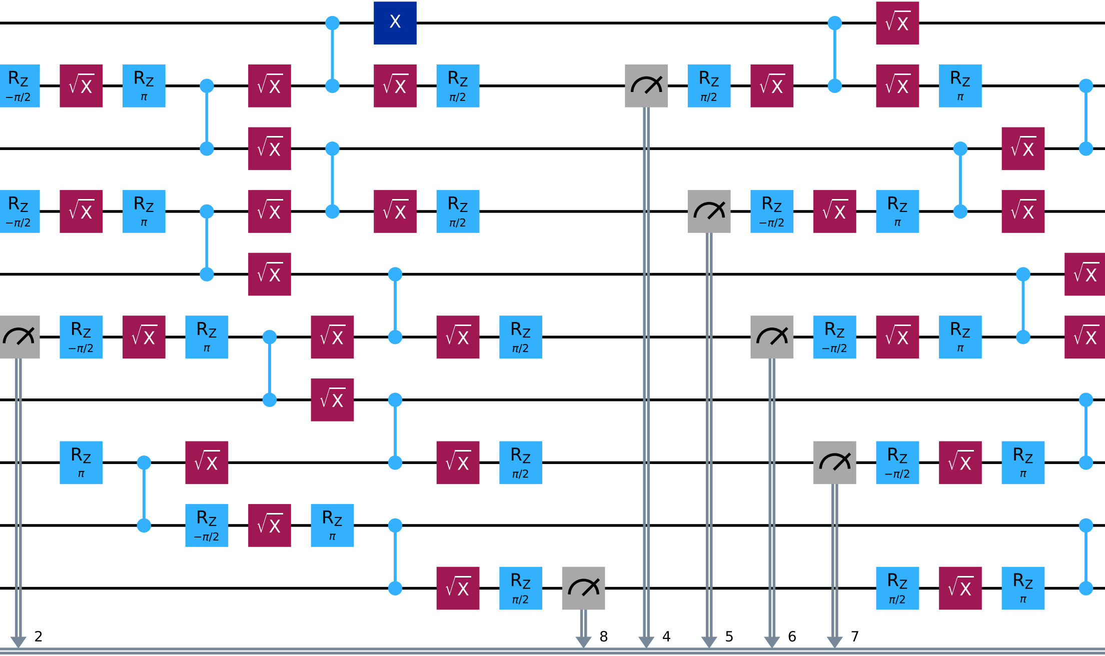
[XZZZX+YZZZY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 16

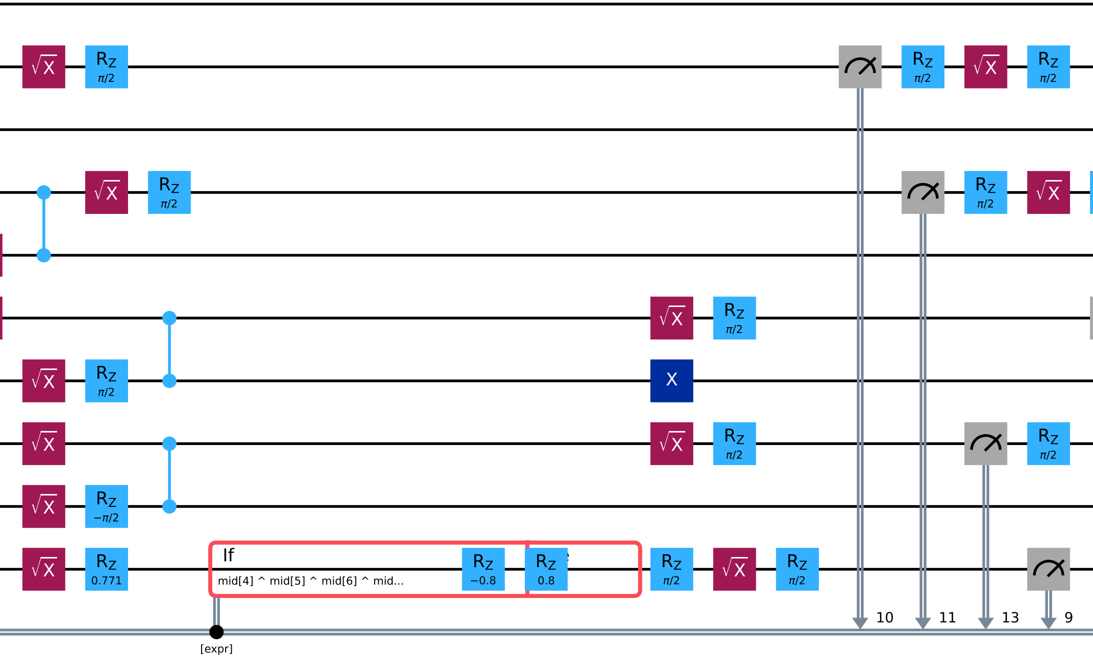


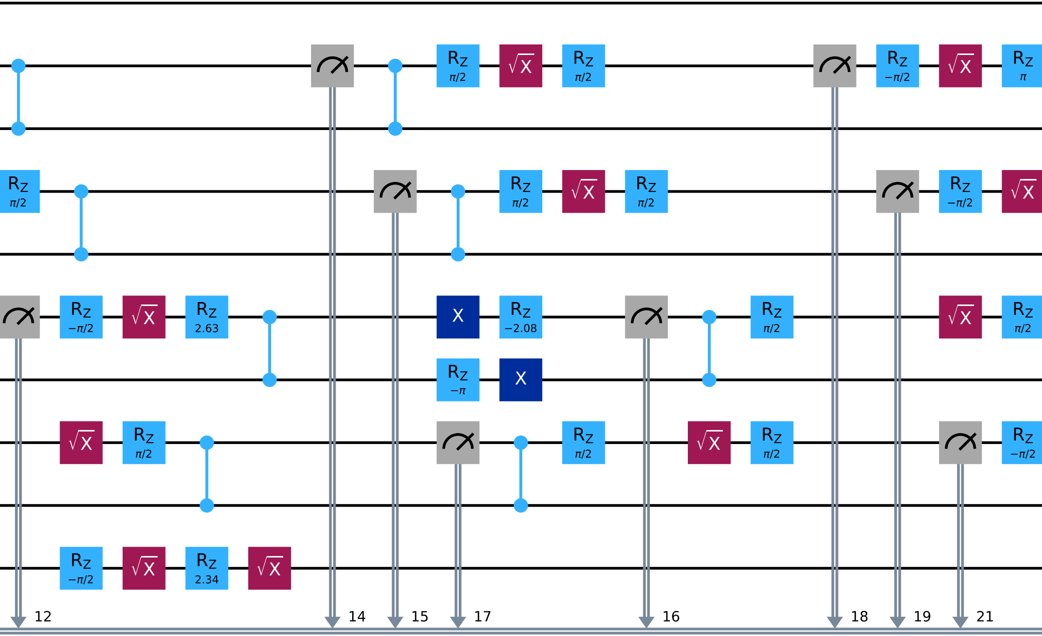
[XZZZX+YZZZY] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_rensselaer (ISA: rz:124 sx:101 cz:52
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 12, total depth 72

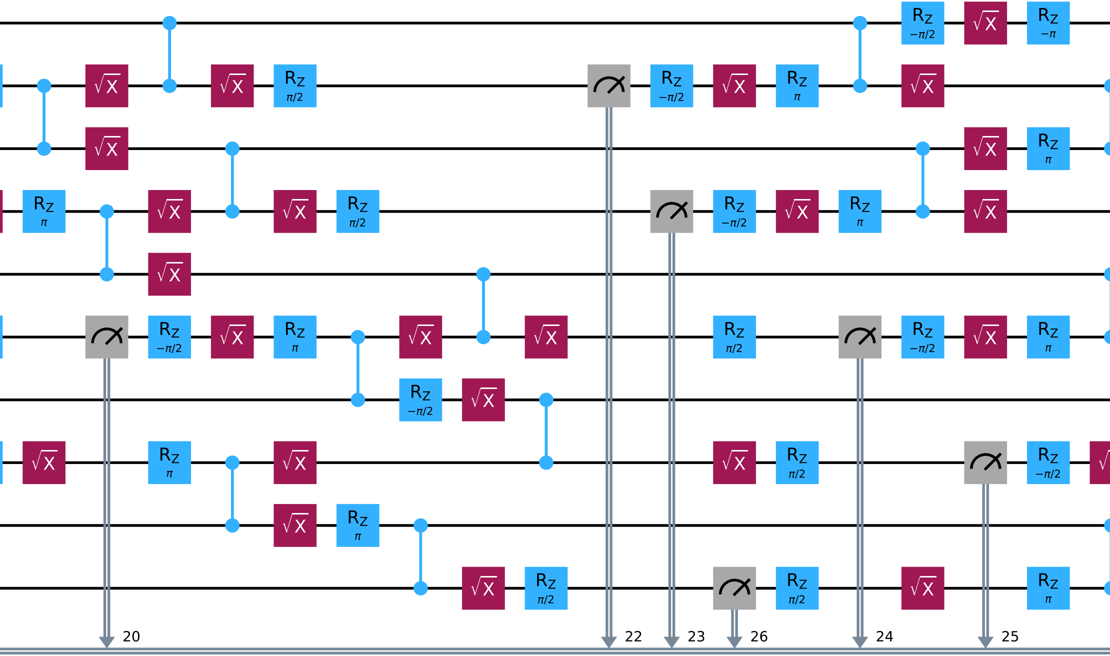
measure:36 if_else:17 x:8)


Global Phase: 3π/4
$q_{18} \mapsto 18$
RZ π/2
√X
RZ π/2
$q_{28} \mapsto 28$
RZ −π/2
√X
RZ π/2
$q_{38} \mapsto 38$
RZ −π/2
√X
RZ 3π/4
X
RZ −π/4
$q_{48} \mapsto 48$
RZ π/2
√X
RZ π/2
$q_{58} \mapsto 58$
RZ π/2
√X
$q_{68} \mapsto 68$
RZ −π/2
√X
RZ π/2
$q_{78} \mapsto 78$
RZ −π/2
√X
RZ 3π/4
X
RZ −π/4
$q_{88} \mapsto 88$
RZ π/2
√X
RZ π/2
RZ −π/2
√X
$q_{98} \mapsto 98$
$q_{99} \mapsto 99$
RZ −π/2
√X
RZ 2.34
√X
mid
36
3
1
0

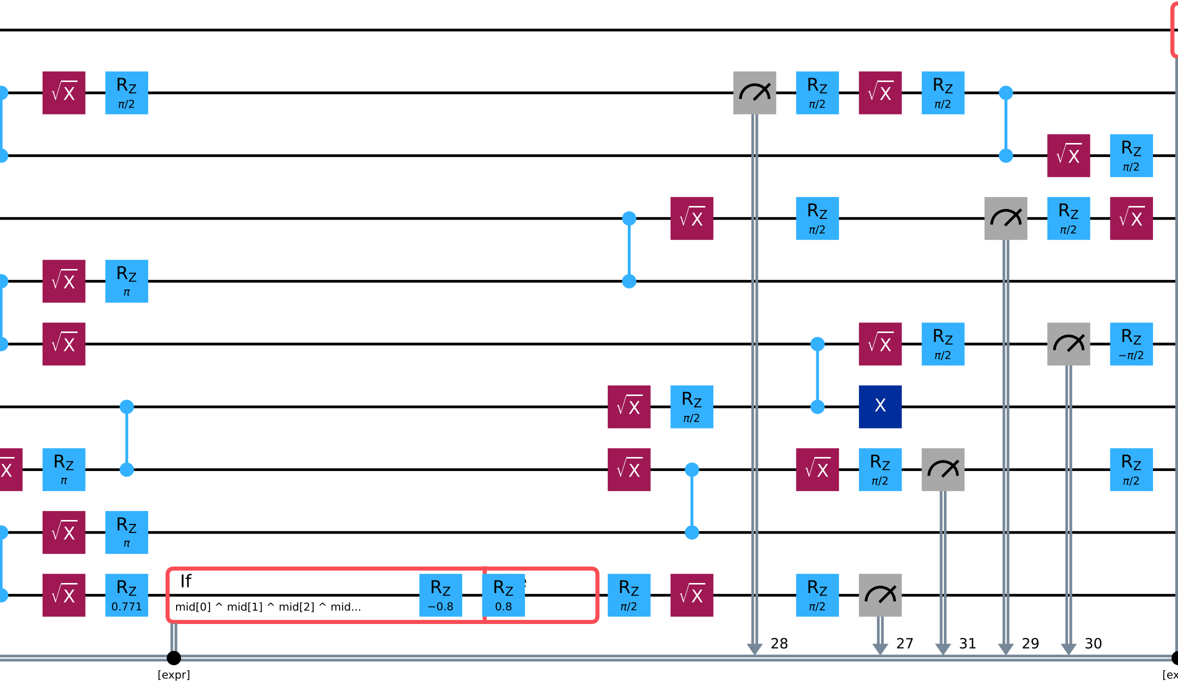

X
√X
Rz
−π/2
π
π/2
2
8
4
5
6
7

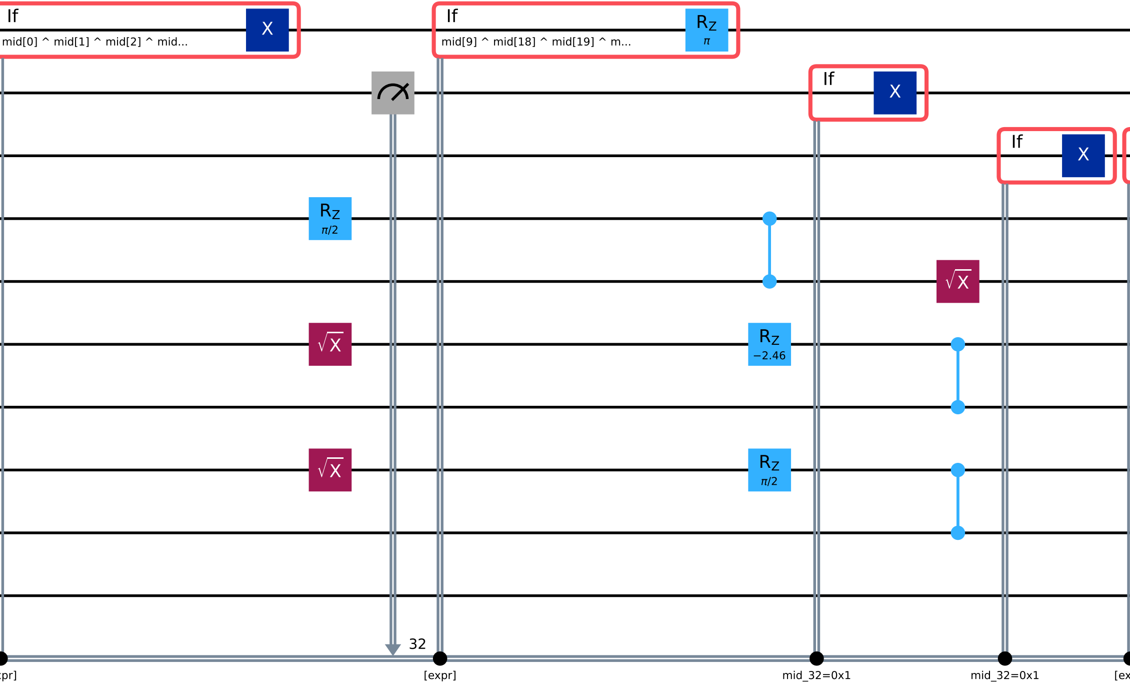

√X
R_Z π/2
√X
R_Z π/2
√X
R_Z π/2
√X
R_Z π/2
√X
R_Z π/2
√X
R_Z −π/2
√X
R_Z 0.771
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
R_Z −0.8
R_Z 0.8
[expr]
√X
R_Z π/2
X
√X
R_Z π/2
R_Z π/2
√X
R_Z π/2
R_Z π/2
√X
R_Z π/2
R_Z π/2
√X
R_Z π/2
10
11
13
9

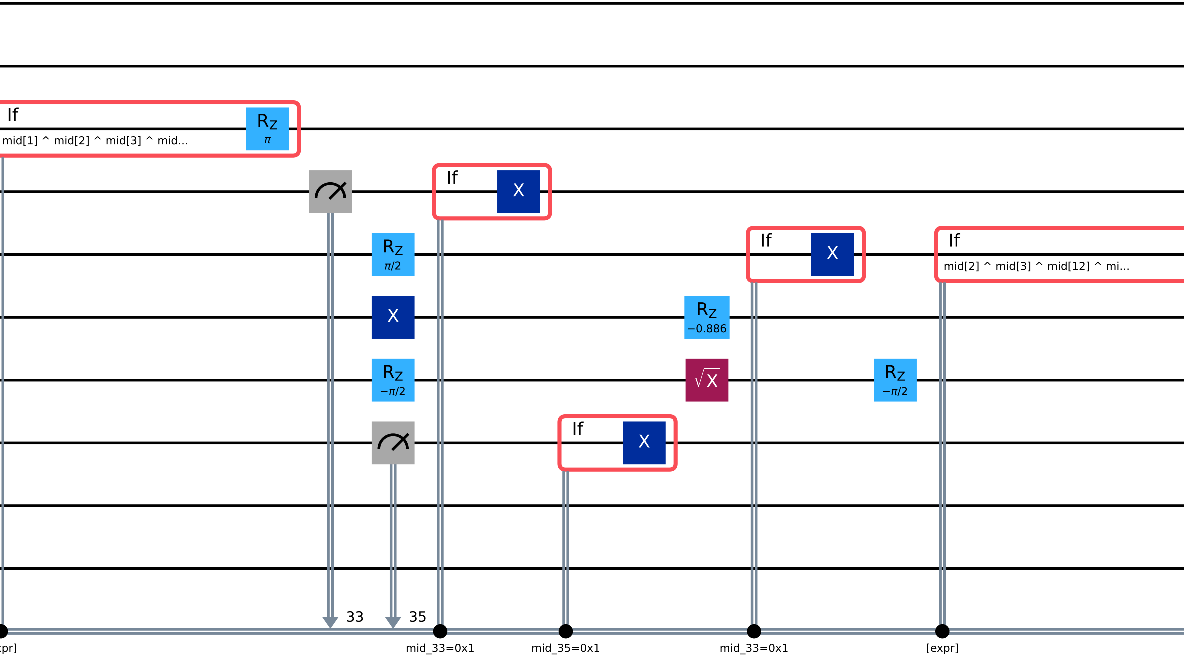

RZ π/2
√X
RZ π/2
RZ −π/2
√X
RZ π
RZ π/2
RZ π/2
√X
RZ π/2
RZ −π/2
√X
RZ −π/2
√X
RZ 2.63
X
RZ −2.08
RZ π/2
√X
RZ π/2
RZ −π
X
√X
RZ π/2
RZ π/2
√X
RZ π/2
RZ −π/2
RZ −π/2
√X
RZ 2.34
√X
12
14
15
17
16
18
19
21



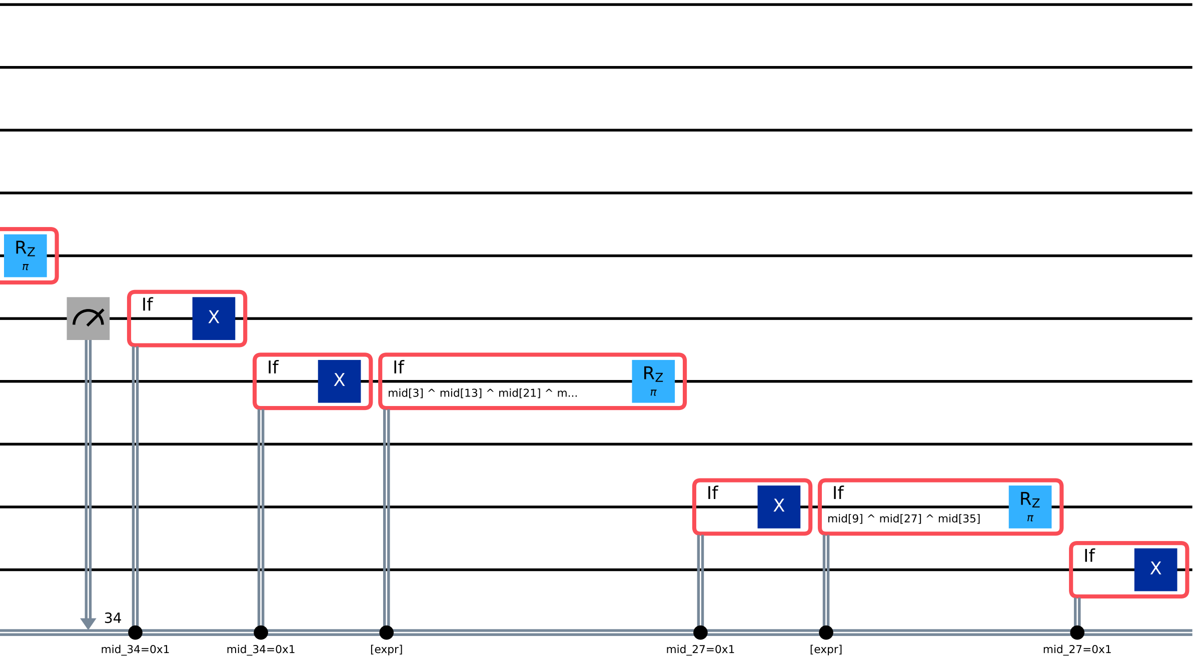
RZ −π/2
√X
RZ −π
√X
√X
RZ π/2
RZ −π/2
√X
RZ π
√X
√X
√X
RZ π
RZ π
RZ π/2
RZ −π/2
√X
RZ π
√X
√X
RZ −π/2
√X
RZ π
√X
RZ π/2
RZ −π/2
√X
RZ π
RZ −π/2
√X
√X
√X
RZ π
RZ π/2
RZ −π/2
RZ π/2
√X
RZ π
√X
√X
√X
RZ π/2
RZ π
√X
RZ π/2
RZ π/2
√X
RZ π
20
22
23
26
24
25


√X
R_Z π/2
R_Z π/2
√X
R_Z π/2
√X
R_Z π/2
√X
R_Z π/2
R_Z π/2
R_Z π/2
√X
√X
R_Z π
√X
√X
R_Z π/2
R_Z π/2
R_Z −π/2
√X
R_Z π/2
X
R_Z π
√X
√X
R_Z π/2
R_Z π/2
√X
R_Z π
√X
R_Z 0.771
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
R_Z −0.8
R_Z 0.8
R_Z π/2
√X
R_Z π/2
28
27
31
29
30
[expr]
[ex

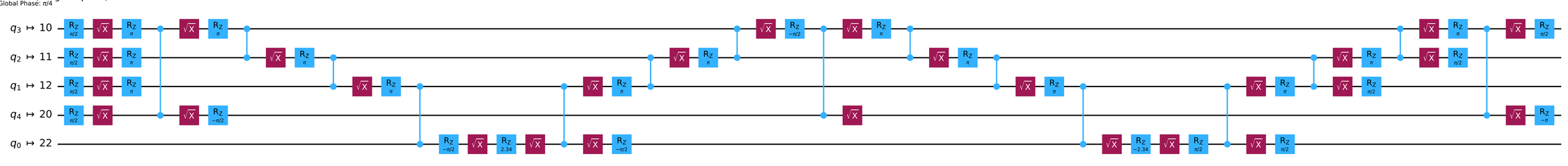

If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
X
If
mid[9] ^ mid[18] ^ mid[19] ^ m...
R_Z
π
If
X
If
X
R_Z
π/2
√X
√X
32
R_Z
−2.46
R_Z
π/2
√X
pr]
[expr]
mid_32=0x1
mid_32=0x1
[ex



If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
R_Z
π
R_Z
$\pi/2$
X
R_Z
$-\pi/2$
33
35
If
X
If
X
R_Z
-0.886
$\sqrt{X}$
If
X
R_Z
$-\pi/2$
If
mid[2] ^ mid[3] ^ mid[12] ^ mi...
pr]
mid_33=0x1
mid_35=0x1
mid_33=0x1
[expr]

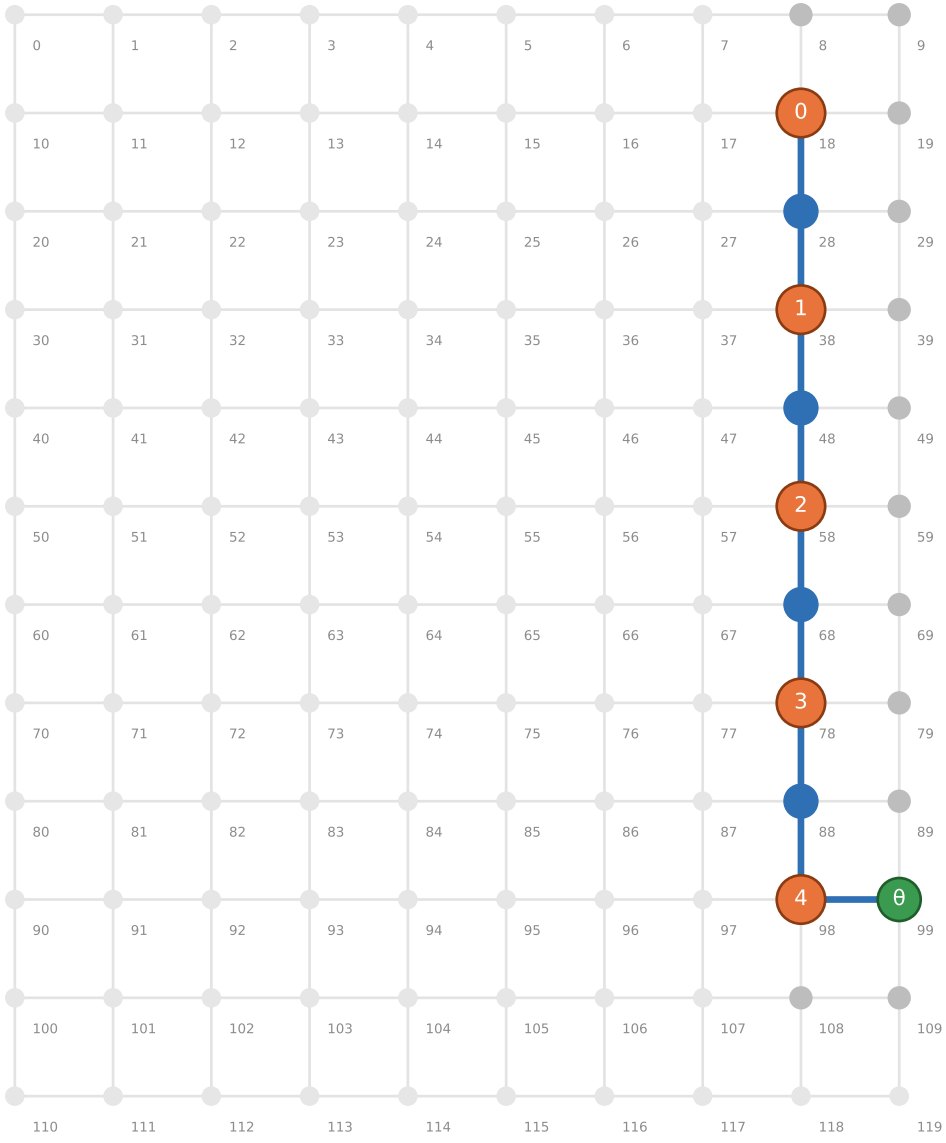

RZ
π
If
X
If
X
If
mid[3] ^ mid[13] ^ mid[21] ^ m...
RZ
π
If
X
If
mid[9] ^ mid[27] ^ mid[35]
RZ
π
If
X
34
mid_34=0x1
mid_34=0x1
[expr]
mid_27=0x1
[expr]
mid_27=0x1

[XZZZX+YZZZY] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:30 sx:28 cz:15)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 15, total depth 52

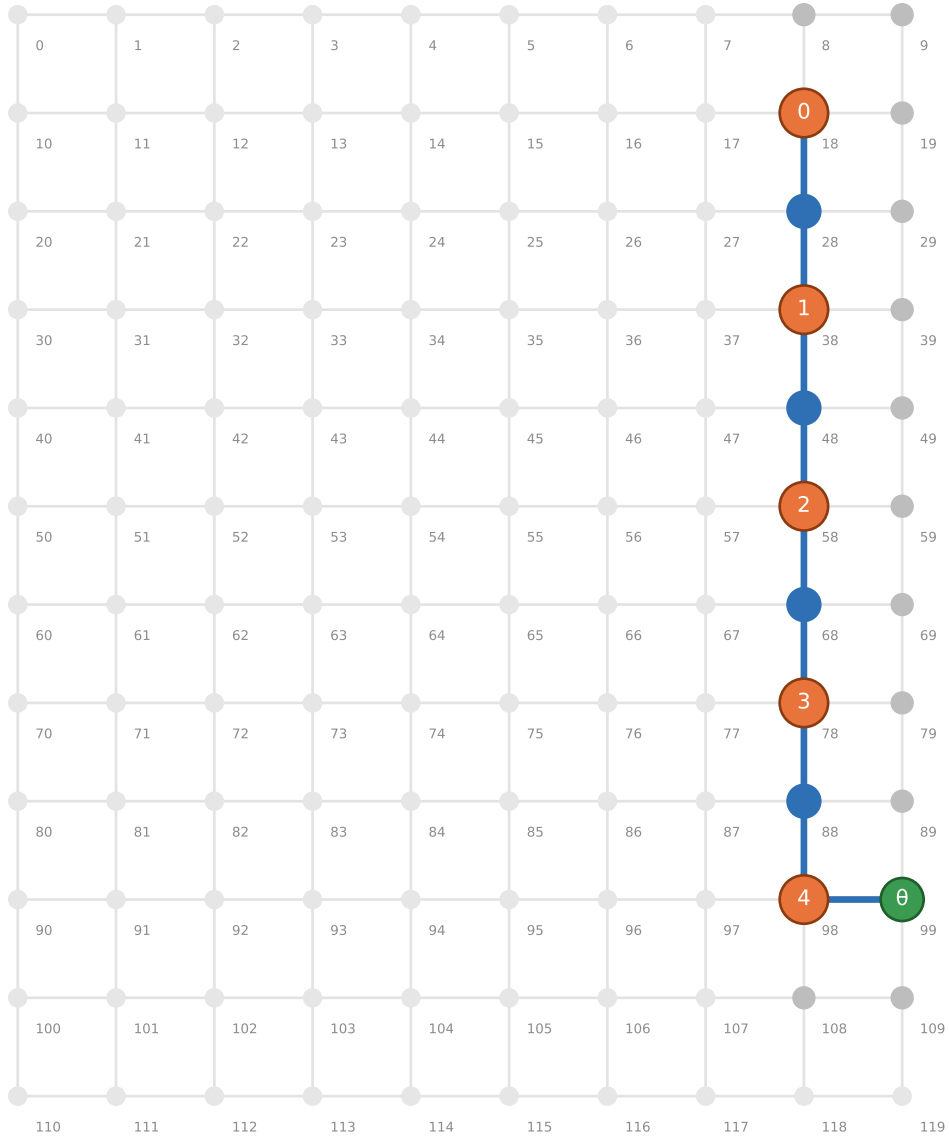


[XZZZX+YZZZY] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

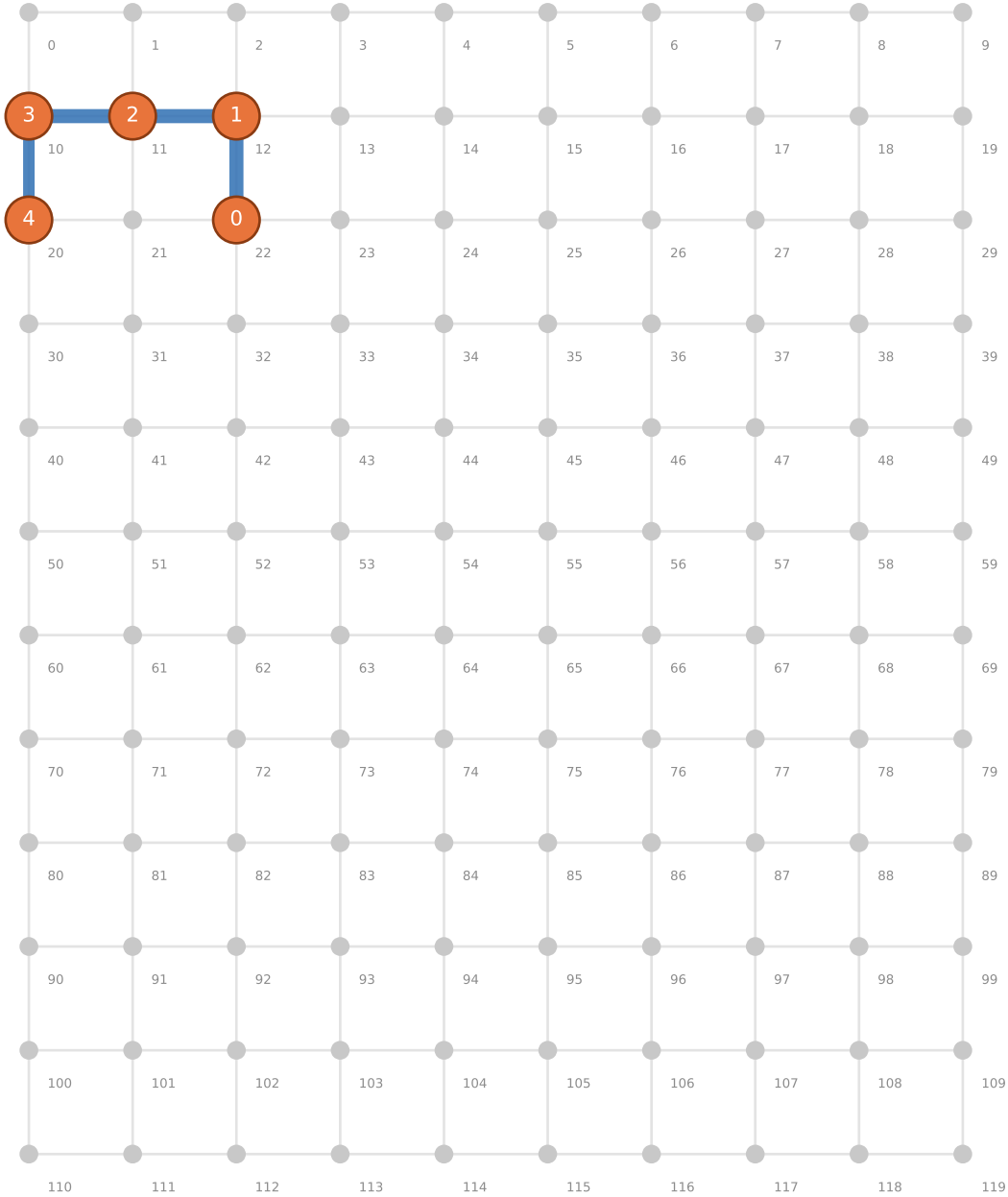
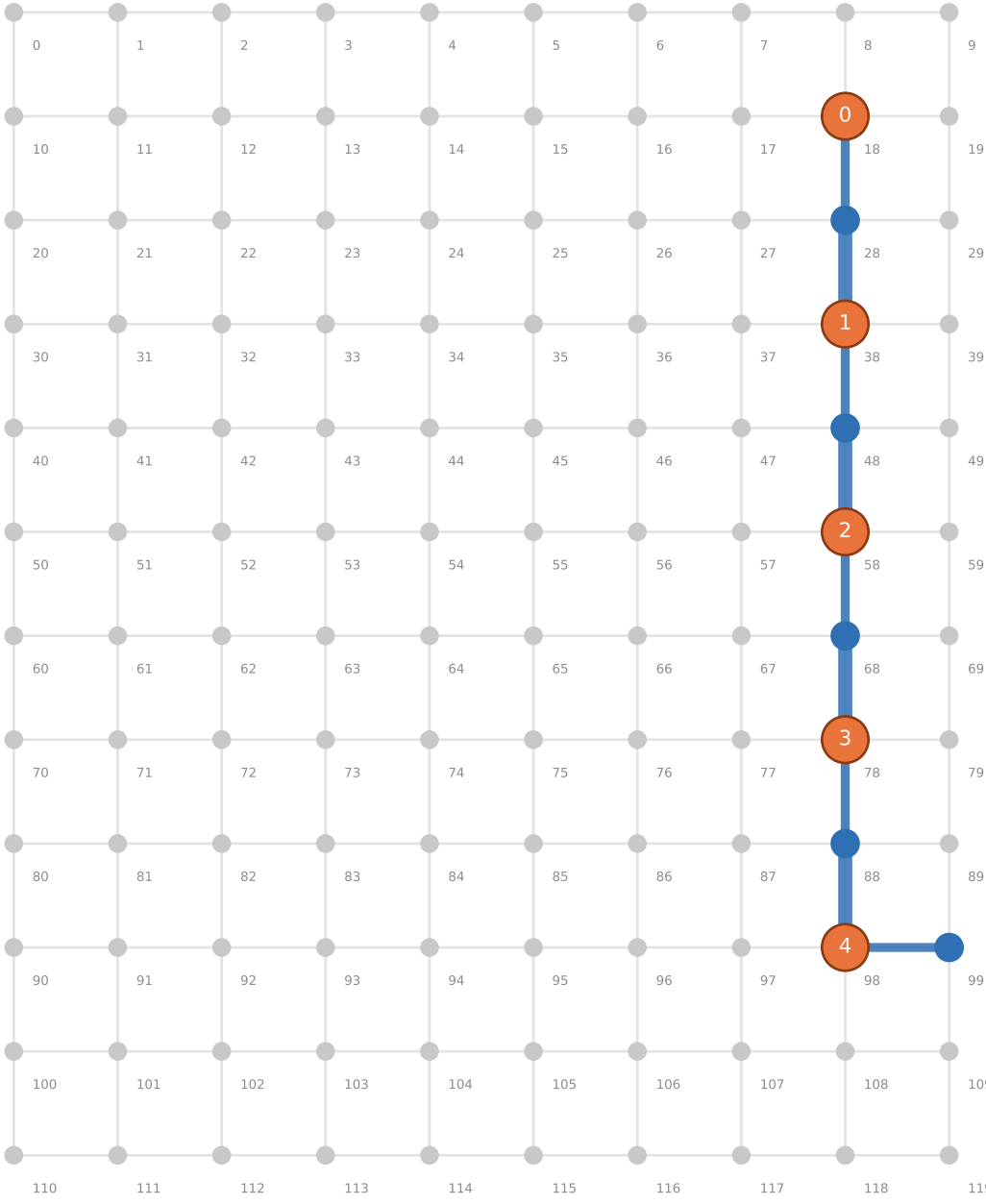
term 0: X0 Z1 Z2 Z3 X4
chain length 9, 4 ancillas, 0 |0> fillers



term 1: Y0 Z1 Z2 Z3 Y4
chain length 9, 4 ancillas, 0 |0> fillers



[XZZZX+YZZZY] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 12, 11, 10, 20] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 15, 2q gates 15 (-1 vs. untranspiled from routing)



[XZZZX+YZZZY] DYNAMIC circuit (ft) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 12, 2q gates 52